

DEVELOPMENT OF DEEP IMAGE-TO-SCALAR REGRESSORS FOR QUALITY ENHANCEMENT AND SEGMENTATION OF DERMATOLOGICAL MACRO-IMAGES

A THESIS

submitted by

VIPIN V

for the award of the degree

of

DOCTOR OF PHILOSOPHY



**DEPARTMENT OF ELECTRONICS & COMMUNICATION
ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY
KARAikal – 609 609**

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Dedicated to

My parents Mr. Venugopalakrishnan V. and

Mrs. Vijayakumari P. G.,

My wife Mrs. Arya G. S.,

My kid baby Sivadha V. A.,

My research supervisor Dr. Malaya Kumar Nath

CANDIDATE'S DECLARATION

I the undersigned hereby declare that the thesis entitled **DEVELOPMENT OF DEEP IMAGE-TO-SCALAR REGRESSORS FOR QUALITY ENHANCEMENT AND SEGMENTATION OF DERMATOLOGICAL MACRO-IMAGES** is a bonafide record of the research work done by me in partial fulfilment of the requirements for the award of the degree of **Doctor of Philosophy** in Department of Electronics & Communication Engineering of the National Institute of Technology Puducherry, Karaikal during the period 22.02.2021 to 31.08.2023. The contents of this thesis, in full or in parts, have not been submitted by me to any other Institute or University for the award of any degree or diploma.

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THESIS CERTIFICATE

This is to certify that the thesis entitled **DEVELOPMENT OF DEEP IMAGE-TO-SCALAR REGRESSORS FOR QUALITY ENHANCEMENT AND SEGMENTATION OF DERMATOLOGICAL MACRO-IMAGES** submitted by **VIPIN V (EC20D1007)** to the National Institute of Technology Puducherry, Karaikal, for the award of the degree of **Doctor of Philosophy** is a bonafide record of research work carried out by him during the period 22.02.2021 to 31.08.2023 under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

Macroscopic images or macro-images of skin lesions captured using standard digital cameras have wide acceptance in screening dermatology due to their simplicity. Many automated tools for analyzing dermatological macro-images use geometric features of the lesions to characterize them. Hence, accurate segmentation of skin lesions has a critical influence on the reliability of these features. Deep learning (DL)-based segmentation models have emerged as an effective approach for skin lesion segmentation due to their ability to extract features from vast amounts of data. These models are computationally heavy because of their encoder-decoder architecture that performs the image-to-image regression task. Macro-images captured with standard digital cameras often suffer from uneven background illumination, which negatively impacts the performance of lesion segmentation algorithms. In the initial stages of skin cancer, the lesions may not have sufficient intensity difference or contrast from the background region. Additionally, low or extremely high light exposure at the time of capturing the image also reduces the contrast. Low contrast between lesion and background regions of macro-images can also adversely impact segmentation.

This thesis addresses the issue of the computational burden imposed by image-to-image segmentation regressors and the resultant performance degradation caused by uneven background illumination and low contrast of macro-images on the segmentation algorithms. A modified ResNet-based illumination equalization method is proposed using a counter exponential transform (IECET) to equalize the uneven background illumination. An EfficientNet-based modified sigmoid transform (EMST) is used to enhance the contrast between the lesion and background region. The computational burden of image-to-image segmentation regressors is solved by proposing a new image-to-scalar regression architecture named deep threshold prediction network (DTP-Net).

A modified ResNet regressor is used to automate the selection of the operational parameter of the IECET. A knowledge base is created by including the value components obtained from the HSV color space of the corrupted macro-images and its corresponding ideal values of the operational parameter. These values are selected based on the maximum peak signal-to-noise ratio (PSNR) between the pristine images and outputs of the IECET. The modified ResNet regressor learns the adaptive selection of the operational parameter from the prior knowledge base. In EMST, a modified sigmoid transform is applied to the value component in HSV color space to improve the contrast between the lesion and the background skin. The crossover point in the modified sigmoid transform divides the macro-image into lesion and background. The crossover

point is predicted by a modified EfficientNet regressor to exclude manual intervention and subjectivity. For training the modified EfficientNet regressor, a set of value components extracted from the HSV color space representation of the macro-images in the training dataset are fed as the input. The corresponding set of ideal crossover points at which the values of Dice similarity coefficient (DSC) between the ground-truth images and the segmented output images obtained from Otsu's thresholding (OT) are maximum are defined as the target. The DTP-Net is trained using grayscale macro-images, which are fed as input, and the corresponding gray-level threshold values at which the DSC between the segmented and the ground-truth images are maximized are defined as target. The segmentation threshold predicted by the DTP-Net is used to segment the lesion from the macro-images.

The algorithms are tested and evaluated on publicly available databases, including the University of Waterloo skin cancer, MED-NODE, SD-260, and SD-198. The performance of the image-to-scalar regressors is evaluated by root mean square error (RMSE). The segmentation accuracy of the segmentation models is evaluated by the widely accepted metric DSC, which measures the overlap between predicted and ground-truth segmentation masks.

After equalizing the macro-images with IECET, the segmentation results exhibited by three DL-based segmentation models (such as DeepLabV3+, FCN, and U-Net) demonstrated superior performance in terms of DSC compared to the results obtained using alternative illumination correction algorithms. Particularly, U-Net showed a significant increase in DSC values, from 0.66 ± 0.26 to 0.75 ± 0.21 . When the images are enhanced with EMST, the DSC of the segmented lesions obtained through OT increases from 0.68 ± 0.34 to 0.81 ± 0.17 . Applying the EMST algorithm to the macro-images significantly improved the segmentation accuracy of OT compared to 15 state-of-the-art (SOA) contrast enhancement methods. The segmented results generated by DTP-Net showed higher DSC than the 11 SOA threshold estimation algorithms. Moreover, the DTP-Net with an image-to-scalar architecture proves to be computationally feasible compared to DL-based lesion segmentation models with encoder-decoder architecture (such as FCN-8s, U-Net, SegNet, and DeepLabV3+). Integrating the pre-processing stage (IECET and EMST) with the DTP-Net demonstrated a substantial increase in DSC score, rising from 0.84 ± 0.19 to 0.89 ± 0.09 . The innovative approach with the preprocessing stage and DTP-Net holds great potential for integration into automated tools for skin cancer detection from dermatological macro-images.

Keywords: Convolutional neural network; Deep learning; Image-to-scalar regressor; Macro-images; Segmentation; Skin cancer

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Dr. Harigovindan V. P., Internal Member and Associate Professor in the Department of Electronics and Communication Engineering. I am sincerely grateful to each of them for their unwavering support, valuable insights, and constructive suggestions throughout my research. I deeply thank each committee member for their invaluable support and guidance.

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NOTATIONS

TS_{ABCD}	Total score for the ABCD rule
w_i	Weight assigned to skin attributes
s_i	Dermatologist-assigned score
i	number of attributes
$\sum$	Summation
D	Domain
D_s	Source domain
D_t	Target domain
T	Task
$P(x)$	Probability distribution
$f_t(x)$	Predictive function
$\hat{y}(k)$	Predicted value by the regressor
$y(k)$	Original value
N	Number of images
$\hat{X}$	Binary ground-truth
$\hat{Y}$	Binary segmented image
$\cap$	Intersection operator
$\cup$	Union operator
H	Hue component in HSV color image
S	Saturation component in HSV color image
V	Value component in HSV color image
$\Gamma(.)$	CET and its associated operations
δ	Slope control parameter (SCP)
(m, n)	Pixel location
$RI(m, n)$	Pixel values of reference image
$IC(m, n)$	Pixel values of illumination-corrected image

$M_i \times N_i$	Dimensions of the image
$\hat{S}_i$	Ideal SCP value
$\hat{R}_i$	SCP value predicted by the modified ResNet regressor
W_i	Model's weights
λ	Regularization parameter
α	Shape parameter
β	Crossover point
$\hat{A}_i$	Ideal crossover point
$\hat{P}_i$	Predicted crossover point by the modified EfficientNet regressor
μ_b	Mean
σ_b^2	Variance
ϵ	Constant
x_i	Input features
$\hat{x}_i$	Normalized activations
R	Red channel in RGB color image
G	Green channel in RGB color image
B	Blue channel in RGB color image
I	Input image
A_i	Actual threshold value
P_i	Predicted threshold value by DTP-Net

ABBREVIATIONS

AECLII	Adaptive enhancement method for correcting low-illumination images
AGIC	Average gradient of illumination component
AHE	Adaptive histogram equalization
AMBE	Absolute mean brightness error
AUC	Area under the curve
CAD	Computer-aided diagnosis
CDF	Cumulative distribution function
CE	Contrast enhancement
CEHCM	Color image enhancement method using heat conduction matrix
CET	Counter exponential transform
CLAHE	Contrast limited adaptive histogram equalization
CNN	Convolutional neural network
CS	Contrast stretching
CT	Computed tomography
CVCEA	Contextual and variational contrast enhancement algorithm
DE	Discrete entropy
DFT	Discrete Fourier transform
DL	Deep learning
DS	Decorrelation stretch
DSC	Dice similarity coefficient
DTP-Net	Deep threshold prediction network
EME	Measure of enhancement
EMST	EfficientNet-based modified sigmoid transform
FC	Fully connected
FCN	Fully convolutional network

FCRN	Fully convolutional residual network
FET	Fuzzy entropy thresholding
FGLEI	Fusion of globally and locally enhanced images
FIE	Fusion-based image enhancement
FLM	Feature-linking model
FPR	False positive rate
FSIM	Feature similarity index
GAN	Generative adversarial network
GCCA	Gray world color constancy algorithm
GEM	Generalized equalization model
GMSD	Gradient magnitude similarity deviation
Grad-CAM	Gradient-weighted class activation mapping
GSIM	Gradient similarity index
HCM	Heat conduction matrix
HE	Histogram equalization
HF	Homomorphic filter
HSDDT	Histogram slope difference distribution based threshold
IARC	International agency for research on cancer
IAT	Illumination-adaptive transformer
IDFT	Inverse discrete Fourier transform
IECET	Illumination equalization method using counter exponential transform
IGM	Internal generative mechanism
IoU	Intersection over union
IT	Isodata thresholding
JI	Jaccard index

KMET	Kapur's maximum entropy threshold
LDR	Layered difference representation
LLNet	Low light network
LOE	Lightness-order error
MAE	Mean absolute error
MBConv	Mobile inverted bottleneck convolution
MCC	Matthew's correlation coefficient
MCET	Minimum cross-entropy thresholding
MET	Minimum error thresholding
MP	Morphological postprocessing
MRI	Magnetic resonance imaging
MSE	Mean squared error
MST	Modified sigmoid transform
NCI	National cancer institute
NIQE	No-reference image quality assessment
OT	Otsu's thresholding
PDMET	Poisson's distribution-based minimum error thresholding
PSNR	Peak signal-to-noise ratio
QMC	Quality assessment metric of contrast
RCNN	Region-based CNN
ReLU	Rectified linear unit
RICE	Robust image contrast enhancement
RMSE	Root mean square error
SA	Shading attenuation
SCP	Slope control parameter

SD	Standard deviation
SDD	Slope difference distribution
SGDM	Stochastic gradient descent with momentum
SHE	Shell histogram equalization
SMI	Spatial mutual information
SMIRANK	Spatial mutual information rank
STCRD	Shading and texture constrained retinex-based devignetting
SVM	Support vector machine
TL	Transfer learning
TPR	True positive rate
Type-2-FT	Type-2 fuzzy-based thresholding
UV	Ultraviolet
VEOT	Valley-emphasized Otsu's thresholding
ViT	Vision transformer
WAHE	Weighted approximated histogram equalization
WHO	World health organization
WPCCA	White patch color correction algorithm

CHAPTER 1

INTRODUCTION

Medical imaging plays a crucial role in modern healthcare by providing valuable visual information that aids in diagnosing, treating, and monitoring various medical conditions. It involves the utilization of advanced technologies to capture detailed images of both internal and external structures of the body. Medical imaging techniques include X-ray, computed tomography (CT), magnetic resonance imaging (MRI), ultrasound, dermatological photography, and nuclear medicine [1]. Each modality offers distinct advantages and is tailored to specific clinical scenarios. These imaging technologies aid healthcare professionals in visualizing organs, tissues, bones, blood vessels, and abnormalities with remarkable accuracy, facilitating the identification of diseases, injuries, and other medical conditions [2].

The skin is the largest organ of the human body, serving as a protective barrier between our internal organs and the external environment. It plays a crucial role in regulating body temperature, sensing touch & pain, and synthesizing vitamin D [3]. Skin diseases are prevalent worldwide and represent a significant health concern affecting individuals across all age groups. They have a considerable impact, with estimates suggesting that approximately 30% to 70% of the global population is affected [4, 5].

In recent years, the field of dermatology has witnessed significant advancements in imaging technology, leading to a shift in focus towards non-invasive imaging modalities. While dermatologists still rely on biopsies for histological examination of skin tissues, non-invasive imaging methods have emerged as potential alternatives, capable of supplementing or even replacing the need for skin biopsies in certain scenarios. Unlike medical conditions affecting other organs, the distinctive characteristics of skin diseases can be directly observed by the naked eye and optical devices. Many abnormalities and malignancies exhibit visible features that not only aid dermatologists in diagnosing suspicious skin lesions more effectively but can also be utilized by advanced computer-aided diagnosis (CAD) systems as clinical decision support systems [6].

1.1 HUMAN SKIN BIOLOGY

Human skin biology encompasses the complex structure, functions, and processes of the skin, which is the largest organ in the human body. The skin serves as a vital protective barrier, shielding the internal organs from external threats, regulating body temperature, and facilitating sensory perception. Fig. 1.1 illustrates the schematic diagram showcasing the three layers that constitute the skin: the epidermis, which forms the top layer, the dermis located in the middle layer, and the hypodermis, which represents the deepest layer [7].

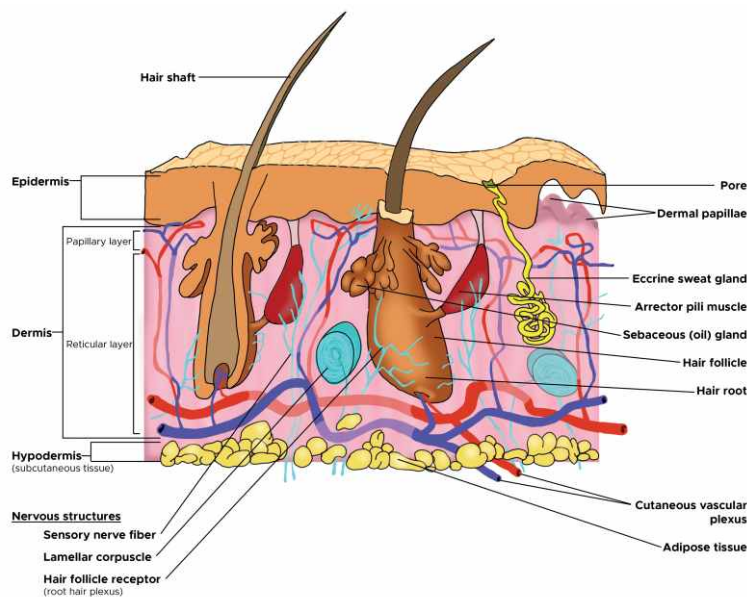


Figure 1.1: Diagram illustrating the arrangement and stratification of the layers within human skin [3].

1. **Epidermis:** It is the outermost layer of the skin and primarily acts as a protective shield. It is composed of several layers of cells, including keratinocytes, which produce the protein keratin, a key component in forming the skin's protective barrier. The epidermis also contains melanocytes, responsible for producing the pigment melanin that gives color to the skin and provides some protection against harmful ultraviolet (UV) radiation. Additionally, the epidermis contains specialized cells called langerhans cells, which play a role in immune responses [7].
2. **Dermis:** It lies beneath the epidermis and provides structural support to the skin. It comprises of connective tissue, blood vessels, nerves, hair follicles, and sweat glands. Collagen and elastin fibers within the dermis provide strength, elasticity, and resilience to the skin. Blood vessels supply nutrients and oxygen to the

skin, while nerves enable the perception of touch, temperature, and pain. Hair follicles are responsible for hair growth, and sebaceous glands secrete sebum, (an oily substance) that lubricates and moisturizes the skin [7].

3. **Hypodermis:** It is also known as the subcutaneous tissue or subcutis. It is the deepest layer of the skin. It consists of fat cells (adipocytes) and connective tissue that provide insulation, cushioning, and energy storage for the body. The hypodermis also helps to regulate body temperature [7].

1.2 SKIN CANCER AND TYPES

Abnormal growth of skin cells can lead to the development of skin cancer, which is the most commonly diagnosed form of cancer worldwide [8]. The growth of skin cells can be broadly categorized as melanoma and benign cancer [9, 10]. Melanoma, characterized as the most severe type of skin cancer, involves the excessive growth of abnormal cells that disrupt the normal functioning of melanocytes, the cells responsible for producing melanin in the skin. These abnormal cells proliferate and spread uncontrollably. These potentially impact the functionality of surrounding tissues and organs. In contrast, benign growths refer to the proliferation of normal cells that divide and grow without interfering with the functions of melanocytes or posing a cancerous risk [9].

The national cancer institute (NCI) [11] has developed a comprehensive classification system for melanoma and benign lesions, as depicted in Fig. 1.2. According to NCI, melanoma is broadly classified as superficial spreading melanoma, nodular melanoma, acral lentiginous melanoma, and lentigo maligna melanoma.

1. Superficial spreading melanoma: This is the most prevalent type of melanoma, comprising approximately 70% of cases worldwide. It typically manifests as a flat or slightly raised lesion that gradually extends across the skin's surface and progressively penetrates to deeper layers.
2. Nodular melanoma: This is a more aggressive type that accounts for approximately 15-30% of melanoma cases. It is characterized by the presence of dome-shaped nodules or bumps on the skin. Compared to other types of melanoma, nodular melanoma exhibits faster growth and progression.
3. Acral lentiginous melanoma: It primarily affects the palms, soles of the feet, and areas under the nails. It is more commonly observed in individuals with darker skin tones. Unlike other types of melanoma, acral lentiginous melanoma is not typically associated with sun exposure.

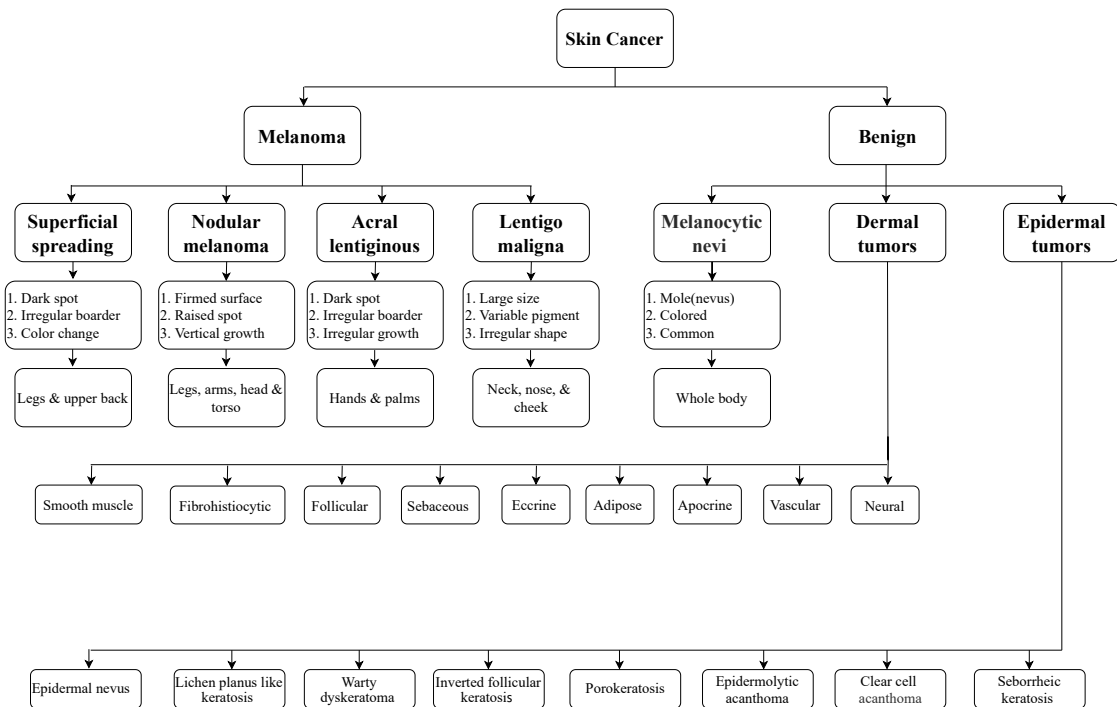


Figure 1.2: Classification of skin cancer.

4. **Lentigo maligna melanoma:** This frequently arises from a pre-existing skin condition known as lentigo maligna. Lentigo maligna is characterized by the presence of a large, flat, and slowly expanding patch of discolored skin. Over time, it develops within the area affected by lentigo maligna. It is important to closely monitor and diagnose lentigo maligna to detect any potential progression to lentigo maligna melanoma.

On the other hand, benign lesions are further categorized into melanocytic nevi, dermal tumors, and epidermal tumors.

1. **Melanocytic nevi:** It is commonly known as moles, which originate from melanocytes, the pigment-producing cells in the skin. These nevi can be either developed after birth or present at birth. They are typically round or oval in shape and may vary in color from brown to black. Melanocytic nevi are usually harmless and do not pose any significant health risks. However, specific characteristics such as size, shape, or color changes may need closer examination by a dermatologist.
2. **Dermal tumors:** It refers to benign growths, that originate from the skin's deeper layers, specifically in the dermis. These tumors can include various types, such as fibromas, lipomas, neurofibromas, and dermatofibromas. Dermal tumors are

typically non-cancerous and do not spread to other body parts.

3. Epidermal tumors: These are benign growths that originate from the outermost layer of the skin, called the epidermis. These tumors are typically harmless and do not pose significant health risks.

According to the recent epidemiological facts and figures released by the world health organization (WHO), malignant melanomas account for approximately 1 in 5 skin cancers reported, with approximately 3,25,000 cases and 57,000 deaths due to melanoma occurred globally in 2020 [12]. A new study by scientists from the international agency for research on cancer (IARC) reported an increase in the trend of new melanoma cases per year and is predicted to be more than 50% from 2020 to 2040 [12]. It is reported that melanoma incidence rates have been increasing during the last 50 years in fair-skinned peoples [13]. This significant increase is likely due to increased exposure of vulnerable populations to UV radiations, which is a strong and ubiquitous risk factor for melanoma, emitted naturally from the sun but also from artificial sources [14]. According to [14], more than three-quarters of all newly diagnosed melanoma cases can be attributed to UV radiation. According to Lee et al. [15], individuals of Caucasian descent typically face greater vulnerability to skin cancer than those with darker skin tones. Conversely, individuals with brown and black skin tones generally possess a natural ability to tolerate higher levels of sun exposure without experiencing sunburn or significantly raising their risk of developing skin cancer.

Five years relative survival rate for melanoma skin cancer detection in the early-stage (less than 0.76 mm thick) is 96%, whereas, in the advanced stage, the survival rate falls to 5% [16]. Also, the cost of treatment for melanoma in the advanced stage is 30 times more costly than the cost of treatment in the early stages [17]. With the rising incidence rates in specific subsets of the general population, early melanoma screening is beneficial.

1.3 MEDICAL DIAGNOSIS

In the past, the diagnosis of skin cancer primarily relied on clinical examination using the naked eye, with biopsy and histopathology serving as confirmatory procedures [18]. However, the biopsy is an invasive technique that can be painful and carry the risk of infection and scarring. As a result, the development of non-invasive diagnostic methods has attracted significant interest.

Approximately a quarter of melanoma cases are attributed to tissue deformations in moles [16]. The transition of a mole to melanoma is often characterized by clinical in-

dications such as accelerated growth/hypertrophy, irregular boundaries, abnormal pigmentation, itching sensation, and tissue breakdown. Visual examination of superficial-spreading melanoma typically involves the application of the ABCD rule [19], as depicted in Fig. 1.3.



Figure 1.3: Illustration of the ABCD rule for classifying a lesion as either melanoma or benign. These images belong to the University of Waterloo skin cancer database [20] (<https://uwaterloo.ca/vision-image-processing-lab/research-demos/skin-cancer-detection>, Accessed on 25-08-2023).

The ABCD rule, the mnemonic represents asymmetry (A), borders (B), color (C), and diameter (D) [19, 21]. Utilizing the ABCD scale, the lesion is assessed for asymmetry, irregular border shape, multiple colors, and diameter greater than 6 mm. Additionally, the skin lesion is monitored over time to detect any changes or evolution. By assigning scores to each attribute, as shown in Table 1.1, a total score is calculated using Eq. 1.1 to evaluate the risk of skin cancer.

$$TS_{ABCD} = \sum_{i=1}^4 w_i s_i \quad (1.1)$$

In Eq. 1.1, TS_{ABCD} represents the calculated total score for the ABCD rule, i is

the number of attributes (asymmetry, border irregularity, color variation, and diameter), w_i corresponds to the weight assigned to each attribute, and s_i represents the dermatologist-assigned score for each attribute. The summation $\sum$ indicate the sum of the weights and scores, respectively.

Table 1.1: ABCD rule with the score and the weight of each attribute with the expected range [22].

Mnemonics	Attributes	Specification/Property	Score	Weight	Range
A	Asymmetry	One half doesn't match the other half	0-2	1.3	0-2.6
B	Border	Uneven border	0-8	0.1	0-0.8
C	Color	Diversity in lesion color	1-6	0.5	0.5-3
D	Diameter	Larger than 6 mm	1-5	0.5	0.5-2.5

The classification of lesions into benign, clinically doubtful, and malignant cases is determined by TS_{ABCD} value in Eq. 1.2.

$$\begin{cases} TS_{ABCD} < 4.75 & \text{lesion is benign} \\ 4.75 \leq TS_{ABCD} \leq 5.45 & \text{lesion is clinically doubtful} \\ TS_{ABCD} > 5.45 & \text{lesion is malignant} \end{cases} \quad (1.2)$$

Diagnosing melanoma through subjective examination is considered a basic and primitive method. Dermoscopy, also known as dermatoscopy, involves the use of a dermoscope (or dermatoscope) to examine skin lesions. Fig. 1.4 shows two commercially available dermoscopes. This non-invasive imaging technique utilizes polarized light to provide magnified visuals of skin lesions, assisting dermatologists in evaluating pigmented skin lesions based on morphological characteristics. It is reported that the diagnostic decisions made using dermoscopes are more specific compared to those made through crude subjective examination [23].



Figure 1.4: Commercially available dermoscopes. (Image courtesy: <https://www.canfieldsci.com/imaging-systems/categories/medical-dermatology>, Accessed on 25-08-2023)

Ascierto et al. [24] reported an average agreement of 87.3% among experienced

dermatologists who employed a dermoscope to make diagnostic decisions. However, Engasser et al. noted that only 48% of dermatologists in the USA use dermoscope for melanoma screening. Lack of knowledge, training, experience, interest, and cost was identified as the primary reasons for not using a dermoscope by the clinicians [25].

Dermatological photography [26] is an emerging imaging modality extensively used for visualizing skin lesions. There are a variety of devices that can be used for capturing dermatological photographs. These devices include smartphone/tablet attachments, point-and-shoot cameras, and interchangeable lens system cameras. Often, dermatological photographs are captured with ordinary digital cameras. Even mobile cams can be used to capture dermatological photographs. Mainly, dermatological photographs are of three types. They are contextual images, macroscopic images (macro-images), and micro images.

Contextual images offer information regarding the anatomic location of the lesions. Macroscopic images, also known as macro-images, are close-up views of the skin lesions. These images are captured to visualize the surface characteristics, texture, color, and other macroscopic features of the skin lesion. Micro images are highly magnified images that provide a detailed view of the microscopic features of the skin lesion captured with digital dermoscopes. Because of the simplicity, flexibility, and availability of high-resolution digital cameras and limited access to dermoscopes, the macro-images are slowly replacing the dermoscopy images, especially in the screening dermatology [26–30].

1.3.1 Computer-aided diagnosis system

Screening skin lesions using visual inspection is challenging for a dermatologist due to the fine-grained variations in the appearance of these lesions [31]. Fig. 1.5 illustrates sample dermatological macro-images, underlining the challenges in visually distinguishable melanoma and benign lesions.

Ascierto et al. [24] reported moderate inter-observer agreement between the experienced dermatologists to diagnose melanoma visually, using a dermoscope. Argenziano et al. [32] found that the inter-observer agreement of dermatologists on the assessment of individual ABCD features had a fair agreement. Losina et al. [33] evaluated the cost-effectiveness of various melanoma screening techniques used in the United States. They reported that despite the rising in melanoma cases, the screening of the general population is not practically feasible. They also reported that one-time visual melanoma screening of the general population older than 50 years and bi-annual screening of siblings of patients with melanoma are very cost-effective. According to their report, screening of the general population are time-consuming



Figure 1.5: Sample dermatological macro-images highlighting the challenges of distinguishable melanoma and benign skin cancer. These images belong to the University of Waterloo skin cancer database [20] (<https://uwaterloo.ca/vision-image-processing-lab/research-demos/skin-cancer-detection>, Accessed on 25-08-2023).

and not cost-effective. In developing and underdeveloped countries, access to a dermatologist for screening of skin cancer is a prevalent issue. Coates et al. reported that the average waiting period for an appointment with a dermatologist is 33.9 days [34].

To avoid subjectivity in the diagnosis and save the time spent by the dermatologists, CAD systems based on dermatological photographs are widely used for the detection of skin cancer [29, 30, 35–39]. Recently, Birkenfeld et al. [40] has proved the potential of the wide-field dermatological photo-macrographs to be used as a tool for identifying the type of suspicious skin lesions at the pre-screening level. The prospect of dermatological macro-images to detect melanoma is already evidenced by many researchers [29, 30, 41–44]. Saleem et al. [45] has demonstrated that support vector machine (SVM) classifier based on textural, intensity, and shape features of lesions distinguish melanoma and nevus lesions with higher accuracy on dermatological photo-macrographs (98.06%) than on dermoscopy images (93.95%).

Skin lesion segmentation serves as the fundamental step for detailed analysis and extracting essential features from dermatological images. Through precise delineation of lesion boundaries, CAD systems can extract specific attributes including size, shape, color, and texture, which are crucial for accurate lesion characterization and subsequent classification [46–49]. This segmentation process enables the precise localization of relevant features within the lesion, facilitating the identification of discriminative patterns that aid in distinguishing between benign and malignant lesions [50, 51]. By extracting these features, CAD systems provide dermatologists with quantitative and objective information, which greatly assists in the diagnostic decision-making process.

Skin lesion segmentation using an encoder-decoder structure has emerged as a popular deep learning (DL) approach for accurately identifying and delineating skin lesions from medical images. This technique leverages the power of neural networks to learn and extract relevant features from the input images automatically. During training, the manually segmented ground-truth images are used as targets. These ground-truth images served as the reference for the network to learn and compare its predictions with the desired segmentation. By minimizing the error between the predicted and ground-truth segmentations, the model learns to segment skin lesions accurately.

In 2015 Long et al. [52] proposed a fully convolutional network (FCN) for performing pixel-wise image segmentation by leveraging the power of convolutional neural networks (CNNs). Inspired by the FCN architecture [52], Ronneberger et al. [53] introduced the U-Net network designed explicitly for biomedical image segmentation. This encoder-decoder architecture has gained significant popularity among researchers. The U-Net model incorporates skip connections, which involve concatenating the feature maps from the encoder and decoder sub-networks. Zhou et al. [54] introduced an enhanced version of the U-Net architecture called U-Net++, incorporating nested skip connections. This nested skip connection is designed further to improve the information flow between the encoder and decoder for facilitating better feature extraction and segmentation.

The unevenness of the background illumination is one of the major distortions in macro-images of skin lesions. Glaister et al. [55] have reported that unevenness of background illumination on images of skin lesions negatively impacts the performance of lesion segmentation and classification algorithms. Lee et al. [56] have suggested illumination correction for enhancing image quality and improving the classification of skin lesions from dermoscopy images by DL models. Barata et al. [57] have noted that illumination correction improves the classification of multisource images of the skin lesions, increasing the sensitivity of a bag-of-features deduced from 1-D RGB histograms, from 71.00% to 79.70% and the specificity from 55.20% to 76.00%. Hence, illumination correction or devignetting is vital in the automated analysis of the macro-images of skin lesions.

During the initial stages of skin cancer, the lesions may not have sufficient intensity difference or contrast from the background region. Additionally, low or extremely high light exposure at the time of capturing the image also reduces the contrast [58]. Low contrast between lesion and background regions adversely impacts segmentation [48, 59]. Contrast enhancement (CE) is an essential procedure to overcome this challenge in CAD systems [39, 59]. There is a common misbelief that CE is not essential when DL models are used for segmenting skin lesions. Yu et al. [60] have shown that even

a well-established fully convolutional residual network (FCRN) fails to produce accurate segmentation results and detect melanoma from low-contrast images. Masoudi et al. [61] and Hesamian et al. [62] have evidenced that the CE can increase the segmentation and classification accuracies of DL models in medical image analysis.

In recent years, DL-based methods have gained widespread popularity in the field of image processing, particularly in tasks such as quality enhancement of images, semantic segmentation, and classification. A notable contribution to the use of DL in the area of image quality enhancement comes from Lore et al. [63]. They introduced a sophisticated deep autoencoder model named low light network (LLNet), designed to enhance and denoise low-light images adaptively.

Vision Transformer (ViT) was another recently developed research area gaining significant attention that focuses on transformer-based architectures for supervised tasks in computer vision. Cui et al. [64] proposed the illumination-adaptive transformer (IAT) to address the challenges of illumination variations in image enhancement tasks. This method leverages a transformer-based architecture to adapt the illumination conditions within an image. It enables effective enhancement across a wide range of lighting conditions.

Reliance on datasets with paired images was the significant drawback of encoder-based DL and ViT-based image quality enhancement algorithms. These models required both low-quality input images and their corresponding high-quality counterparts for training, which could impose limitations on the availability and preparation of datasets. To address this issue, Jiang et al. [65] proposed the first unsupervised learning method called EnlightenGAN. EnlightenGAN employed a generative adversarial network (GAN) framework for enhancement. EnlightenGAN first uses a global-local discriminator structure to handle spatially varying light conditions in the input image. Secondly, it utilizes a self-feature retention loss and self-regularization attention mechanism to preserve the original image content features throughout the enhancement process.

1.4 MOTIVATION

Macro-images of skin lesions captured using standard digital cameras have gained wide acceptance in dermatology screening due to their simplicity, flexibility, and availability. The accurate segmentation of skin lesions plays a crucial role in computing the area and border features of the lesion in the automated tools used for analyzing dermatological macro-images of skin lesions. Accurately segmenting skin lesions in dermatological images is challenging due to the significant variability in color, texture, shape, and size of the lesion. Additionally, lesions can have similar

colors and textures to the surrounding skin, making it even more challenging to accurately delineate their boundaries. The quality of the macro-image profoundly impacts the performance of lesion segmentation algorithms, including DL-based methods. In dermatological macro-images, uneven background lighting caused by nonstandard acquisition leads to significant distortion. Additionally, during the early stages of skin cancer, the lesion may not exhibit sufficient intensity difference or contrast compared to the surrounding background region. Moreover, capturing the image under low or excessively high light exposure can further reduce the contrast. Uneven background illumination and low contrast between lesion and background skin significantly impact the performance of lesion segmentation algorithms.

In dermatological images, the available illumination correction algorithms often reduce the contrast between the background and the lesions. They can also introduce color distortions or drastic changes in the color tone. Also, the existing CE methods often fail to selectively enhance the contrast between the lesions and the background. These methods either increase the brightness of the lesion and background equally or do not significantly improve the quality of the macro-images. Equally enhancing lesion and background regions simply increases the brightness rather than increasing the intensity difference between the regions. Such inconsistent enhancement results in over-segmentation or under-segmentation of lesions. Recently, GAN, autoencoder-based, and ViT-based enhancement methods have been introduced to improve image quality. However, these techniques require extensive training data and impose a heavy computational burden.

Dermatological macro-images consist of highly zoomed views containing only lesions and background regions. Consequently, localizing lesions in these images becomes a relatively simple thresholding problem. However, existing intensity-based thresholding algorithms do not offer adequate accuracy and fail to retain consistent performance on different dermatological macro-images. In recent years, DL-based segmentation models for skin lesion segmentation have gained popularity, often employing an encoder-decoder architecture. These models perform image-to-image regression tasks, using the original images as inputs and manually segmented ground-truth images as targets. However, these image-to-image regression models are computationally heavy compared to image-to-scalar regression models. Thus, segmentation of the skin lesions using the DL models with encoder-decoder architecture is computationally heavy.

The application of DL-based image-to-scalar regression models, which are computationally lighter than image-to-image regression models, for predicting the threshold and subsequently segmenting lesions from macro-images remains largely unexplored.

This presents an opportunity for further exploration and investigation.

This research aims to develop deep image-to-scalar regressors for quality enhancement and segmentation of dermatological macro-images, which is computationally feasible. The proposed methods can be integrated into the segmentation module in automated tools that detect skin cancer from dermatological macro-images.

1.5 OBJECTIVES

The following objectives have been formulated to fulfill the motivation of this research:

1. To design an illumination correction algorithm for macro-images of skin lesions that can make the background illumination uniform without reducing the contrast between the background & the lesion and inducing color distortion or drastic changes in the color tone.
2. To formulate a CE algorithm for improving the contrast between the lesion and the background skin in dermatological macro-images. This enhancement technique aims to enhance the visibility of the lesion, thereby facilitating accurate lesion segmentation.
3. To propose a computationally-feasible DL-based image-to-scalar regressor model to predict the optimal threshold value from the input dermatological macro-image that can accurately segment the lesion.

By accomplishing these objectives, this research aims to make significant advancements in the field of dermatology. It seeks to enhance segmentation accuracy, which plays a crucial role in automated tools for skin cancer detection. This research will contribute to the development of improved and reliable automated tools in the early detection of skin cancer.

1.6 HIGHLIGHTS AND CONTRIBUTIONS

The following contributions are made in this research to achieve the objectives.

- An efficient illumination equalization using counter exponential transform (IECET) is introduced. It has an operational parameter named slope control parameter (SCP), that controls the trade-off between the contrast and uniformity of the background luminance in the equalized image. To eliminate subjectivity and manual intervention associated with the selection of the SCP, a

modified ResNet regressor is used to automate its selection. The IECET suppresses the nonuniformity of the background lighting without altering the natural color tone of the image and preserving the contrast between the lesion and the background. A counter exponential transform (CET) is designed and demonstrates its potential in illumination correction applications on macroscopic images. For the first time, the ResNet model is modified as an image-to-scalar regressor. The IECET is fully automated and partially explainable, as the IECET uses the CET to equalize the background lightning, and the modified ResNet is used just to automate the selection of SCP.

- An EfficientNet-based modified sigmoid transform (EMST) is proposed to improve the contrast between lesion and background regions in dermatological macro-images. The crossover point in the modified sigmoid transform divides the macro-image into lesion and background. It is predicted by a modified EfficientNet regressor to exclude manual intervention and subjectivity. Unlike traditional sigmoid transforms, EMST provides provisions to fix the boundary that divides pixel values into lesion and background regions, allowing for precise adjustment of the enhancement level.
- This research introduces a deep threshold prediction network (DTP-Net), for accurately predicting the segmentation threshold in dermatological macro-images. The proposed DTP-Net performs image-to-scalar regression. Here, the original images are fed as inputs, and a simple column vector of scalar values representing optimum values of the intensity threshold is used as targets during training. The DTP-Net can learn the difference between the lesion and background in the intensity space and accurately predict the threshold to separate the lesion from the background. The computational expense for training the image-to-scalar regressor is less compared to that of the image-to-image regressors.

1.7 STRUCTURE OF THE THESIS

Based on the research motivation and objectives outlined above, this study primarily concentrates on algorithm development to enhance the quality of macro-images. It seeks to facilitate improved segmentation accuracy for lesion segmentation models. A key aspect of this research involves the exclusive design and development of a DL-based image-to-scalar regressor specifically developed to predict the optimal threshold value required for accurate lesion segmentation in macro-images. The following sections provide a comprehensive overview and demonstrate the work conducted throughout this study.

Chapter 2 provides a comprehensive literature review on illumination correction

and CE techniques applied to dermatological images. This chapter emphasizes the significance of image quality in achieving accurate segmentation and classification results in dermatological image analysis. It discusses various segmentation methods, covering both threshold-based and DL-based approaches. The advantages, limitations, and drawbacks associated with these methods are analyzed. Additionally, this chapter covers the methodology for validating the performance of the deep image-to-scalar regressor models, along with specific metrics to evaluate the quality segmentation. Here, detailed information about the publicly available datasets used in this thesis for the experiments are discussed.

Chapter 3 describes the illumination equalization method using a counter exponential transform. This chapter provides a detailed explanation of the methodology, highlighting the steps involved in equalizing the illumination in dermatological images. Additionally, it offers insights into the architecture, training strategy, and dataset preparation followed for the modified ResNet regressor. Furthermore, subjective and objective evaluations are conducted, along with the effects of different illumination correction methods on segmentation of the lesion.

Chapter 4 explains EMST for CE. The detailed procedure adopted for training the modified EfficientNet regressor and details of the dataset preparation are explained. This chapter proceeds with a subjective comparison, evaluating the performance of the proposed EMST against state-of-the-art CE methods. Additionally, the segmented results obtained using Otsu's thresholding (OT) from enhanced images by various CE methods are objectively compared in terms of segmentation accuracy.

Chapter 5 presents the network architecture of DTP-Net, along with the training procedure and dataset preparation. It discusses the gradient-weighted class activation mapping (Grad-CAM) to provide explainability to the performance of the DTP-Net. This chapter compares the accuracy of threshold values predicted by the DTP-Net with state-of-the-art threshold estimation algorithms. The segmentation results obtained from the DTP-Net are compared with those from existing thresholding algorithms and DL-based skin lesion segmentation methods, considering segmentation accuracy and computational time. It provides a comprehensive analysis of the DTP-Net's performance and its comparison with existing methods in skin lesion segmentation.

Chapter 6 offers an overview of an innovative approach for skin lesion segmentation, encompassing subjective and objective comparisons of the lesion segmentation. The evaluations focus on segmentation accuracy and consider different scenarios, including cases without the preprocessing steps, with only illumination correction, and

with both illumination correction & CE stages. This comprehensive analysis provides a detailed assessment of the proposed pipeline's effectiveness and its impact on the overall segmentation performance. This chapter contributes to a comprehensive understanding of skin lesion segmentation by examining various configurations.

Finally, Chapter 7 summarizes the key findings and implications of this thesis, providing an overview of each chapter and highlighting their contributions to the field. Additionally, it identifies areas for further investigation and outlines paths for future research.

CHAPTER 2

RELATED WORK

Skin cancer detection using computer-aided diagnosis (CAD) systems have been the subject of extensive research for over a decade [66]. Numerous techniques have been developed specifically for identifying melanoma [66–69]. In a comprehensive survey, Hasan et al. [70] examined the methods used for analyzing skin lesions using clinical and dermoscopic images from 2011 to 2022. They reported a total of 594 publications available during this period. The lesions should be segmented accurately for computing their area and border features [71]. However, uneven background illumination and low contrast in skin lesion images pose significant challenges in achieving reliable and precise segmentation results [39, 55–57, 59]. To address these challenges, researchers have applied illumination correction and contrast enhancement (CE) techniques as pre-processing steps before performing skin lesion segmentation [70].

This chapter covers the advancements in preprocessing steps, mainly focusing on illumination correction and CE methods, along with a detailed overview of the existing techniques for lesion segmentation. This chapter offers valuable insights into the strengths, limitations, and comparative performance of various illumination correction, CE, and segmentation approaches by reviewing and analyzing the current state-of-the-art algorithms.

2.1 PREPROCESSING

The image preprocessing stage plays a vital role in enhancing the quality of the original digital images of skin lesions, leading to improved accuracy in subsequent segmentation stages within CAD systems [72]. This stage primarily involves illumination correction and CE [73] for improving the overall quality of the macro-images to result in superior segmentation performance and ultimately contributing to the effectiveness of CAD systems in lesion analysis.

2.1.1 Illumination correction

Uneven background illumination in images is characterized by a non-uniform distribution of light intensity throughout the image. It occurs when the illumination source or conditions are inconsistent, resulting in variations of brightness or darkness across

different regions [74]. In macro-images of skin lesions, nonstandard acquisition can cause uneven background illumination, leading to vignetting, a phenomenon observed as darkening in the outer portions, particularly in the corners of macro-images [75, 76]. To mitigate this issue and facilitate the automated analysis of skin lesions, many researchers have adopted illumination correction as a preprocessing step [55–57, 75–80].

Among the existing algorithms for equalizing uneven background illumination on digital images of skin lesions, the homomorphic filter (HF) is the most traditional one. Saleem et al. [45], Abbas et al. [81], Tajeddin and Asl [82], and Salvi et al. [83] have used the HF for equalizing uneven background illumination on digital images of the skin lesions. In homomorphic filtering, the input image is subjected to a logarithmic transform to bring the illumination and reflectance components in multiplicative form into additive form. After computing the 2D discrete Fourier transform (DFT) of the log-transformed image, the low-frequency Fourier coefficients within a user-defined radius are attenuated with the help of a 2D Butterworth filter, leaving the high-frequency Fourier coefficients intact. The spatial domain image is retrieved from the filtered Fourier coefficients via inverse DFT (IDFT), and then an exponential transformation is applied in the spatial domain to nullify the effect of the logarithmic transformation.

Mahbod et al. [79] used the gray world color constancy algorithm (GCCA) to equalize uneven background illumination on digital images of skin lesions. In the first step, of the GCCA, tristimulus values in the RGB image are linearized using a parametric curve. Following this, the illuminant component is estimated using the gray world algorithm. Prior linearization is essential as the gray world algorithm assumes uniform illumination and linear RGB values. Then the color balance of the RGB image is adjusted according to the estimated illuminant component. Finally, the color balance-adjusted is subjected to a gamma correction to reverse the effect of prior linearization. Bagchi et al. [84] also employed a gray world color constancy algorithm to equalize background illumination in dermoscopy images before classifying the lesions using a meta-ensemble comprising EfficientNet and DenseNet models. Cavalcanti et al. [80] employed shading attenuation (SA) by eliminating the low-frequency spatial component of the image to correct uneven illumination before performing Otsu’s thresholding (OT) on macro-images of skin lesions.

The white patch color correction algorithm (WPCCA) used by Schaefer et al. [49] on dermatological images. The procedure of WPCCA is similar to that of the GCCA except for the method used for estimating the illuminant component. In WPCCA, the white patch retinex algorithm estimates the illuminant instead of the gray world algorithm. The white patch retinex algorithm estimates the illumination component in

the RGB image by assuming that the top percentile of brightest red, green, and blue values defined by the user represents the color white.

Shading and texture constrained retinex-based devignetting (STCRD), proposed by Sathish et al. [75], is one of the latest algorithms for equalizing uneven background illumination on digital images of skin lesions. In STCRD, the illumination-corrected image is reconstructed from the gamma-corrected illumination component in hue value saturation (HSV) color space. The retinex decomposition of the value component is formulated as a non-convex optimization problem. Constraints within the cost function are derived from the shading and texture priors. The shading and texture priors are computed from the derivatives of the illumination and texture priors, respectively.

Deep learning (DL) models helpful for illumination correction of skin lesion images are also available. Recently, Salvi et al. [83] proposed a DL model for equalizing uneven background illumination on digital images of skin lesions in which a generative adversarial network (GAN) was trained using the output images of the HF with optimum parameter selection.

However, it is important to note that despite the existence of several algorithms explicitly designed for illumination correction of dermatological images, their applicability is limited due to certain reported limitations. Shamsudeen and Raju [85] conducted an experiment on fundus photographs which showed that the HF method was not effective in subduing the uneven illumination in the background. In addition, Ardizzone et al. [86] have shown that the HF introduces illumination artifacts on the edges of the foreground. According to Lee and Plataniotis [87], the gray world algorithm is not effective in scenarios where an image contains significant areas of uniform colors. This is because it cannot differentiate between whether the dominant color in the image is due to the illuminant or is an intrinsic color of the scene. The GCCA [79] and WPCCA [88, 49] map the input images into a completely different color tone because of their gray world and white patch assumptions. The STCRD [75] significantly reduces the difference in contrast between the lesion and the surrounding background skin. When the HF is not a complete success in suppressing the nonuniformity of the background lighting in images, a GAN model trained using the output images of the HF [83] will not produce better illumination correction performance than the HF.

Table 2.1 comprehensively outlines the various illumination correction techniques utilized by researchers in the context of dermatological images. Additionally, it provides a concise overview of the datasets employed and the evaluation measures used.

Table 2.1: Review of illumination correction methods used as a preprocessing step in automated skin cancer analysis using dermatological images.

Author & year	Method	Database	Performance measure	Remarks
Saleem et al. [45] (2019)	HF	University of Waterloo, ISIC 2017	DSC, JI	1. Used a hybrid texture for the segmentation of lesions.
Abbas et al. [81] (2011)	HF	320 clinical images	TPR, FPR	1. Localized region-based active-contour model was adopted to localize lesions.
Tajeddin & Asl [82] (2018)	HF	PH2 dataset	DSC, JI, Accuracy	1. Contour propagation model used with a novel two-component speed function. 2. Performance depends on the selection of optimum features.
Salviet et al. [83] (2022)	Optimized HF	HAM10000	JI	1. Trained a GAN for color constancy in dermatological images. 2. Used U-Net to evaluate the performance of GAN.
Mahbod et al. [79] (2020)	GCCA	ISIC 2018	Accuracy	1. Proposed multi-scale fusion ensemble CNN for lesion classification.
Bagchi et al. [84] (2020)	GCCA	ISIC 2019	AUC	1. Used multi-class meta-ensemble DL models for the classification of lesions.

Table 2.1 continued from previous page

Author & year	Method	Database	Performance measure	Remarks
Cavalcanti et al. [80] (2013)	SA	Dermis	Avg. error	1. Otsu's thresholding was used for the segmentation of lesions.
Schaefer et al. [49] (2011)	WPCCA	100 dermoscopic images	Avg. error	1. An iterative segmentation algorithm was adopted to segment lesions.
Sathish et al. [75] (2022)	STCRD	University of Waterloo	AGIC, LOE	1. Significant reduction in contrast between lesion and surrounding background skin.

2.1.2 Contrast enhancement

Contrast enhancement (CE) is a digital image processing technique used to improve the visual content of the output image by increasing the perceptual difference between the dark and bright regions. Its objective is to distinguish the details in an image for computer/machine vision applications [89].

In an image, contrast refers to the difference in intensity or color between neighboring pixels. A high-contrast image significantly differs between the dark and bright areas, making it visually more appealing and easier to interpret. In literature, CE methods are broadly classified into two categories: global CE and local CE [90–92].

1. Global CE techniques adjust the contrast of the entire image uniformly. It modifies the histogram of the image, which represents the distribution of pixel intensities. The most common methods include histogram equalization (HE) and contrast stretching (CS). HE redistributes the pixel intensities to maximize the contrast, while CS expands the range of pixel values to cover the full dynamic range of the display.
2. Local CE techniques aim to enhance contrast selectively in different regions of the image, considering the local characteristics of the image content. These methods often use image filtering operations to enhance edges and details. Popular local CE techniques include adaptive histogram equalization (AHE), which applies HE to small regions of an image independently, and various image enhancement filters, such as unsharp masking and the Laplacian filter.

In the early stages of skin cancer, the lesions may not have sufficient intensity difference or contrast from the background region. Additionally, low or extremely high light exposure while capturing the image also reduces the contrast [58]. To overcome this challenge, CE is an essential preprocessing step in automated systems used to analyze images of skin lesions [39, 59]. Decorrelation stretch (DS) [93], fusion of globally and locally enhanced images (FGLEI) [59], histogram equalization (HE) [94], and contrast-limited adaptive histogram equalization (CLAHE) [95] are few popularly used CE techniques for enhancing the contrast of dermatological images.

Khan et al. [96] applied the DS [93] to enhance the contrast of dermoscopy images and was first proposed by Soha and Schwartz [93] to enhance the color difference in images. In DS, the inter-channel correlation among the input pixels is removed, thus known as the ‘decorrelation stretch’.

Afza et al. [59] used FGLEI to enhance the contrast of the dermoscopy images

used for skin lesion classification via a DL framework. In FGLEI, the standard deviation (SD) of neighborhood pixels are subtracted from each pixel. The product of this difference and a scalar value termed ‘contrast gain’ is added to the mean of the neighborhood pixels to enhance the local information. A top-hat filter is applied on the input image. The input image is subtracted from the sum of the outputs of the local enhancement and the top-hat filter to obtain the final output. The FGLEI algorithm often over-enhances the bright background and gives an unnatural output image.

Amoabedini et al. [47], Schaefer et al. [49], and Gomez et al. [97] applied HE [94] on the normalized RGB color channels to enhance the color contrast of dermoscopy images. Pennisi et al. [98] employed HE [94] on the Y channel of the input image represented in YC_bC_r color space. Based on the input image histogram, the HE [94] technique reassigns the intensity value of pixels so that the histogram of the resultant image is uniformly distributed. The HE occasionally reduces the number of pixel intensities, resulting in detail loss and an unnatural appearance [99, 100].

Pizer et al. [95] introduced CLAHE. In CLAHE, the whole image is split into small rectangular tiles and processed separately. The optimal threshold value is computed on each tile, and the histogram values below the threshold are adjusted. The contrast of each rectangular tile is modified. Thus, region-specific enhancement is achieved in CLAHE, which is impossible in HE-based methods where the enhancement is applied to the entire image. Premaladha et al. [101] and Kim et al. [102] adopted CLAHE [95] for enhancing the contrast of lesions in dermoscopy images. Sarkar et al. [103] utilized CLAHE to enhance the low-frequency features of dermoscopy images.

Apart from the techniques discussed above, several enhancement methods are available in the literature to enhance the quality of panoramic images. Xu et al. [104] proposed a combination of HE and white balance integrated into a general convex optimization framework called the generalized equalization model (GEM). In this method, model parameters control the balance between enhanced contrast and natural tone. Its outcome always depends on the optimum selection of model parameters and varies with different input images.

Katircioglu [105] proposed a color image enhancement method using a heat conduction matrix (CEHCM). In CEHCM, the RGB color image is transformed into the HSI color space. The ‘ I ’ channel represents the intensity component that needs significant enhancement. A feature map called the heat conduction matrix (HCM) is computed from the heat conduction equation of solids applied to each pixel in the original ‘ I ’ channel. HCM value is calculated for each center pixel using a 3×3 mask that passes over all the pixels on the ‘ I ’ channel. The magnitude of elements in HCM can

be positive, negative, or zero. Elements with a slight negative magnitude in HCM indicate that the pixels in the original ' I ' channel need a significant enhancement. In contrast, a small positive magnitude of elements in HCM indicates the need to reduce the pixel value in the ' I ' channel. Elements with high positive or negative magnitude with respect to the user-defined threshold HCM value indicate that these are the object's edges and must preserve the pixel values in the original ' I ' channel. Next, the original ' I ' channel is replaced with the updated ' I ' channel in the HSI model. Finally, an enhanced image is obtained by converting HSI to RGB color space.

Spatial mutual information rank (SMIRANK) is a global CE method proposed by Celik [106] to enhance low-illuminated images. The gray values in the input images are sorted as a set of all gray levels from '0' to '255'. Then normalized 2D joint spatial histogram of the input image is calculated. From the 2D joint spatial histogram, spatial mutual information (SMI) is computed, and corresponding gray-level scores are estimated. At last, the page rank algorithm is applied to find the gray rank vector. A mapping function is derived from the rank vector, which maps the input gray-level to the output.

Fu et al. [107] proposed a fusion-based image enhancement (FIE) technique for enhancing weakly illuminated images. It decomposes the image into reflectance and illumination components, then derives luminance-improved and contrast-enhanced inputs using sigmoid and adaptive histogram equalization. By fusing these inputs with appropriate weights, the method produces an adjusted illumination component which is concatenated with the reflectance component, resulting in an enhanced image that balances local contrast improvement and natural appearance.

Celik and Tjahjadi [108] employed a 2D histogram-based image enhancement strategy based on the contextual information from the neighboring pixels of the input image. This technique is named as contextual and variational CE algorithm (CVCEA). An output histogram is obtained by minimizing the sum of the Euclidean norm difference of the input and uniform distributed histograms. The enhanced image is generated by mapping diagonal elements of the input histogram onto the corresponding diagonal elements of the output histogram. The main limitation of this method is that it failed to handle large histogram values properly and generated artifacts due to over-stretching.

Lee et al. [109] developed a CE method based on layered difference representation (LDR) of 2D histograms. The LDR consists of a transformation function and a difference vector arranged in a pyramid-like layer structure. Intra-layer optimization computes dissimilarity vectors per layer by highlighting frequently

occurring gray-level differences in the input image. At the same time, inter-layer accumulation merges these vectors to form the transformation function. By applying this function, the input image's gray levels are mapped to output gray levels, resulting in an enhanced version of the image.

Arici et al. [110] proposed a weighted approximated HE (WAHE), which minimizes a cost function. In this method, a modified histogram is created from the weighted average of the input histogram and uniformly distributed histogram. Specifically designed penalty terms are introduced to adjust the level of CE. Contouring and grainy noises are formed in uniform regions of the enhanced images, when narrow pixel values are transformed into a more comprehensive range. Black and white stretching made the dark areas darker and the bright spots brighter on the enhanced images.

Gu et al. [111] proposed an automatic robust image CE (RICE) method, which is a modified version of WAHE [110]. This framework has two steps: First, find an intermediate histogram ' h ' from the input histogram ' h_i ', equalized histogram ' h_{eq} ', and S-shaped transfer function based brightness preserving (STBP) processed histogram ' h_{sig} '; by minimizing a weighted distance $\|h - h_i\| + \phi\|h - h_{eq}\| + \psi\|h - h_{sig}\|$; Second, the parameters (ϕ, ψ) are automatically selected by the idea of saliency preservation. Saliency detection is done with the help of quality assessment metric of contrast (QMC) from the image signature model.

Ding et al. [112] proposed the shell HE (SHE) technique. It is implemented based on the dimensionality reduction method, which decomposes 3D RGB color space to 256 RGB shells. Then, histogram equalization is applied on shells, which coincides with the iso-luminance plane in the 3D RGB color space. However, while enhancing the contrast, this method failed to preserve the hue component of the image.

Inspired by gamma-band oscillations, Zhan et al. [113] proposed a feature-linking model (FLM) for image enhancement. The FLM consists of three elements: membrane potential, the threshold, and the action potential. A uni-channel input image that acts as a stimulus is fed to the network. The stimulation induces feed-forward input and feedback linking, which combines to produce a membrane potential. Leaky integrators represent the threshold for the neurons. An action potential is generated when the membrane potential of a neuron exceeds the threshold value, called the firing of neurons. The firing time of each neuron is stored in the time matrix. The transformation function is computed from the time matrix, which maps the input image to the enhanced output image.

Wang et al. [114] proposed an adaptive enhancement method for correcting

low-illumination images (AECLII). This method addresses the issue of local over-enhancement caused by uneven enhancement and lack of adaptability in parameter settings of low-illumination image enhancement techniques. The RGB image is transformed to HSV color space, and the ‘ V ’ component is extracted. Using a multiscale Gaussian function, the illumination intensity is estimated from the ‘ V ’ component. A correction function based on the Weber-Fechner law is constructed, and two images are generated by adaptively adjusting the parameters of the enhancement function based on the illumination intensity distribution. These images are then fused to get the enhanced ‘ V ’ component. Finally, the enhanced ‘ V ’ component is combined with the ‘ H ’ and ‘ S ’ components, and the output image in HSV color space is converted back to RGB color space to obtain the enhanced version of the input image.

Table 2.2 and 2.3 discusses the CE methods available in the literature for enhancing the contrast of dermatological and panoramic images. These studies provide a concise overview of the datasets utilized as well as the evaluation measures for CE. The existing CE methods discussed have their limitations. Most of the CE methods available either increase the brightness of lesion and background equally or do not significantly improve the quality of the dermatological macro-images. Equally enhancing lesion and background regions increases the brightness rather than increasing the intensity difference between the regions.

Table 2.2: Review of CE methods for enhancing the quality of dermatological images.

Author & year	Method	Database	Performance measure	Remarks
Khan et al. [96] (2021)	DS	ISIC 2016, ISIC 2017, HAM1000	Sensitivity, Precision, F1-score	1. Lesions are segmented by Mask RCNN. 2. Extreme learning machine is used for classification.
Afza et al. [59] (2022)	FGLEI	PH2, ISIC 2016, HAM1000	Accuracy, Sensitivity, Precision, F1-score, AUC	1. Superpixel mapped-based lesion segmentation is performed on preprocessed dermoscopy images. 2. Pretrained ResNet50 is used for feature extraction. 3. Modified Grasshopper optimization is used to select optimal features. 4. Naive Bayes classifier is used for classifying the lesions.
Amoabedini et al. [47] (2018)	HE	University of Waterloo	Sensitivity, Specificity, Precision, Approx. correlation	1. Local Radon transform is used for segmentation of lesions from the HE images. 2. ABCD features are extracted from the segmented lesions.
Schaefer et al. [49] (2011)	HE	Private images	Border error	1. HE dermoscopy image is applied with iterative segmentation. 2. Time-consuming procedure.
Gomez et al. [97] (2008)	HE	Private images	Error %	1. HE image is applied with independent histogram pursuit, for lesion segmentation.

Table 2.2 continued from previous page

Author & year	Method	Database	Performance measure	Remarks
Pennisi et al. [98] (2016)	HE	PH2	Accuracy, Sensitivity, Precision, F1-score	1. HE is applied to the Y channel of the input image represented in YC_bC_r color space. 2. Delaunay triangulation is used to segment the lesion. 3. Adaboost algorithm is used to classify the lesions.
Premaladha et al. [101] (2016)	CLAHE	992 dermoscopy images from the internet	Accuracy, Sensitivity, Specificity, Kappa value	1. CE image is applied with the fusion of several thresholding methods to segment the lesions. 2. Extracted features are given to ensemble-based hybrid Adaboost algorithm for classification.
Kim et al. [102] (2013)	CLAHE	Private images	Subjective evaluation	1. CLAHE is applied to the blue channel, and then Fuzzy C-mean clustering is done to segment the lesion.
Sarkar et al. [103] (2019)	CLAHE	PH2, MED-NODE, DermIS	Accuracy, AUC, Sensitivity, Specificity	1. Preprocessed images are classified using deep depth-wise separable residual convolutional network.

Table 2.3: Review of CE methods for enhancing the quality of panoramic images.

Author & year	Method	Database	Performance measure	Remarks
Xu et al. [104] (2014)	GEM	Indoor images, Real world set, HDR set	Subjective evaluation	1. Used a combination of HE and white balance integrated into a general convex optimization framework. 2. Specific model parameters control the balance between enhanced contrast and natural tone. 3. Outcome always depends on the optimum selection of model parameters, which vary for different input images.
Katircioglu [105] (2020)	CEHCM	TID 2008	Saturation, Error, PSNR, NIQE	1. Heat conduction equation in solids is used in 'I' channel of HSI color space. 2. CEHCM requires a larger processing time.
Celik [106] (2016)	SMIRANK	TID 2013, RGB-NIR, CSIQ	Subjective evaluation	1. SMIRANK uses page rank algorithm to find the gray rank vector. 2. A mapping function is derived from the rank vector, to map the input gray level to the output. 3. Output depends on the optimum selection of the hyperparameter.
Fu et al. [107] (2016)	FIE	Private images	GMSD, NIQE	1. FIE technique enhances the contrast of weakly illuminated images. 2. Computational complexity is high.

Table 2.3 continued from previous page

Author & year	Method	Database	Performance measure	Remarks
Celik and Tjahjadi [108] (2011)	CVCEA	Natural images	AMBE, DE, EME	1. 2D histogram is constructed using the contextual information from the neighboring pixels of the input image. 2. CVCEA failed to handle large histogram values properly and generated color artifacts due to over-stretching.
Lee et al. [109] (2013)	LDR	USC-SIPI database, Berkeley image data	AMBE, DE, EME, Pixel distance	1. LDR uses intra-layer optimization and inter-layer accumulation blocks. 2. Intra-layer optimization block computes the dissimilarity vector per layer from the 2D histogram. 3. Inter-layer accumulation block merges all dissimilarity vectors from different layers to form a transform function. 4. Enhanced version of the input image is obtained by mapping input gray levels to the output gray levels using the transform function.
Arici et al. [110] (2009)	WAHE	Private images	AMBE, Entropy, EME	1. A modified histogram is created from the weighted average of the input histogram and uniformly distributed histogram. 2. Specifically designed penalty terms are introduced to adjust the level of CE. 3. Contouring and grainy noises are formed in uniform regions. 4. Black and white stretching made the dark areas darker and the bright spots brighter in enhanced images.

Table 2.3 continued from previous page

Author & year	Method	Database	Performance measure	Remarks
Gu et al. [111] (2015)	RICE	Kodak database	FSIM, GSIM, IGM	1. Developed by modifying WAHE. 2. Saliency detection is done with the help of the QMC from the image signature model.
Ding et al. [112] (2014)	SHE	Private images	Mean, SD	1. Implemented based on the dimensionality reduction method. 2. Fails to preserve the hue component in enhanced image.
Zhan et al. [113] (2016)	FLM	BSD, DRIQ, LIVE	Contrast, Spatial freq., Gradient	1. A neural network-based feature-linking mode is designed for CE of panoramic images. 2. FLM is computationally complex.
Wang et al. [114] (2019)	AECLII	6 test images	Entropy, Gradient	1. AECLII is an adaptive image enhancement method. 2. Addresses the issue of local over-enhancement caused by uneven enhancement and lack of adaptability in parameter settings. 3. Overall brightness of the images are increased rather than increasing the contrast.

2.2 SKIN LESION SEGMENTATION

Skin lesion segmentation algorithms play a crucial role in the analysis of skin lesions. Researchers commonly rely on two prominent approaches: thresholding algorithms and deep learning (DL)-based algorithms. Thresholding algorithms are widely recognized for their computational efficiency but often suffer from lower segmentation accuracy. In contrast, DL-based algorithms exhibit remarkable accuracy in segmenting skin lesions, at the expense of significant computational resources [70]. These approaches present a trade-off between efficiency and accuracy, highlighting the need for further advancements in skin lesion segmentation techniques to strike a balance between computational requirements and the precise localization of lesions.

2.2.1 Threshold-based segmentation

Almost all threshold estimation algorithms use the grayscale version of the image. The basic procedure for most histogram-based image threshold estimation algorithms is practically similar. At first, an array of unique gray levels present in the image and the histogram are constructed. The threshold is increased from the minimum gray level to the maximum gray level present in the image. The array of unique gray levels is split into two segments with respect to the threshold. Similarly, the histogram is also divided into two segments. At each threshold value, the cost function is computed from the segments of the gray level array and the histogram. The gray level at which the cost function maximizes/minimizes is considered as the optimal threshold.

Otsu's thresholding (OT) [115] is one of the most widely used threshold estimation algorithms for binary image segmentation. In this algorithm, the threshold value for which 'inter-class variance' reaches the maximum is considered to be optimum. Pixels with values greater than the threshold value are considered one class, and those less than & equal to the threshold value are considered another class. Cavalcanti et al. [116], Amin et al. [117], Fan et al. [118], Reshma et al. [119], Iyatomi et al. [120], and Premaladha et al. [101] have used the OT algorithm to segment lesions from dermoscopy images. The OT technique is the most widely used threshold-based segmentation strategy employed for skin lesion segmentation, surpassing other popular techniques. This approach is automatic and computes the threshold faster [70].

Xing et al. [121] proposed a modified OT [115] called valley-emphasized OT (VEOT). In VEOOT [121], 'inter-class variance' for various threshold values are computed by a 'valley metric' of the histogram. The valley metric is calculated using the second derivative of the histogram, and it objectively indicates the possibility that the threshold corresponds to a valley point in the histogram.

Isodata thresholding (IT) [122] is a traditional threshold estimation method, similar to OT [115]. Here, the optimal threshold value is determined by an iterative search. It assumes a threshold (in most cases, it represents the average intensity value of the image) for dividing the image into two classes. The mean values of pixels for each class are computed to determine the composite mean. The composite mean value of the mean intensities of the two classes is considered as threshold. The threshold is incremented, and the steps are repeated until the threshold is greater than the composite mean. Vandaele et al. [123] and Rajab et al. [124] used the IT algorithm to segment the skin lesions.

Histogram slope difference distribution based threshold (HSDDT) algorithm [125, 126] finds an optimal threshold by analyzing the distribution of slopes computed from the histogram of the image. The algorithm starts by smoothing the histogram using Fourier coefficients and used for computing slopes by fitting straight lines before and after each histogram point. The slope difference distribution (SDD) is derived from the computed slopes, and histogram valleys are identified using the first derivative of SDD. The optimal threshold is selected based on the gray level corresponding to the valley point with a higher magnitude of the first derivative of the SDD. The HSDDT algorithm adapts well to images with uneven illumination, making it useful for image segmentation tasks.

As already mentioned, most threshold estimation techniques differ in terms of the cost function used to track the optimum threshold. Kittler and Illingworth [127] used an estimate of the misclassification error associated with grayscale image segmentation into foreground and background regions as a cost function to determine the optimal threshold. This method is often referred to as minimum error thresholding (MET). The MET algorithm [127] is based on a parametric assumption that the image's histogram is bimodal and the pixels within the foreground and background have a normal distribution.

Pal and Bhandar [128] used Poisson's distribution of pixels captured with standard digital cameras and modified the model for computing the error of misclassification. It is referred as Poisson's distribution-based minimum error thresholding (PDMET).

Kapur et al. [129] proposed a different cost function to track the optimal threshold. In Kapur's maximum entropy threshold (KMET) [129], calculates the histogram of the image and normalizes it to obtain probabilities of pixel intensities. The cumulative distribution function (CDF) is computed by summing up the normalized histogram values. The CDF represents the cumulative probability of the pixel intensities. For each possible threshold value, the image is segmented into two classes (foreground

and background). Then, the entropies of both classes are calculated using their respective CDFs. The value that maximizes the sum of entropies for both foreground and background classes is selected as the optimal threshold.

In the entropy-weighted OT (EOT) algorithm [130], the objective function combines Otsu’s algorithm and KMET. In minimum cross-entropy thresholding (MCET) algorithm [131], unlike KMET [129] method that maximizes the entropy, MCET minimizes the cross-entropy between the original image and the thresholded binary image. MCET aims to find the threshold that leads to the least information loss during image segmentation. By minimizing the cross-entropy, this method can be effective in segmenting images with complex intensity distributions, resulting in accurate and adaptive thresholding.

Tizhoosh [132] proposed a type-2 fuzzy-based thresholding (Type-2-FT). It employs type-2 fuzzy sets, where the membership function is a fuzzy set, allowing a more flexible representation of uncertainty than traditional type-1 fuzzy sets. The method involves defining type-2 fuzzy sets for foreground and background based on histogram information, considering primary and secondary membership grades. The threshold value, which maximizes the ‘ultra-fuzziness’ of the type-2 membership function is considered the optimum. Yuksel et al. [133] applied the Type-2-FT for segmenting the skin lesions on dermoscopy images. Fuzzy entropy thresholding (FET) [134] [135] is another fuzzy-based model applied for image thresholding applications. In this model, the threshold value at which the fuzzy entropy is minimum is selected as the optimal threshold value.

2.2.2 Deep learning-based segmentation

Recently, many researchers have used DL frameworks to segment skin lesions from dermatological images. The DL-based segmentation models consist of an encoder-decoder architecture. The encoder-decoder structure consists of two main components: an encoder and a decoder. The encoder captures the high-level features and semantic information from the input image. It typically consists of multiple convolutional and pooling layers, which progressively reduce the spatial dimensions of the input while increasing the number of feature maps. This process helps extracting hierarchical representations of the image, capturing both global and local features of the skin lesions. Conversely, the decoder aims to recover the spatial information and generate a pixel-wise segmentation map. It consists of up-convolutional layers, also known as deconvolutions, which gradually increase the spatial dimensions while reducing the number of feature maps. The decoder combines the high-level semantic information from the encoder with the recovered spatial information to generate a detailed segmentation map.

The encoder-decoder structure is designed to address the challenge of capturing both contextual information and fine-grained details necessary for accurate segmentation. The model can effectively identify and differentiate skin lesions from the background by combining the encoder’s ability to capture global context with the decoder’s ability to restore spatial details [136]. A dataset of annotated dermatological images of skin lesions is required to train an encoder-decoder model for skin lesion segmentation. Medical experts manually annotate the images, where each pixel is labeled as either a lesion or background skin [137]. The model is trained using the input image and the expert segmented ground-truth as the target.

In medical image analysis, obtaining extensively annotated datasets can be challenging and time-consuming, resulting in the limited availability of diverse training samples [138]. To overcome this challenge, image augmentation techniques are employed to expand the training dataset artificially [139]. Table 2.4 overviews standard augmentation techniques, including rotation, scaling, flipping, translation, shearing, and resizing. These transformations introduce geometric variations that simulate different viewpoints and orientations of objects within the images. Exposing the DL models to such variations makes them more robust to changes in object position, scale, and rotation, which improves their ability to generalize for unseen data [140].

Table 2.4: Geometrical transforms used for image augmentations [139].

Transform	Description
Rotation	Rotates the image by a specific angle θ .
Flipping	Mirrors the image horizontally or vertically.
Scaling	Resizes the image to a larger or smaller size.
Shearing	Applies a shearing transformation (skewing the image).
Translation	Shifts the image horizontally or vertically.

Transfer learning (TL) has gained widespread adoption in DL research [141, 142]. It is a method of learning a new task by utilizing the knowledge from a related task already learned [143]. The success of DL models generally depends on the availability of large-scale data used for training. Thus, insufficient training data may adversely impact the performance of the DL models [142, 144]. TL is introduced in training DL models to solve the problems mentioned earlier [144].

A source domain D_s , needs to learn a task T_s , and the target domain D_t , needs to learn a task T_t . The TL helps to learn a predictive function $f_t(x)$ from D_t using the knowledge of D_s and T_s , where $D_s \neq D_t$ or $T_s \neq T_t$ [145].

A domain D can be represented using feature space X and a probability distribution

$P(x)$, where $X=\{x_1, x_2, \dots, x_m\}$, and m is the number of features in X .

$$D = \{X, p(x)\} \quad (2.1)$$

For a domain D , task T comprises of a label space Y and a predictive function $f_t(x)$.

$$T = \{Y, f_t(x)\} \quad (2.2)$$

The predictive function $f_t(x)$ is learned from feature-label pair $\{x_i, y_i\}$, where $x_i \in X$ and $y_i \in Y$.

For a given source domain, D_s , and learning task T_s , TL improves the predictive function $f_t(x)$ for the target domain D_t based on the knowledge of D_s and T_s , such that the condition, $D_s \neq D_t$ or $T_s \neq T_t$ is satisfied.

In TL, a pre-trained model is often trained on a large-scale dataset (like ImageNet [146]) and is employed as either a feature extractor or a starting point for a new task. Rather than training the new model from scratch, the pre-trained model serves as the backbone, and only a subset of its layers are replaced or fine-tuned to adapt to the specific task at hand. This strategy significantly reduces the training time and computational resources required to train the models. Moreover, TL enables improved generalization capabilities, especially when the model is trained with limited data [142].

Researchers have extensively utilized various state-of-the-art pre-trained CNNs for TL in the literature. These include widely recognized CNN architectures, such as: AlexNet [147], GoogLeNet [148], InceptionNet [149], ResNet [150], VGG-16 [151], DenseNet [152], SqueezeNet [153], MobileNet [154], NASNet [155], XceptionNet [156], and EfficientNet [157]. These pre-trained models serve as powerful feature extractors and provide a solid foundation for TL, enabling researchers to leverage their learned representations for various tasks.

Kaymak et al. [158] conducted a comparative analysis focusing on the segmentation of skin lesions using fully convolutional networks (FCNs) [52]. The researchers investigated and evaluated four FCN architectures: FCN-AlexNet, FCN-8s, FCN-16s, and FCN-32s. To assess the performance of these architectures, authors employed TL using ImageNet and PASCAL VOC datasets. From this study, FCN-8s exhibited significantly superior segmentation performance compared to the remaining three architectures (FCN-AlexNet, FCN-16s, and FCN-32s).

Lin et al. [159] conducted a comparative study between U-Net and C-means

clustering methods for the segmentation of skin lesions from the dermoscopy images. They reported that U-Net showed better segmentation performance than the C-means clustering methods.

Sahin et al. [160] proposed Bayesian optimized SegNet for segmenting the skin lesions from the dermoscopy images. The SegNet consists of an encoder and a decoder. The encoder produces low-resolution feature maps. The decoder transforms low-resolution feature maps from the encoder to full input resolution feature maps. Finally, full input resolution feature maps from the decoder are used for pixel-wise classification. The hyperparameters are optimized using Bayesian optimization. From this study, Bayesian hyperparameter optimization in the SegNet increased the segmentation accuracy compared to the baseline network.

Goyal et al. [161] proposed ensemble DL methods using mask region-based CNN (RCNN) and DeepLabV3+ to accurately segment skin lesions from dermoscopy images. They developed three ensemble methods: ensemble-add, ensemble-compare (large), and ensemble-compare (small). Authors reported that their proposed ensemble methods exhibited better segmentation results than the FCNs [158], U-Net [159], and SegNet [160].

Table 2.5 presents a compilation of the thresholding methods employed by researchers for performing binary segmentation. On the other hand, Table 2.6 showcases a selection of widely utilized DL-based segmentation methods for segmenting lesions from dermatological images. Both tables concisely summarize the datasets used and the evaluation measures employed in the respective studies.

The thresholding algorithms discussed in the literature do not offer adequate accuracy and fail to retain consistent performance. In contrast, DL-based segmentation models discussed in this chapter use an encoder-decoder architecture that performs an image-to-image regression task, which is computationally heavy.

Table 2.5: Review of threshold-based segmentation methods.

Author & year	Method	Database	Performance measure	Remarks
Cavalcanti et al. [116] (2013)	OT	DermIS	XOR error, Sensitivity, Specificity	1. OT is applied on the macro-images of skin lesions to segment the lesion. 2. Segmented lesion is refined by active-contour method and morphological operation.
Amin et al. [117] (2020)	OT	ISIC 2018, PH2	Accuracy, Sensitivity, Specificity	1. Normalized dermoscopy image is segmented using OT. 2. Features are extracted from the segmented lesion using AlexNet & VGG 16 and are given to ensemble classifier.
Fan et al. [118] (2017)	OT	ISIC 2016, PH2	Error, DSC	1. Input dermoscopy images are enhanced. 2. Enhanced images are applied with saliency combined with OT to automatic segmentation of dermoscopy images.
Reshma et al. [119] (2020)	OT	ISIC 2016	Accuracy, Sensitivity, Specificity	1. Preprocessed dermoscopy images are segmented using OT. 2. ABCD features are extracted and given to the SVM classifier, to classify lesions.
Iyatomi et al. [120] (2008)	OT	Private images	Border error, Sensitivity, Specificity	1. Lesion area is extracted by OT. 2. Selected features are fed to neural network for classifying lesions. 3. Computational complexity is high.
Xing et al. [121] (2020)	VEOT	22 cell images, Internet images	DCS, JI	1. For each threshold, inter class variance is computed by valley metric of the histogram. 2. Valley metric indicates the possibility that threshold value corresponds to a valley point in the histogram.

Table 2.5 continued from previous page

Author & year	Method	Database	Performance measure	Remarks
Vandaele et al. [123] (2020)	Isodata	ISIC 2018	Accuracy, DSC, MCC	1. Topological image modification (TIM), and topological image processing (TIP) are applied to dermoscopy images. 2. Processed images are applied to isodata thresholding for segmentation of lesions.
Rajab et al. [124] (2004)	Isodata	Private images	Mean error	1. Skin lesion is segmented using Isodata and neural network edge detector. 2. It is slow and computationally expensive.
Wang et al. [126] (2016)	HSDDT	Private images	Precision, Recall, F1-measure	1. HSDDT is applied to segment cells or nanoparticles. 2. Performance of HSDDT depends on the calibration of the parameters.
Kittler and Illingworth [127] (1986)	MET	Private images	Subjective evaluation	1. MET estimated misclassification error associated with grayscale image segmentation. 2. Image's histogram is assumed to be bimodal.
Pal and Bhandar [128] (1993)	PDMET	Private images	Entropy	1. Assumes pixel values for the images captured with standard digital cameras followed Poisson's distribution. 2. Model is modified for computing the error of misclassification.
Kapur et al. [129] (1985)	KMET	Private images	Subjective evaluation	1. Identifies the optimal threshold by maximizing the entropy of the binary image.

Table 2.5 continued from previous page

Author & year	Method	Database	Performance measure	Remarks
Truong and Kim [130] (2018)	EOT	Private images	Pixel counts	1. The objective function combines OT and KMET. 2. The method requires more processing time.
Li and Lee [131] (1993)	MCET	Private images	Subjective evaluation	1. Finds the threshold value by minimizing the cross-entropy between the original image and the binary image.
Tizhoosh [132] (2005)	Type-2-FT	10 images	Mean, SD	1. Type-2 FT, the input image is fuzzified w.r.t the threshold value using a generalized bell-type fuzzy membership function.
Bandyopadhyay et al. [135] (2020)	FET	MSRA 10K, DUT-OMRON, Caltech 256	PSNR, MSE	1. Threshold corresponding minimizes fuzzy entropy is selected as optimal threshold. 2. FET can classify pixels with low signal-to-noise ratio (SNR).

Table 2.6: Review of DL-based skin lesion segmentation methods.

Author & year	Method	Database	Performance measure	Remarks
Kaymak et al. [158] (2020)	FCN	ISIC 2017	DSC, JI, Accuracy	1. Used TL by fine-tuning on four different FCN architectures (such as FCN-AlexNet, FCN-8s, FCN-16s, and FCN-32s) for segmentation. 2. FCN-8s obtained DSC of 0.83. 3. Long training time.
Lin et al. [159] (2017)	U-Net	ISIC 2017	DSC, JI	1. Performance of U-Net and clustering algorithm for lesion segmentation are studied. 2. Effect of pre/post processing is studied for U-Net. 3. U-Net with pre/post processing reported the highest DSC and JI of 0.77 and 0.62, respectively.
Sahin et al. [160] (2022)	SegNet	ISIC 2016, ISIC 2017	DSC, JI, Accuracy, Sensitivity, Specificity	1. Optimized SegNet is used for localizing lesions on dermoscopy images. 2. Hyperparameters are optimized using Bayesian optimization.
Goyal et al. [161] (2020)	Ensemble method	ISIC 2017, PH2	DSC, JI, MCC,	1. Developed ensemble methods based on mask RCNN and DeeplabV3+. 2. Ensemble method exhibited better segmentation results than FCNs, U-Net, and SegNet. 3. Computationally expensive.

2.3 DATABASES

The datasets utilized in this study include the University of Waterloo skin cancer database [20], MED-NODE [162], SD-260 [163], and SD-198 [164]. These datasets provide a diverse range of skin lesion images, enabling comprehensive evaluation and analysis of the proposed algorithms.

The University of Waterloo skin cancer dataset [20] contains 206 macro-images, out of which 119 images are of melanoma lesions, and 87 are nevus lesions. Expert-segmented ground-truth is available for all the macro-images in the University of Waterloo skin cancer dataset [20]. The MED-NODE [162] database encompasses 70 melanoma and 100 nevus lesion images. However, it does not provide expert-segmented ground-truth annotations. The SD-260 [163] contains images of 260 different skin diseases. Only 60 melanoma and 60 nevus images from SD-260 [163] are considered, and the remaining images of skin diseases like acne vulgaris, eczema, psoriasis, etc., are excluded. The SD-198 [164] have the same melanoma images as in SD-260 [163], so only 34 nevus images are considered from SD-198 [164] to prepare the custom dataset. Expert-segmented ground-truth were absent in MED-NODE [162], SD-260 [163], and SD-198 [164] databases. Table 2.7 presents the details of the four databases used in this thesis.

Table 2.7: Detailed summary of the databases.

Database	Melanoma	Nevus	Total	Expert segmented ground-truth	Resolution	Format
University of Waterloo [20]	119	87	206	Available	Ranges from 550×360 to 1649×1043	JPEG
MED-NODE [162]	70	100	170	Not available	1649×1043	JPEG
SD-260 [163]	60	60	120	Not available	1649×1043	JPEG
SD-198 [164]	0	34	34	Not available	1649×1043	JPEG

Three dermatologists and two experts in medical imaging participated in producing the annotated ground-truth for macro-images selected from the MED-NODE [162], SD-260 [163], and SD-198 [164] databases. To reduce the interoperator variability, the consensus segmentation of the three dermatologists and two medical imaging experts are taken as the final annotated ground-truth. The manually segmented ground-truth images already available in the database are rechecked and verified by dermatologists and medical imaging experts.

2.4 PERFORMANCE MEASURES

The performance of the proposed deep image-to-scalar regression models and the segmentation accuracy of the segmentation models are evaluated quantitatively. This evaluation involves measuring and assessing the models' performance using specific met-

rics and criteria. The performance of the deep image-to-scalar regression models is typically evaluated using the root mean square error (RMSE) [165]. While the segmentation accuracy of the models are assessed using the Dice similarity coefficient (DSC) [166]. It measures the overlap between predicted and ground-truth segmentation masks. These quantitative evaluations provide objective measures to assess the effectiveness and accuracy of the proposed models in their respective tasks.

RMSE [165] is the square root of the mean of squared errors between the original values and the predictions made by the regressor model. RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^N (y(k) - \hat{y}(k))^2} \quad (2.3)$$

where, $\hat{y}(k)$ represents the predicted value by the regressor, $y(k)$ represents the original value, and N refers to the number of images in the dataset. The Eq. (2.3) quantifies the average discrepancy between the predicted and actual values, providing a measure of the overall accuracy of the regressor model's predictions. RMSE is a widely adopted metric in regression tasks because of its effectiveness in measuring the difference between predicted and actual values. The key reasons for its prevalent usage are [167],

1. **Emphasis on prediction accuracy:** RMSE quantifies the average prediction error between the model's predicted and actual values. Minimizing RMSE corresponds to improving the accuracy of the regression model's predictions, making it a desirable metric for assessing performance.
2. **Sensitivity to outliers:** RMSE is sensitive to outliers, penalizing large prediction errors more than other metrics such as mean absolute error (MAE). This sensitivity of RMSE helps to identify and address instances where the model performs poorly on extreme or atypical data points, providing a robust evaluation of the model's performance.
3. **Differentiability:** RMSE is a differentiable metric, which is beneficial in optimizing regression models using gradient-based optimization algorithms.

The DSC [166] is a prevalent metric to evaluate the segmentation accuracy of different segmentation algorithms with the manually segmented ground-truth. It helps to confirm the reproducibility of segmentation algorithms. The DSC value varies in a range from '0' and '1'. A higher DSC value indicates a better segmentation performance. DSC is derived from the Jaccard index (JI). JI is also known as intersection

over union (IoU) of the segmented result against the ground-truth mask [158]. The relationship between DSC and JI [166] is given in In Eq. (2.4).

$$DSC(\hat{X}, \hat{Y}) = \frac{2JI(\hat{X}, \hat{Y})}{1 + JI(\hat{X}, \hat{Y})} \quad (2.4)$$

$$JI(\hat{X}, \hat{Y}) = \frac{|\hat{X} \cap \hat{Y}|}{|\hat{X} \cup \hat{Y}|} \quad (2.5)$$

In Eq. (2.4) and Eq. (2.5), the variables $\hat{X}$ and $\hat{Y}$ represent the binary ground-truth and the binary segmented image, respectively. In Eq. (2.5), the symbols $\cap$ and $\cup$ denote intersection and union operators, respectively.

The researchers have widely adopted DSC as a standard evaluation metric in the majority of scientific publications for medical image segmentation tasks [168–170]. The reasons for its widespread usage are as follows [171]:

1. Sensitivity to false positives and negatives: The DSC demonstrates sensitivity to both false positives and false negatives, thereby providing a balanced measurement of segmentation accuracy. It takes into account under-segmentation (false negatives) and over-segmentation (false positives), allowing for a comprehensive assessment.
2. Robustness to class imbalance: The background class often dominates the pixel distribution in many segmentation tasks. The DSC is less sensitive to class imbalance than other metrics like pixel accuracy, making it a more reliable measure.

2.5 DISCUSSION

Preprocessing steps, specifically illumination correction and CE, seek to improve image quality. These steps play a crucial role in enhancing additional processing features, thereby improving the performance of classical and DL-based classification and segmentation methods.

Given the limitations, it is crucial to develop illumination correction algorithms that can equalize uneven background illumination without reducing the contrast between the background and the lesion or inducing color distortion. Also, building an algorithm that can selectively adjust the contrast between the lesion and the background. Localization of lesions on dermatological macro-images can be

approached as a thresholding problem, and accurate threshold values can be predicted using an image-to-scalar regressor. Such a regressor is computationally lighter than DL models with encoder-decoder architectures used for lesion segmentation from macro-images.

CHAPTER 3

AN AUTOMATED ILLUMINATION CORRECTION ALGORITHM BASED ON COUNTER EXPONENTIAL TRANSFORM AND MODIFIED RESNET FOR DERMATOLOGICAL MACRO-IMAGES

In this chapter, a novel method called illumination equalization using counter exponential transform (IECET) as a solution to address the issue of uneven background illumination in dermatological macro-images is discussed. Uneven background illumination can negatively impact the performance of lesion segmentation and classification algorithms. Traditional methods for equalizing illumination often reduce the contrast between the background and the lesion and can even introduce color distortion or drastic changes in the color tone.

A counter exponential transform (CET) is introduced to effectively equalize the background illumination of the macro-images while preserving the contrast between the background and the lesion. The impact of IECET on lesion segmentation is evaluated by inputting illumination-corrected macro-images into state-of-the-art deep learning (DL)-based segmentation models. This helps to assess the effectiveness of the IECET method in enhancing the performance of existing segmentation techniques.

This chapter provides an in-depth analysis of the IECET method and its implications for skin lesion segmentation. Experimental results are presented and compared with existing illumination correction techniques to demonstrate the advantages of the proposed approach in accurately delineating skin lesions while mitigating the challenges posed by uneven background illumination.

3.1 METHODOLOGY

Fig. 3.1 depicts the steps involved in the IECET. To normalize the background lighting, the CET method is applied on the value component of the image represented in the HSV color space. Hence, the RGB image is transformed to HSV color space as described by [172]. The image in HSV color space consists of hue (H_i), saturation (S_i), and value (V_i) components. The CET is applied to the normalized value component $V_i(m, n)$. The value component represents background lightness in HSV color model.

Therefore, equalizing the value component of the image is an effective way to reduce nonuniformity of the background lighting. In HSV color model, the chroma component comprises dominant color information. As the value component directly does not account for color information, equalizing the value component does not produce any inadvertent color distortions. Equalizing red, green, and blue channels altogether or hue, and saturation components along with the value component will produce color distortions on the illumination-equalized image [173].

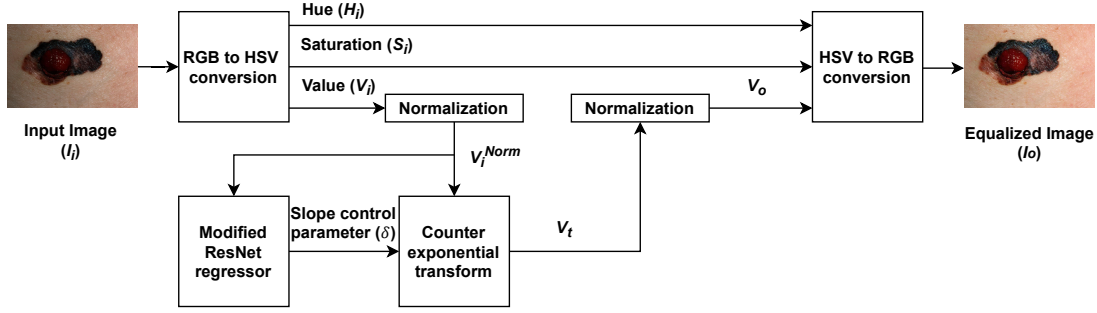


Figure 3.1: A schematic of the steps in the IECET.

The HSV color model representation of the illumination equalized image is created by replicating the hue and saturation channels from the input image and combining them with the value channel obtained after applying CET as:

$$H_o = H_i, S_o = S_i, \text{ and } V_o = \Gamma(V_i), \text{ and } I_o = \{H_o, S_o, V_o\} \quad (3.1)$$

In Eq. (3.1), H_i , S_i , and V_i , respectively are hue, saturation, and value components of the input image, $I_i = \{H_i, S_i, V_i\}$. Similarly, H_o , S_o , and V_o , are hue, saturation, and value components of the output image, $I_o = \{H_o, S_o, V_o\}$, respectively. The operator, $\Gamma(\cdot)$ represents the CET and its associated operations. In CET, the value component of the input image V_i is stretched to a range [0 1] via a simple normalization step before applying the CET. The normalized value component is,

$$V_i^{Norm}(m, n) = \frac{V_i(m, n) - V_i^{min}}{V_i^{max} - V_i^{min}} \quad (3.2)$$

In Eq. (3.2), V_i^{min} and V_i^{max} respectively indicate minimum and maximum gray levels in the value component. The normalization step helps to expand dynamic range of the value component and enhance the contrast as in a traditional contrast stretching operation. Applying the CET on the normalized value component $V_i^{Norm}(m, n)$ yields the transformed value component,

$$V_t(m, n) = 1 - e^{-\left(\frac{V_i^{Norm}(m, n)}{\delta}\right)} \quad (3.3)$$

In Eq. (3.3), the variable ‘ δ ’ is a user-defined parameter named ‘slope control parameter (SCP)’. It controls the slope of the exponential function in CET. The CET shifts darker regions in the input image where the background lightning is poor to a level on par with the illumination level at other regions of the image. The SCP decides the range of darker or low intensity values that is shifted to the upper level. Before reinserting the equalized value component ‘ V_t ’ back to the output image $I_o = \{H_o, S_o, V_o\}$ it has to be renormalized to $[0 \ 1]$. The renormalized value component V_o obtained from the equalized component V_t is,

$$V_o(m, n) = \frac{V_t(m, n) - V_t^{min}}{V_t^{max} - V_t^{min}} \quad (3.4)$$

In Eq. (3.4), V_t^{min} and V_t^{max} indicate minimum and maximum levels in the equalized value component V_t , respectively. The illumination equalized image in HSV color space is converted back to RGB color space [172].

3.1.1 Selection of the slope control parameter (SCP)

A new ResNet regressor with a modified architecture is constructed and further trained to automatically select the optimum value of SCP (δ) from the normalized value component (V_i^{Norm}). The ResNet regressor is modified by changing the last fully connected (FC) layer to output a single scalar value of the SCP and replacing the softmax and classifier layers with a regression layer of the ResNet-50 baseline network [150]. The output of the regression layer represents the optimum value of the SCP of the given input image.

The architecture of ResNet-50 [150] consists of a sequence of multiple bottleneck residual blocks, each consisting of several convolutional layers and shortcut connections. Fig. 3.2 shows a bottleneck residual building block for the ResNet-50 model [150]. In the bottleneck residual block, the input is represented as ‘ a ’, and the intended underlying mapping is denoted as ‘ $H(a)$ ’. This is achieved by having the sequence of interconnected nonlinear layers fit an additional mapping represented as $F(a) := H(a) - a$. Consequently, the initial mapping can be recast into $F(a) + a$. This architecture reduces the computational complexity of the network while maintaining its representational power. The first two layers of the bottleneck block perform feature compression, reducing the feature maps and the computation required to process them. The third layer performs feature expansion, increasing the feature maps and the representational power of the block. Using bottleneck blocks in ResNet-50 allows for a deeper and wider network, compared to traditional residual blocks, without significantly increasing the computational cost. Additionally, the reduced number of filters in the first two layers helps to reduce overfitting since the

network is forced to learn more abstract representations of the data [150].

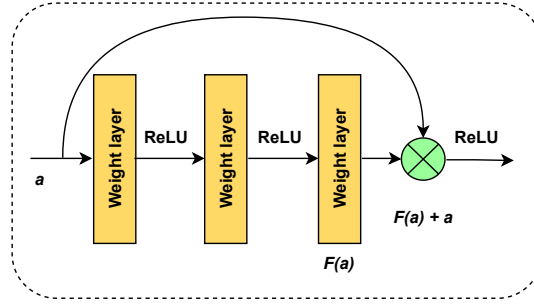


Figure 3.2: A bottleneck residual building block of ResNet-50 [150].

In ResNet-50 [150], the filter size for the first convolutional layer is typically 7×7 , which is large enough to extract large-scale features from the input image. As the network goes deeper, the filter size is reduced to 3×3 in order to capture increasingly fine-grained features. The number of filters in each convolutional layer is also increased as the network goes deeper, allowing the network to learn more complex features. The number of filters is specified by the depth of the layer, and it determines the number of feature maps that are produced by the layer. Each residual block typically consists of multiple convolutional layers, succeeded by batch normalization and activation layers. The filters in the residual block learn to extract features from the input, and the shortcut connections allow information to be bypassed from one or more layers, which helps to reduce the vanishing gradients problem [150].

Fig. 3.3 shows the network architecture of the modified ResNet regressor, which consists of five groups of 49 convolutional layers and one FC layer. In the residual block, the convolutional layers use kernel sizes of both 1×1 and 3×3 to capture features from the input image at varying resolutions. After every convolution operation, the rectified linear unit (ReLU) activation function is applied [147]. The output feature dimension is halved after average and maximum pooling with a stride of 2. The output dimension of the FC layer corresponds to the output scalar value, which is set to 1. Each residual block is repeated multiple times, which is represented by a numerical digit appearing below each block of Fig. 3.3

3.1.1.1 Training process for the modified ResNet

As part of the training phase, the modified ResNet regressor is fed with a collection of normalized value components that are derived from the HSV color model representation of the macro-images from the training data. These normalized value components serve as the input to the network. In this context, the target refers to a set of ideal SCP values that correspond to each macro-image in the training dataset. The ideal SCP value for a particular macro-image is determined by finding the SCP value that pro-

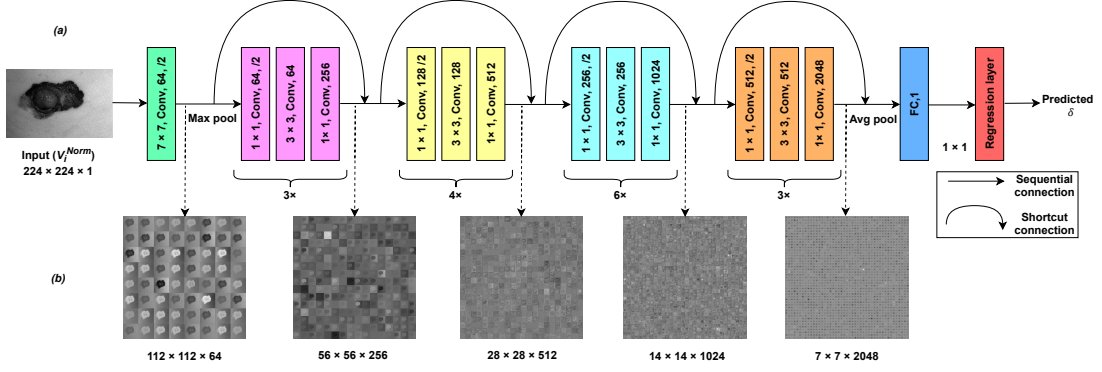


Figure 3.3: Layer representation of modified ResNet architecture, its operations, and feature maps: (a) schematic illustration of representative layers of the modified ResNet network, (b) feature maps obtained at different layers of the trained modified ResNet, from the input V_i^{Norm} image, to predict the optimal SCP (δ).

duces the highest peak signal to noise ratio (PSNR), when the IECET algorithm is applied to the illumination-corrected image. PSNR (given in Eq. (3.5) [174]) is a widely used measure of image quality, often used to evaluate the effectiveness of illumination correction algorithms [175, 176]. It is based on comparing the illumination-corrected image with the reference image and computing the amount of distortion or error between them. PSNR is expressed in decibels (dB). A higher PSNR value indicates a better quality image [177], as the distortion or error between the reference image and illumination-corrected image is smaller.

$$PSNR = 10 \log_{10} \left(\frac{(2^k - 1)^2}{MSE} \right) \quad (3.5)$$

where ‘ k ’ is the number of bits per pixel, and MSE is the mean squared error between the reference and illumination-corrected images and represented in Eq. (3.6)

$$MSE = \frac{1}{M_i \times N_i} \sum_{m=1}^{M_i} \sum_{n=1}^{N_i} [RI(m, n) - IC(m, n)]^2 \quad (3.6)$$

where $RI(m, n)$ and $IC(m, n)$ are the pixel values of the reference and illumination-corrected images, respectively, at location (m, n) , and $M_i \times N_i$ represents the dimensions of the image.

As reported in [178], the existing datasets of dermatological macro-images do not contain unevenly illuminated images. Therefore, a new dataset with unevenly illuminated dermatological macro-images is created. For creating the new dataset, 152 high quality macro-images from the skin cancer database, University of Waterloo, Canada [20] and 147 high quality macro-images from the MED-NODE

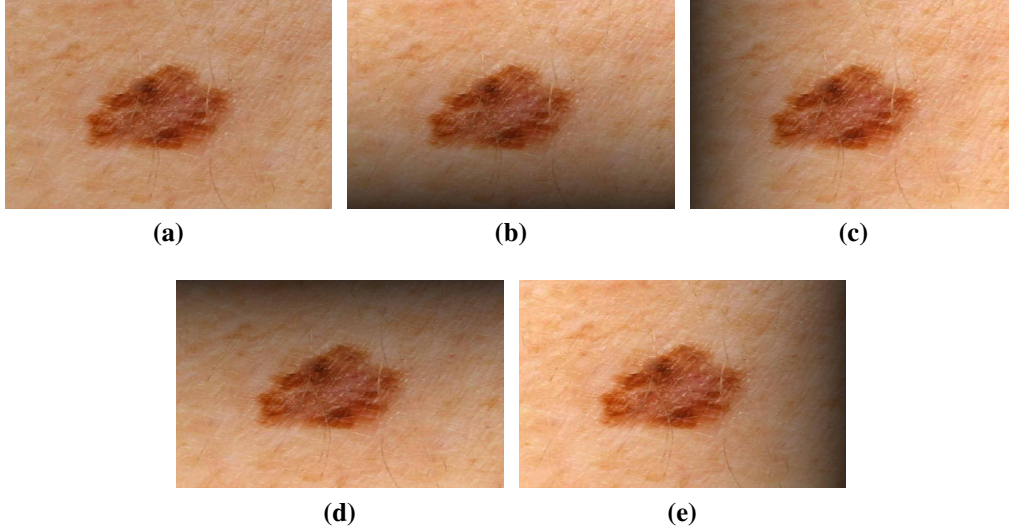


Figure 3.4: Representation of simulated unevenly illuminated images generated by adding illumination distortion on four sides of the reference image: (a) reference image, (b)-(e) simulated images.

dataset [162], with an image size of 224×224 , were chosen as the reference images. Uneven illumination commonly occurs on one side of the image [178]. According to this, uneven background lightning is simulated using the procedure described by Xie et al. [178]. Fig. 3.4 displays a set of simulated unevenly illuminated images. Four unevenly illuminated images are produced by introducing illumination distortion to one side of the reference image and applying the same degree of distortion to the other three sides. By adjusting the degree of distortion on each side of the four sides, 12 unevenly illuminated images from a single reference image are generated. In total, 3,588 uneven background illuminated macro-images are generated from the 299 high quality reference images in the newly created dataset. The dataset is divided such that 80% of the images from the dataset are allocated for training, whereas 05% for validation, and the remaining 15% are reserved for testing.

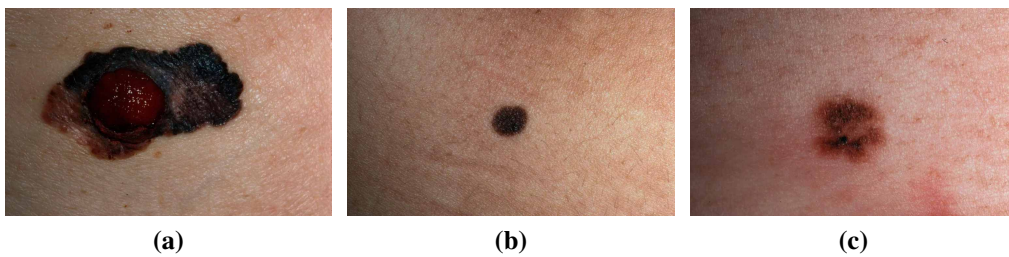


Figure 3.5: Test images: (a) image 1, (b) image 2, and (c) image 3.

Three test images depicted in Fig. 3.5 illustrate the process of making the training dataset. To produce illumination-corrected images for each unevenly illuminated macro-image, the SCP value is adjusted from 0.05 to 3 with an incremental step size

of 0.0525. The PSNR value is then recorded for each of the resulting illumination-corrected images. The ideal SCP value is determined by identifying the SCP value that produces the maximum PSNR value for the illumination-corrected image. This process is carried out on all simulated macro-images with uneven illumination in the training data images, and the resulting optimal/ideal SCP values are recorded.

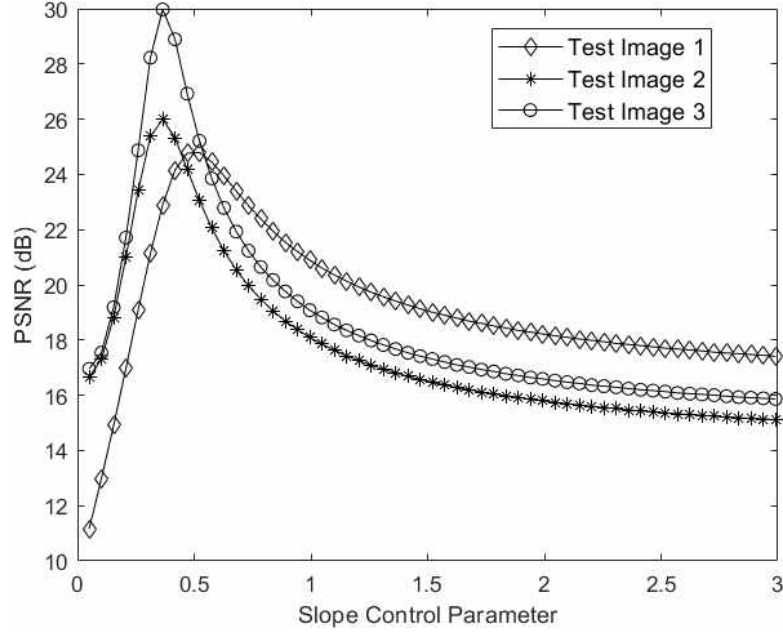


Figure 3.6: Influence of SCP on PSNR values.

The graph depicted in Fig. 3.6 illustrates how the PSNR values change across three test images for various SCP values. As depicted in Fig. 3.6, the maximum PSNR of 24.83 dB is obtained for test image 1 at an SCP value of 0.487. This suggests that the ideal SCP value for test image 1 is 0.487. Similarly, for test image 2, the maximum PSNR obtained is 26 dB at an SCP value of 0.362. Test image 3 also shows the ideal SCP value of 0.362, with the maximum PSNR obtained being 29.96 dB. The maximum PSNR values and their corresponding ideal SCP values for the three test images are presented in Table 3.1.

Table 3.1: The ideal SCP values and corresponding maximum PSNR values for three test images.

Test image	Max. PSNR (dB)	Ideal SCP
Image 1	24.83	0.487
Image 2	26.00	0.362
Image 3	29.96	0.362

During training, the modified ResNet regressor is fed with value components of the macro-images, with the corresponding ideal SCP values as the target from the training dataset. The regressor learns to predict the optimal SCP for an input image by

analyzing the relationship between the input value components and their corresponding ideal SCP values. During the training stage of the modified ResNet regressor, the loss function employed is the half mean squared error [179], and is expressed in Eq. (3.7).

$$loss = \frac{1}{2} \sum_{i=1}^L (\hat{S}_i - \hat{R}_i)^2 \quad (3.7)$$

In Eq. (3.7), $\hat{S}_i$ denotes the ideal SCP value corresponding to the i^{th} input image, $\hat{R}_i$ represents the SCP value predicted by the modified ResNet regressor, and L refers to the total number of input images employed to train the network.

To improve the performance of DL models for limited training datasets, transfer learning (TL) [180, 142] and data augmentation [139] techniques are commonly used [181]. In TL, pre-trained weights from the ImageNet dataset [146] are used to initialize the model, which is then fine-tuned for the image-to-scalar regression task using the newly created dataset. This approach can significantly reduce the training time and computational resources needed to train the modified ResNet regressor [142]. Data augmentation is implemented in the model training pipeline to avoid overfitting. Data augmentation includes implementing different geometric transformations to the images used for training. These transformations include horizontal and vertical flips, random translations in x and y directions within the range of [-30% 30%]. By applying these transformations, four sub-images are generated from each training image, thereby increasing the diversity and quantity of the training dataset. This technique can help the model to learn more versatile and generalizable features, resulting better performance for unseen data [140].

A sixfold cross-validation technique is implemented to evaluate the performance of the modified ResNet regressor. Around 600 images that are randomly selected are included in each fold. Five folds are combined for training, and one fold is used for testing. Approximately 3,000 images are used for training, 150 images are used for validation, and the remaining 150 images are used to test the performance of the regressor. The model's performance is evaluated on six different subsets, and the final result denotes the average performance across all six subsets.

The hyperparameter configuration used for training the modified ResNet regressor is presented in the Table 3.2 and is selected based on experimental results. The model is trained incrementally using stochastic gradient descent with momentum (SGDM) [182] optimizer, with a momentum value of 0.9 applied in each step. The training is performed with a batch size of 32 and an initial learning rate of 0.03. The modified ResNet regressor is trained for 16 epochs, and the learning rate is reduced by

a factor of 0.1000 after ten epochs. During the training process, early stopping [183] and L_2 regularization [182] techniques are incorporated to avoid overfitting.

Table 3.2: Hyperparameter configurations used for training the modified ResNet regressor.

Parameter	Value
Image size	$224 \times 224 \times 3$
Optimizer	SGDM
Momentum	0.9
Initial learning rate	0.03
Max. number of epochs	30
Mini-batch size	32
Validation frequency	05
L_2 regularization parameter (λ)	0.0005

In this study, early stopping [183] with a patience of 5 epochs is implemented. After every epoch, the model’s performance on a validation set is evaluated, and if there is no improvement for 5 consecutive epochs, then the training process is stopped. L_2 regularization works by adding a penalty term to the loss function [182]. This penalty term encourages the model’s weights to be small, making them less prone to overfitting. The penalty term in L_2 regularization is proportional to the square of the magnitude of the model’s weights. As a result, larger weights will incur a larger penalty, which encourages the model to learn smaller weights. The L_2 penalty term can be written as [182],

$$L_2 = \lambda ||W_i||^2 \quad (3.8)$$

where ‘ λ ’ is the regularization parameter, and $||W_i||^2$ is the squared magnitude of the model’s weights. The regularization parameter, ‘ λ ’, controls the strength of the regularization and is empirically chosen as 0.0005.

The IECET algorithm is simulated using Matlab 2022a on Windows 10 OS-based computer system. The system is equipped with an AMD Ryzen 5 processor with a clock frequency of 3.7 GHz. Additionally, it has an Nvidia GeForce GTX 1650 GPU and 8 GB RAM.

3.2 RESULTS AND DISCUSSION

The IECET algorithm is compared against four state-of-the-art illumination correction techniques, namely homomorphic filter (HF) [83], gray world color constancy algorithm (GCCA) [79], white patch color correction algorithm (WPCCA) [49], and shading and texture constrained retinex-based devignetting (STCRD) [75]. These algorithms are commonly used to equalize uneven background

lightning in digital images of skin lesions.

Fig. 3.7 shows scatter plots of the training and testing stages, which demonstrate the selection accuracy of the modified ResNet regressor. The scatter plot shown in Fig. 3.7 (a) displays the SCP values chosen for the training images. The plot shows a strong correlation between the chosen SCP values and the trend line, indicating that the modified ResNet regressor is able to accurately select SCP values that are close to the ideal ones for the training images. The scatter plot presented in Fig. 3.7 (b) illustrates the accuracy of the modified ResNet regressor in selecting SCP values for the test images. The plot reveals that most of the selected SCP values are in agreement with the trend line, with a few outliers slightly deviating from it. These findings suggest that the modified ResNet regressor accurately selects SCP values that are close to the ideal SCP values for the test images.

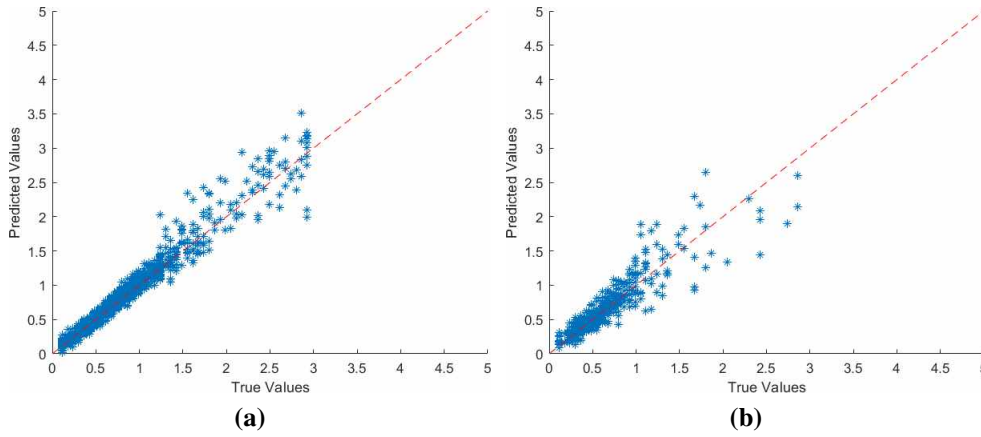


Figure 3.7: Scatter plots for training and testing stages of modified ResNet regressor: (a) training stage, (b) testing stage.

The effectiveness of the SCP selection by the modified ResNet regressor is assessed using the root mean square error (RMSE). The RMSE obtained by the proposed modified ResNet regressor is reported to be 0.16.

3.2.1 Subjective validation

The existing algorithms used for the illumination correction of digital images of the skin lesions are HF [45, 81–83], GCCA [79], WPCCA [88, 49], and STCRD [75]. The perceptual quality of the output images of the proposed IECET method is compared here with that of the HF, GCCA, WPCCA, and STCRD. The output images of various illumination correction algorithms produced from three test images are shown in Fig. 3.8. In the output images of the HF (Fig. 3.8 (d), (e), and (f)), GCCA (Fig. 3.8 (g), (h), and (i)), and WPCCA (Fig. 3.8 (j), (k), and (l)), the left halves of the images look darker than the right halves. The HF, GCCA, and WPCCA do not

equalize the background lightning completely. On the output images of GCCA (Fig. 3.8 (g), (h), and (i)), and WPCCA (Fig. 3.8 (j), (k), and (l)), the entire background regions have become darker than those in the input images (Fig. 3.8 (a), (b), and (c)). The lesions are generally darker than the background, as seen in the macro-images provided in Fig. 3.8 (a), (b), and (c). When the background skin become darker because of the equalization, in fact, the contrast between lesion and skin diminishes. Reduced contrast between lesion and skin will impact segmentation accuracy. In the output images of the STCRD (Fig. 3.8 (m), (n), and (o)), the uniformity of the background regions is relatively better than those in the output images of the HF (Fig. 3.8 (d), (e), and (f)), GCCA (Fig. 3.8 (g), (h), and (i)), and WPCCA (Fig. 3.8 (j), (k), and (l)). However, the lesion-to-background contrast is diminished in the output images of the STCRD (Fig. 3.8 (m), (n), and (o)). As already mentioned, diminished contrast of the lesion and background will affect the segmentation accuracy. In the output images of the IECET (Fig. 3.8 (p), (q), and (r)), the uniformity of the background regions is relatively better than those in the output images of the HF (Fig. 3.8 (d), (e), and (f)), GCCA (Fig. 3.8 (g), (h), and (i)), and WPCCA (Fig. 3.8 (j), (k), and (l)). In addition, the contrast among lesions and background regions in the output images of the IECET (Fig. 3.8 (p), (q), and (r)) is better than that in the output images of the STCRD (Fig. 3.8 (m), (n), and (o)). The above observations regarding the perceived quality of the illumination equalized macro-images have been verified on all 150 test images. The subjective validation of the macro-images equalized by IECET reveals that the IECET can equalize the nonuniformity of the background lighting on macro-images without compromising lesion-to-background contrast.

3.2.2 Objective validation

PSNR is used for objective evaluation of illumination-corrected images produced by different methods. PSNR values in decibels (dB) are computed by comparing the illumination-corrected images to the reference images. The illumination correction methods are applied to simulated images generated from the reference images, as described in Section 3.1.1.1. Table 3.3 shows the PSNR values obtained for the three test images and the mean PSNR for all test images considered. It can be observed that the proposed IECET method outperforms all other existing illumination correction methods on the three sample images and also obtains the best mean PSNR values on 150 test images.

The illumination correction is expected to improve the localization [184] of skin lesions on macro-images offered by the automated algorithms. Hence, the performance of the IECET is assessed and compared with that of other existing devignetting algo-

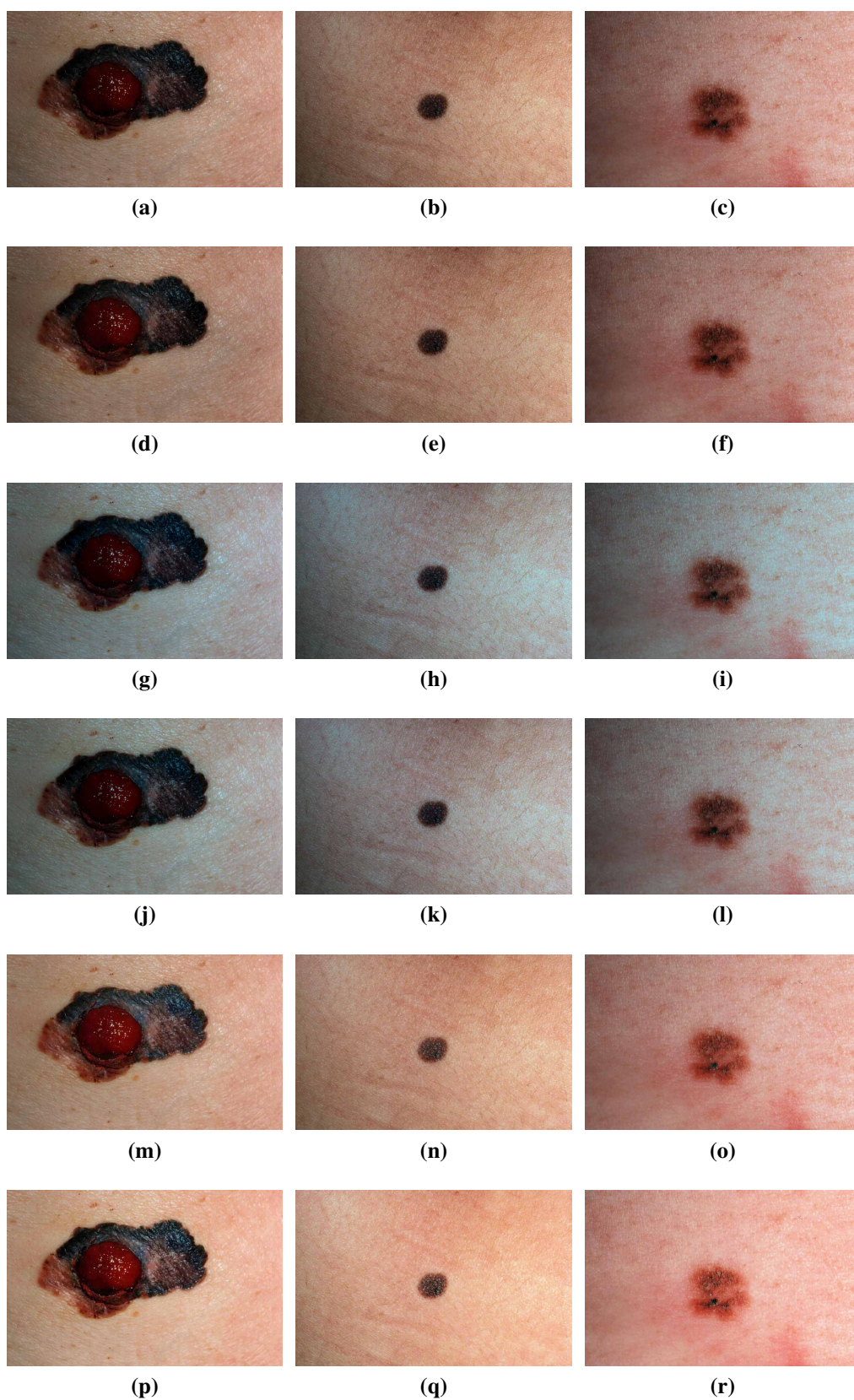


Figure 3.8: Illumination-corrected output for three test images using various algorithms: (a)-(c) Uneven illuminated images, (d)-(f) HF, (g)-(h) GCCA, (j)-(l) WPCCA, (m)-(o) STCRD, and (p)-(r) IECET.

Table 3.3: PSNR value in dB computed between the reference and illumination-corrected images obtained from various illumination correction methods.

Method	Test image 1	Test image 2	Test image 3	Summary on 150 test images (Mean $\pm$ SD)
HF (2019) [45]	14.25	13.20	13.59	11.92 $\pm$ 3.15
GCCA (2020) [79]	14.05	12.68	12.84	12.73 $\pm$ 1.62
WPCCA (2011) [49]	14.16	13.09	13.42	13.00 $\pm$ 1.79
STCRD (2022) [75]	20.87	19.18	20.11	20.36 $\pm$ 3.61
IECET	24.83	26.00	29.96	26.21 $\pm$ 2.17

rithms in terms of the segmentation accuracy offered by the automated approaches. Lin et al. [185] assessed the performance of image enhancement techniques in terms of segmentation accuracy on welding defect images. Following the validation protocol used by Lin et al. [185], performance of the IECET is assessed and compared with that of other existing devignetting algorithms in terms of the segmentation accuracy offered by three DL-based segmentation methods namely, DeepLabV3+ [161], FCNs [158], and U-Net [159]. Objective evaluation for segmentation accuracy is performed Dice similarity coefficient (DSC) [166].

Fig. 3.9 show the segmentation results of three DL-based segmentation methods on output images with various illumination correction techniques. From Fig. 3.9, it can be observed that on test image 1 with uneven background lightning (Fig. 3.9 (a)), U-Net segmentation resulted in over-segmentation, where a part of the skin is segmented as lesion. Whereas, DeepLabV3+ segmentation resulted in under-segmentation where a part of the lesion is incorrectly segmented as skin. While FCN yielded better segmentation results when compared with the expert-segmented ground-truth. On output image of HF (Fig. 3.9 (d)), U-Net yielded over-segmentation, and DeepLabV3+ produced under-segmented results. Only FCN yielded better segmentation when compared with the expert-segmented ground-truth. Observing segmentation results on output images of GCCA (Fig. 3.9 (g)) and WPCCA (Fig. 3.9 (j)), U-Net produced over-segmented results and FCN yielded under-segmented results. At the same time, DeepLabV3+ obtained better segmentation accuracy when compared with the expert-segmented ground-truth. On output images of STCRD (Fig. 3.9 (m)) and IECET (Fig. 3.9 (p)), all three DL-based segmentation methods yielded significantly better segmentation results when compared with the expert-segmented ground-truth.

In Fig. 3.9, it is observed that on test image 2 (Fig. 3.9 (b)) and HF (Fig. 3.9 (e)), U-Net produced unacceptable over-segmented results, a considerable portion of the skin is wrongly segmented as the lesion. In contrast, FCN yielded under-segmentation. On

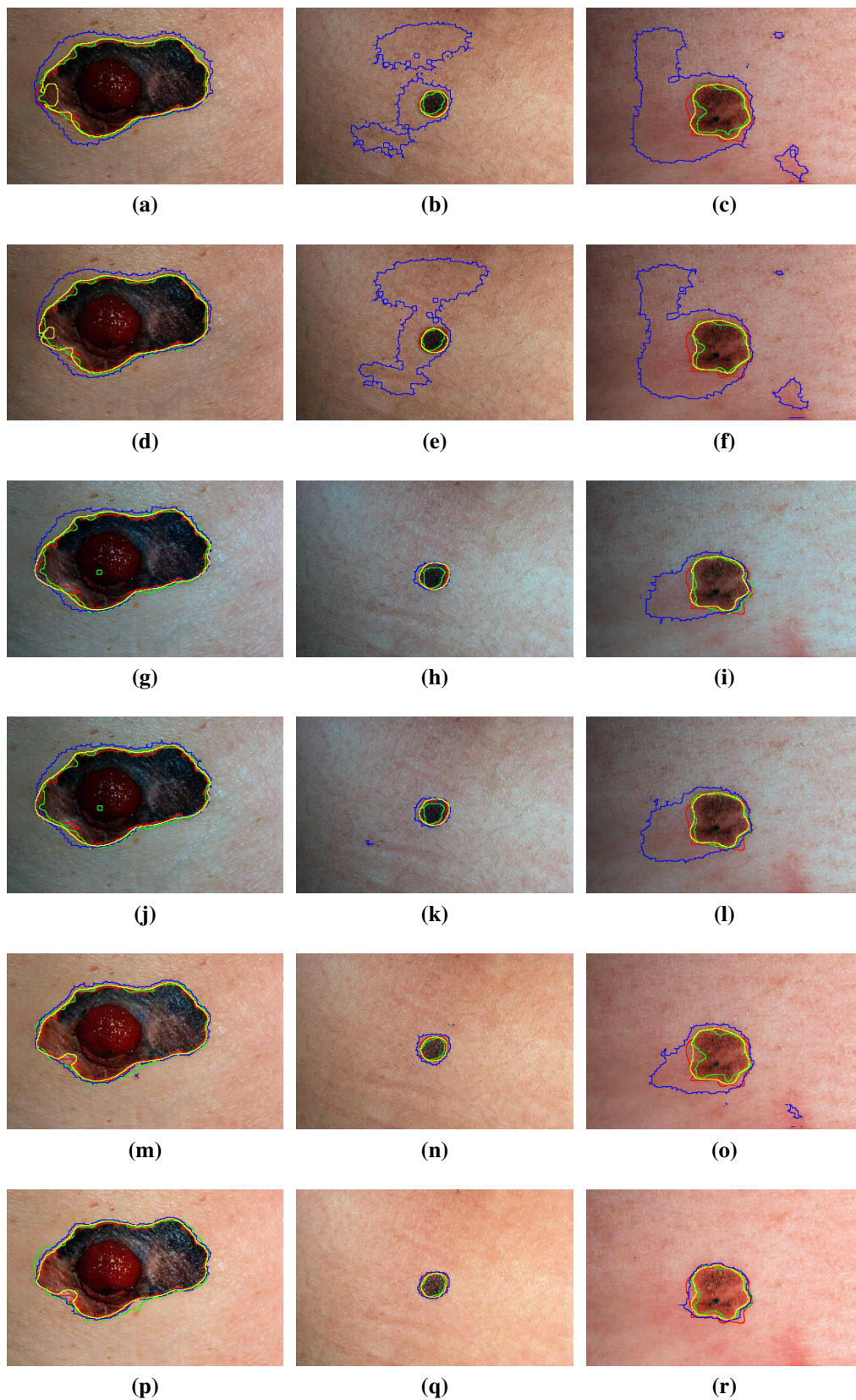


Figure 3.9: Segmented lesions by various DL-based segmentation methods (red contour represents expert-segmented ground-truth, yellow contour represents DeepLabV3+, green contour represents FCN, and blue contour represents U-Net): (a)-(c) Uneven illuminated images, (d)-(f) HF, (g)-(h) GCCA, (j)-(l) WPCCA, (m)-(o) STCRD, and (p)-(r) IECET.

output images produced by GCCA (Fig. 3.9 (h)) and WPCCA (Fig. 3.9 (j)), U-Net and FCN, methods yielded slightly over-segmented and under-segmented results, respectively. On illumination corrected test image 2 using STCRD (Fig. 3.9 (n)), U-Net segmentation yielded slightly over-segmented results, whereas DeepLabV3+, and FCN produced better segmentation results. All three DL-based segmentation methods produced segmentation results that significantly overlap with the expert-segmented ground-truth on illumination-corrected test image 2 produced by the proposed IECET algorithm (Fig. 3.9 (q)).

Fig. 3.9, reveals that on the unevenly illuminated test image 3 (Fig. 3.9 (c)), the U-Net algorithm incorrectly segmented a significant part of the background skin as lesion. On illumination corrected images produced by HF (Fig. 3.9 (f)) and GCCA (Fig. 3.9 (i)), the U-Net produced significantly over-segmented results, where a large part of the skin is incorrectly segmented along with the lesion. It can be observed that U-Net produced over-segmented lesions on images equalized by WPCCA (Fig. 3.9 (l)) and STCRD (Fig. 3.9 (o)). Upon observing the segmentation results produced on the output image equalized with the IECET (Fig. 3.9 (r)), it is found that all three DL-based segmentation methods yielded segmentation results which are having better overlap with the expert-segmented ground-truth. The IECET algorithm consistently improved the segmentation accuracy of the DL-based segmentation methods on the three test images and is validated across 150 test images.

Table 3.4 presents the DSC values obtained from various DL-based segmentation methods on output images of various illumination correction algorithms. The IECET method provided superior segmentation accuracy, as validated by mean DSC values obtained for the three DL-based segmentation methods on the 150 test images.

3.3 CONTRIBUTIONS

The main contributions of this chapter are as follows:

1. The proposed modified ResNet-based illumination equalization method, named IECET, is introduced to address uneven illumination in dermatological macro-images.
2. An automated approach is deployed using a modified ResNet regressor to select the operational parameter SCP in the IECET. This eliminates subjectivity and the need for manual intervention.
3. Illumination-correction as a preprocessing stage significantly improved the segmentation accuracy and corresponding DSC score of DL-based

segmentation models (such as DeepLabV3+, FCN, and U-Net) when applied to the output images of the IECET, outperforming other methods, including HF, GCCA, WPCCA, and STCRD algorithms.

4. The IECET algorithm is fully automated due to the integration of the modified ResNet, which facilitates the adaptive selection of the SCP.

CHAPTER 4

AN EFFICIENTNET-BASED MODIFIED SIGMOID TRANSFORM FOR ENHANCING DERMATOLOGICAL MACRO-IMAGES

The illumination equalization method using counter exponential transform (IECET) is proposed in chapter-3 and aims to equalize the uneven background illumination in macro-images. However, illumination correction alone may not effectively enhance the contrast between the lesion and the background skin. By performing contrast enhancement (CE), the differences in pixel intensities between foreground and background become more prominent. This increased discriminability makes it easier for subsequent segmentation algorithms to delineate the lesion from the background skin accurately.

In this chapter, the EfficientNet-based modified sigmoid transform (EMST) algorithm is introduced for enhancing the contrast between the lesion and background regions in dermatological macro-images. The effectiveness of the proposed EMST algorithm in lesion segmentation from dermatological macro-images is validated by comparing the segmentation accuracy of the results obtained through Otsu's thresholding (OT) on images enhanced with the EMST algorithm and existing state-of-the-art CE methods. By demonstrating superior segmentation performance, this research highlights the potential application of EMST to analyze skin lesions.

4.1 METHODOLOGY

On dermatological macro-images, the lesions appear comparatively darker than the background skin. Hence, segmentation of dermatological macro-images refers to grouping the pixels into background and lesion. The accuracy of the grouping can be improved by enhancing the contrast among grey levels that lie in between the dynamic ranges of the lesion and background. The modified sigmoid transform (MST) offers the flexibility to selectively enhance the contrast among grey levels that lie in between the dynamic ranges of the lesion and background. The selection of the MST is motivated by the above factors.

A schematic diagram of the proposed EMST-based CE method is depicted in Fig. 4.1. The input RGB macro-image is first converted to HSV color space using the

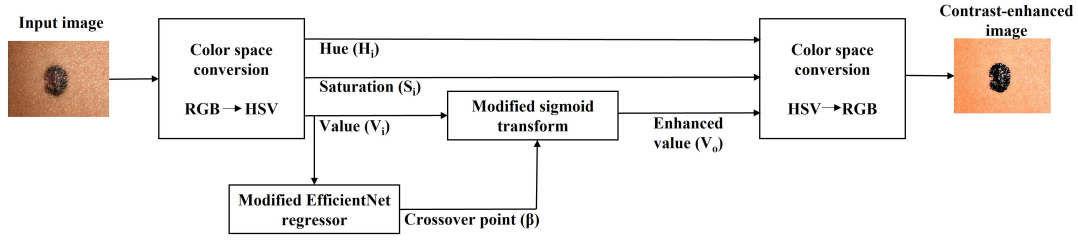


Figure 4.1: A schematic of the proposed EMST-based CE method.

color space transformation model [172]. MST is applied in the HSV color space to enhance the contrast between the lesion and background region on dermatological macro-images. In the HSV color space model, the luma channel/component reflects the spatial intensity profile of the image. In contradiction to the inter-channel correlation in the RGB color space representation, the luma component in the HSV color space model is independent of the chroma component that carries the color information [186]. Madooei et al. [72] have suggested that it is better to perform operations like shading correction on the value channel of the HSV color space than on any other channels of various color space representations. The image contrast depends directly on the spatial variation of the luma/intensity component. Hence, most CE transforms operate on the luma/value component. Applying nonlinear mapping functions used for CE simultaneously on red, blue, and green color channels increases the computational burden, deteriorates inter-channel correlation, and results in enhanced images with processing-induced color artifacts [186].

The image in HSV color space consists of hue (H_i), saturation (S_i), and value (V_i) components. The enhanced value component (V_o) is computed by MST, which is represented in Eq. (4.1). MST is provided with the V_i component, and the corresponding crossover point that divides the macro-image into lesion and background is predicted by the modified EfficientNet regressor.

$$V_o(m, n) = 1/[1 + \exp(\alpha * (\beta - V_i(m, n)))] \quad (4.1)$$

In Eq. (4.1), the symbols α and β denote the MST's shape parameter and crossover point, respectively.

The CE output image in HSV color space is obtained by concatenating the enhanced value component (V_o) obtained from the MST given in Eq. (4.1), with the hue (H_i) and saturation (S_i) components of the input image in HSV color space. The input dermatological macro-images to be enhanced exist in RGB color spaces, and the enhanced output images obtained from the MST are in HSV color space.

Therefore, the CE output images in HSV color space are converted back to RGB color space [172].

4.1.1 Prediction of the crossover point

A modified EfficientNet regressor is trained to predict the optimal crossover point from a given value component (V_i) of the HSV image. The modified EfficientNet regressor is constructed from the EfficientNet-B0 baseline network [157] by replacing the classifier layer with a regression layer. The output from the regression layer is the optimal crossover point. Transfer learning (TL) [142] using ImageNet weights is employed during the training phase to reduce the training time and training data required to train the model. The architecture of the modified EfficientNet regressor is shown in Fig. 4.2.

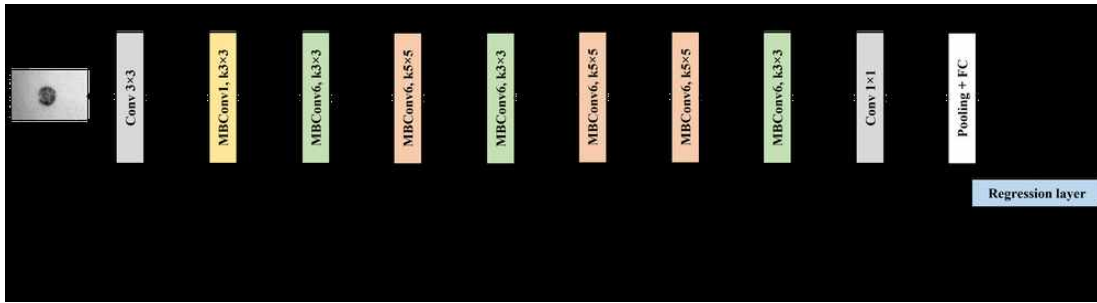


Figure 4.2: The architecture of the modified EfficientNet regressor.

The modified EfficientNet regressor is based on the EfficientNet-B0 convolutional neural network (CNN) model. This particular CNN model is selected based on the evidence that it displayed a good trade-off between computational resources and accuracy [157]. The primary building block of EfficientNet model is the Mobile inverted bottleneck convolution (MBConv). The MBConv is based on ideas borrowed from the MobileNet models [154].

4.1.1.1 Training the modified EfficientNet regressor

A set of normalized value components extracted from the HSV color space representation of the macro-images in the training dataset is fed as input, and the set of corresponding ideal values of the crossover point is defined as the target for training the modified EfficientNet regressor. For a particular macro-image, the value of the crossover point at which the OT [115] algorithm produces maximum Dice similarity coefficient (DSC) [166] from the enhanced image is considered as its ideal value. The dermatological macro-images contain only two distinct regions. The regions are lesion and the background. The lesions are comparatively darker than the background skin, which can be verified from the sample images shown in the first row of Fig. 4.3. As there are only two regions on the macro-images, segmentation of

lesions on macro-images is a typical binary thresholding problem. The OT is one of the traditional algorithms used in two-level image segmentation applications. The selection of the OT is motivated by its popularity in the segmentation of skin lesions. Premaladha et al. [101], Amin et al. [117], Fan et al. [118], Reshma et al. [119], Iyatomi et al. [120], Bansal et al. [187], and Ghalejoogh et al. [188] have used the OT for segmenting the skin lesions.

The three test images shown in the first row of Fig. 4.3 are used to demonstrate the procedure of preparing the training data. The three test images are selected such that their information content and the intensity profile are mutually exclusive. From each macro-image, enhanced images are produced by varying the crossover point from the minimum value to the maximum value of the normalized value component. The lesion is segmented from each enhanced image using the OT, and the value of the DSC is noted. The value of the crossover point at which the OT algorithm produces the maximum DSC value is considered ideal. The procedure is repeated on all macro-images in the training dataset, and ideal values of the crossover points are noted. As already mentioned, the set of normalized value components of the macro-images in the training dataset and the set of corresponding ideal values of the crossover point are fed to the modified EfficientNet regressor for training.

The variation of DSC values for different values of the crossover point on three test images are shown in Fig. 4.3. It can be observed from Fig. 4.3 (g) that a maximum DSC of 0.88 is obtained on image-1 at a crossover point of 0.494. Hence, it can be concluded that the ideal crossover point for image-1 is 0.494. In image-2, the maximum DSC obtained is 0.86 at a crossover point of 0.484, which is evident in Fig. 4.3 (h). Similarly, in Fig. 4.3 (i) the ideal crossover point for image-3 is 0.282, and the maximum DSC obtained is 0.89. Table 4.1 shows the maximum values of DSC and corresponding ideal crossover point values identified for three test images.

Table 4.1: The maximum values of DSC and corresponding ideal values of crossover point identified for three test images.

Test image	Maximum DSC	Ideal crossover point
Image-1	0.88	0.494
Image-2	0.86	0.482
Image-3	0.89	0.282

The dataset used to validate the EMST is created by merging macro-images of melanoma and nevus lesions obtained from the University of Waterloo skin cancer dataset [20], MED-NODE [162], SD-260 [163], and SD-198 [164]. Details about the databases are discussed in Section 2.3 of Chapter 2. The modified EfficientNet regressor is trained with a set of value components extracted from the macro-images

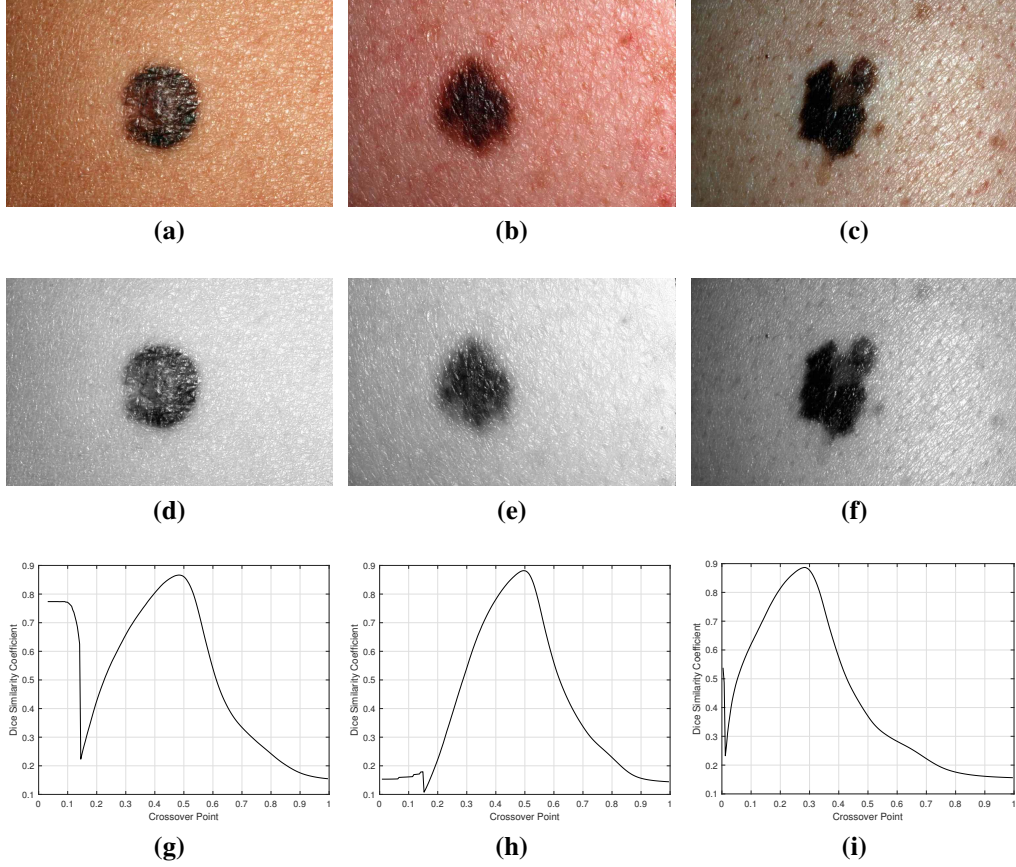


Figure 4.3: Identification of ideal crossover point from macro-images: (a)-(c) test images, (d)-(f) corresponding value components, (g)-(i) graph showing the variation of DSC with respect to the change in crossover point for three test images.

in the dataset and the corresponding set of ideal crossover points as the target. Half of the sum of the squared error [179] is used as the loss function during training of the modified EfficientNet regressor.

$$loss = \frac{1}{2} \sum_{i=1}^K (\hat{A}_i - \hat{P}_i)^2 \quad (4.2)$$

In this context, $\hat{A}_i$ is the ideal crossover point corresponding to the i^{th} input image, and $\hat{P}_i$ is the predicted crossover point by the modified EfficientNet regressor model. In Eq. (4.2), K indicates the number of images used for training. The normalization factor ‘2’ used in the loss function minimizes the influence of squaring the error on the derivative used in the gradient descent algorithm.

For evaluating the performance of the regressor, a 10 fold cross-validation technique is performed, with each fold consisting of approximately 50 images. Nine folds are used for training the regressor, the remaining one for testing. Images in each fold are randomly selected. Approximately 480 images are utilized for the training purpose

and the remaining 50 for testing. Image augmentation techniques are implemented to reduce overfitting and improve the model's performance by increasing the training images. Various geometric transformations [139] are applied to the training images. The training images undergo vertical and horizontal flips. Additionally, each image is rotated by a random angle θ within the range of $[-45^\circ, 45^\circ]$. Moreover, the training images are scaled or zoomed to 105% of their original size to introduce further diversity into the dataset. Each image in the training phase is augmented to get nine sub-images, making the total number of training images to 4,800. Hyperparameters are configured to yield the optimum performance from the modified EfficientNet regressor network. The modified EfficientNet regressor is trained for 15 epochs with a learning rate of 0.0001. The optimizer used is stochastic gradient descent with momentum (SGDM) [182] and batch size of 16. The hyperparameters used to train the modified EfficientNet regressor are given in Table 4.2 and were selected empirically.

Table 4.2: Hyperparameter configuration used in modified EfficientNet regressor.

Image size	No. of epoch	Learning rate	Batch size	Optimizer
$224 \times 224 \times 1$	15	0.001	16	SGDM

4.2 RESULTS AND DISCUSSION

The proposed EMST is compared with 15 state-of-the-art CE algorithms, namely, decorrelation stretch (DS) [96], the fusion of globally and locally enhanced images (FGLEI) [59], histogram equalization (HE) [47], contrast limited adaptive histogram equalization (CLAHE) [101], generalized equalization model (GEM) [104], heat conduction matrix (CEHCM) [105], spatial mutual information rank (SMIRANK) [106], fusion-based image enhancement (FIE) [107], contextual and variational CE algorithm (CVCEA) [108], layered difference representation (LDR) [109], weighted approximated HE (WAHE) [110], robust image CE (RICE) [111], shell HE (SHE) [112], feature-linking model (FLM) [113], and adaptive enhancement method for correcting low-illumination images (AECLII) [114]. The proposed EMST and all the compared CE algorithms are implemented in MATLAB R2022a on a desktop computer operating with Windows 10 OS with an NVIDIA GeForce GTX 1650 GPU. The system works with an AMD RYZEN 5 processor with a maximum clock frequency of 3.7 GHz.

The accuracy of the crossover point predicted by the modified EfficientNet regressor can be observed from the scatter plots of the training and testing phases shown in Fig. 4.4. By inspecting Fig. 4.4 (a), it is observed that the majority of the predicted crossover points for training images lie on the trend line to the points showing the best fit to the given training data. Thus, it can be concluded that the modified EfficientNet

regressor can predict the crossover point very close to the estimated ideal crossover point for the training images. On test images, the prediction accuracy of the model can be observed from Fig. 4.4 (b). It can be seen that the majority of the data points lie on the trend line, and a few data points fall around the trend line. Thus, it indicates that the modified EfficientNet regressor can predict the crossover point close to the estimated ideal crossover point on the 50 test images. Also, the prediction accuracy of the modified EfficientNet regressor is quantitatively evaluated using the root mean square error (RMSE) [189] between the predicted crossover point and the ideal crossover point.

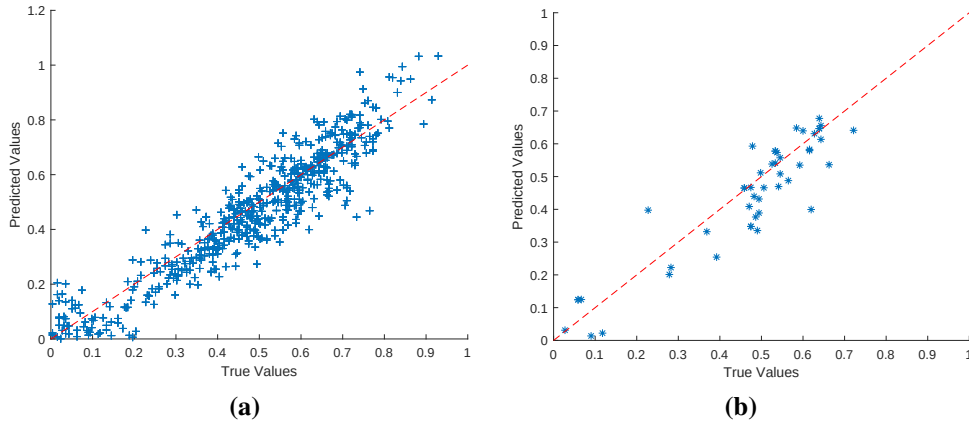


Figure 4.4: Scatter plots of training and testing phases: (a) training phase, and (b) test-phase. The closer the points are to the red line, the better is the model prediction.

Two separate experiments were conducted to understand the influence of image augmentation on the performance of the modified EfficientNet regressor. First, the modified EfficientNet regressor is trained with the original data without image augmentation, which resulted in an RMSE of 0.118 for test data with a training time of 173 seconds per epoch. To check the model performance for more images, the training data were artificially increased by performing different image augmentation on the original images, and these data are used for training the model. From Table 4.3, it is observed that performing image augmentation have significantly improved the performance of the model. After performing image augmentation, RMSE for test data is reduced from 0.118 to 0.078. The training time per epoch is slightly increased to 328 seconds as the number of training data has grown. It can be intuitively inferred that image augmentation created more diverse training data, which improved the performance of the modified EfficientNet regressor.

The EMST-based CE algorithm utilizes the enhanced value component to create the enhanced image in HSV color space. The enhanced value component is obtained from MST fed with the value component, shape parameter, and the regressor predicted crossover point. The shape parameter controls the shape of the sigmoid transform.

Table 4.3: Performance of the modified EfficientNet regressor with image augmentation.

Augmentation	RMSE	Time per epoch (sec.)
No	0.118	173
Yes	0.078	328

This experiment found that the shape parameter value above 20 does not significantly improve the DSC. So, 20 is used as the shape parameter value for the whole experiment. Finally, an enhanced output image is obtained by converting the enhanced image in HSV color space to RGB color space.

4.2.1 Visual analysis

Fig. 4.5 to Fig. 4.7 shows the enhanced output images produced by 15 state-of-the-art CE algorithms and the proposed EMST algorithm on the three test images. By visual inspection of Fig. 4.5, it can be observed that the DS method (Fig. 4.5 (b)) altered the RGB color combination of the test image-1, and the enhanced image lost the natural feel. The HE algorithm (Fig. 4.5 (d)) failed to enhance the image's top left and bottom left corners. With CLAHE (Fig. 4.5 (e)), the lesion got fused with the background, resulting in reduced contrast between the lesion and background. The GEM (Fig. 4.5 (f)), SMIRANK (Fig. 4.5 (h)), and FLM (Fig. 4.5 (o)) enhanced the contrast of the lesion but failed to enhance the top left and bottom left corners of the background skin region on the image when compared with the input test image-1 (Fig. 4.5 (a)). The FGLEI (Fig. 4.5 (c)), CEHCM (Fig. 4.5 (g)) and CVCEA (Fig. 4.5 (j)) does not enhance the image as expected. The FIE (Fig. 4.5 (i)) and AECLII (Fig. 4.5 (p)), algorithms reduce the contrast between lesion and background and makes it difficult to identify lesion from the background. The LDR (Fig. 4.5 (k)), WAHE (Fig. 4.5 (l)), and RICE (Fig. 4.5m) approaches could enhance the contrast of lesion, but failed to enhance the background, which is necessary to detect the lesion from the background. The mean brightness of the image was significantly reduced when applied with SHE (Fig. 4.5 (n)) on test image-1. With the proposed EMST (Fig. 4.5 (q)), the contrast between lesion and background is significantly improved, which can be visually observed.

On test image-2, the DS method (Fig. 4.6 (b)) modified the RGB color combination of the images, and it lost the natural feel. FGLEI (Fig. 4.6 (c)) enhanced the contrast of the lesion, but the mean brightness at the top right corner is increased to an unacceptable level. The HE algorithm (Fig. 4.6 (d)) failed to enhance the image's bottom left corner. With CLAHE (Fig. 4.6 (e)), the lesion got fused with the background. The GEM (Fig. 4.6 (f)), SMIRANK (Fig. 4.6 (h)), and FLM

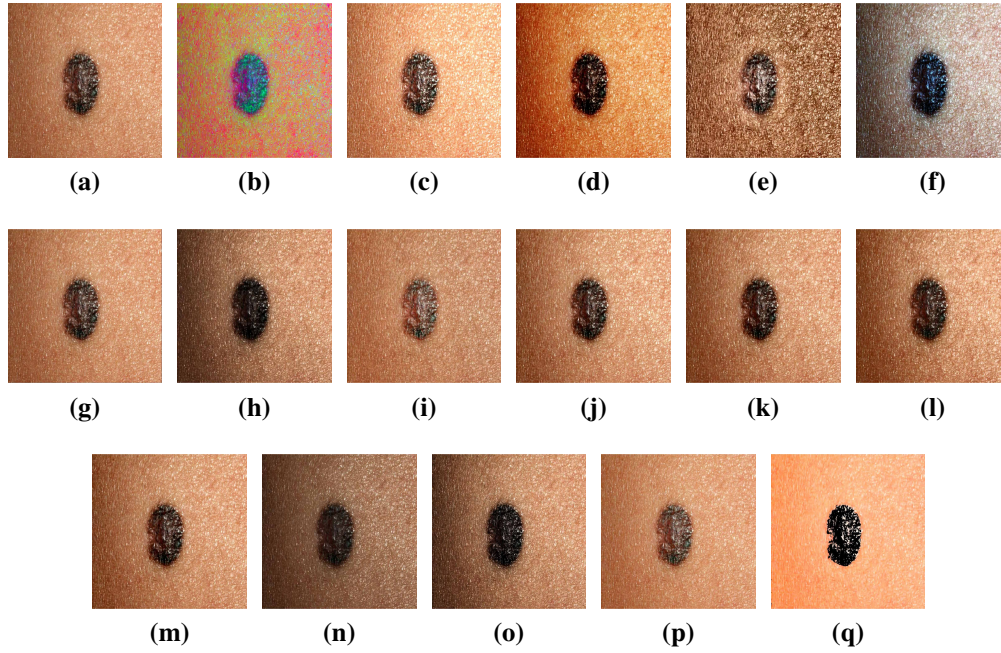


Figure 4.5: CE results on the test image-1: (a) Input image, (b) DS, (c) FGLEI, (d) HE, (e) CLAHE, (f) GEM, (g) CEHCM, (h) SMIRANK, (i) FIE, (j) CVCEA, (k) LDR, (l) WAHE, (m) RICE, (n) SHE, (o) FLM, (p) AECLII, and (q) EMST.

(Fig. 4.6 (o)) methods enhanced the contrast of lesion, but the bottom left corner of the image were under enhanced when compared with the input test image-2 (Fig. 4.6 (a)). The CEHCM (Fig. 4.6 (g)) and CVCEA (Fig. 4.6 (j)) does not enhance the image as expected. The FIE (Fig. 4.6 (i)) and AECLII (Fig. 4.6 (p)), enhanced the whole image. The contrast between the lesion and the background is reduced, making it difficult to identify the lesion from the background. The LDR (Fig. 4.6 (k)), WAHE (Fig. 4.6 (l)), and RICE (Fig. 4.6 (m)) could slightly enhance the lesion. The SHE (Fig. 4.6 (n)) significantly reduced the mean brightness of the output image. The proposed EMST could remarkably enhance the contrast of lesion from the background and is visible in (Fig. 4.6 (q)).

On test image-3, the DS (Fig. 4.7 (b)) introduced noise to the output image, and it lost the natural feel. The FGLEI (Fig. 4.7 (c)) failed to enhance the whole background region. The HE algorithm (Fig. 4.7 (d)) failed to enhance the contrast on left side of the image. With CLAHE (Fig. 4.7 (e)), the lesion got fused with the background. Application of GEM (Fig. 4.7 (f)), SMIRANK (Fig. 4.7 (h)), LDR (Fig. 4.7 (k)), WAHE (Fig. 4.7 (l)), RICE (Fig. 4.7 (m)), and SHE (Fig. 4.7 (n)) significantly reduced the mean brightness of the output image. The CEHCM (Fig. 4.7 (g)) and CVCEA (Fig. 4.7 (j)) failed to enhance the image. The FIE (Fig. 4.7 (i)) and AECLII (Fig. 4.7 (p)) yielded poor contrast between lesion and background. The FLM (Fig. 4.7 (o)) increased the mean brightness of the whole image to an unacceptable

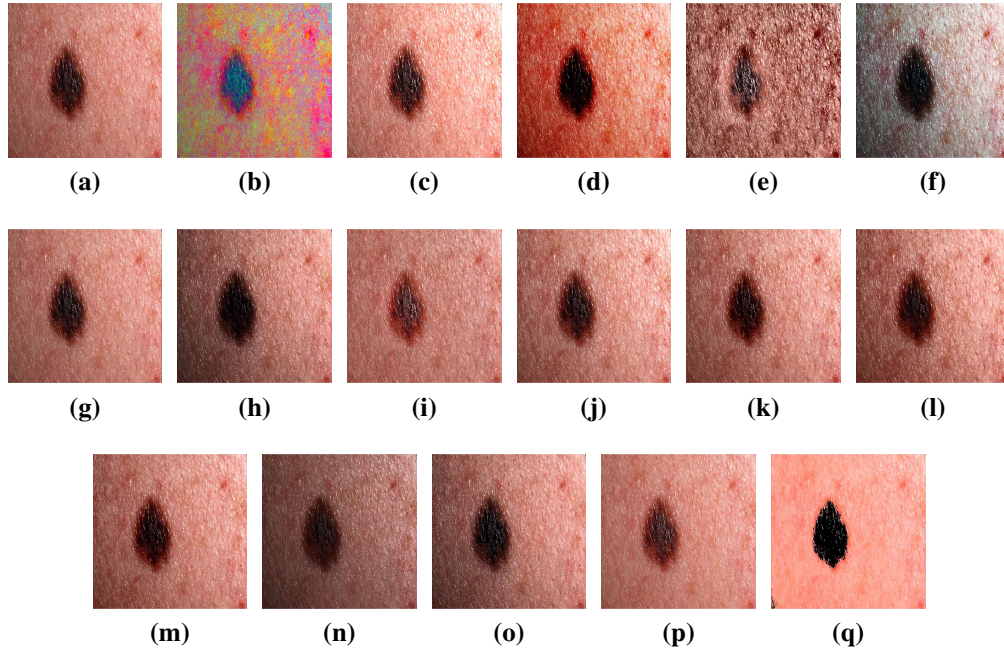


Figure 4.6: CE results on the test image-2: (a) Input image, (b) DS, (c) FGLEI, (d) HE, (e) CLAHE, (f) GEM, (g) CEHCM, (h) SMIRANK, (i) FIE, (j) CVCEA, (k) LDR, (l) WAHE, (m) RICE, (n) SHE, (o) FLM, (p) AECLII, and (q) EMST.

level. The proposed EMST could exceptionally enhance the contrast of lesion from the background and is visible in Fig. 4.7 (q).

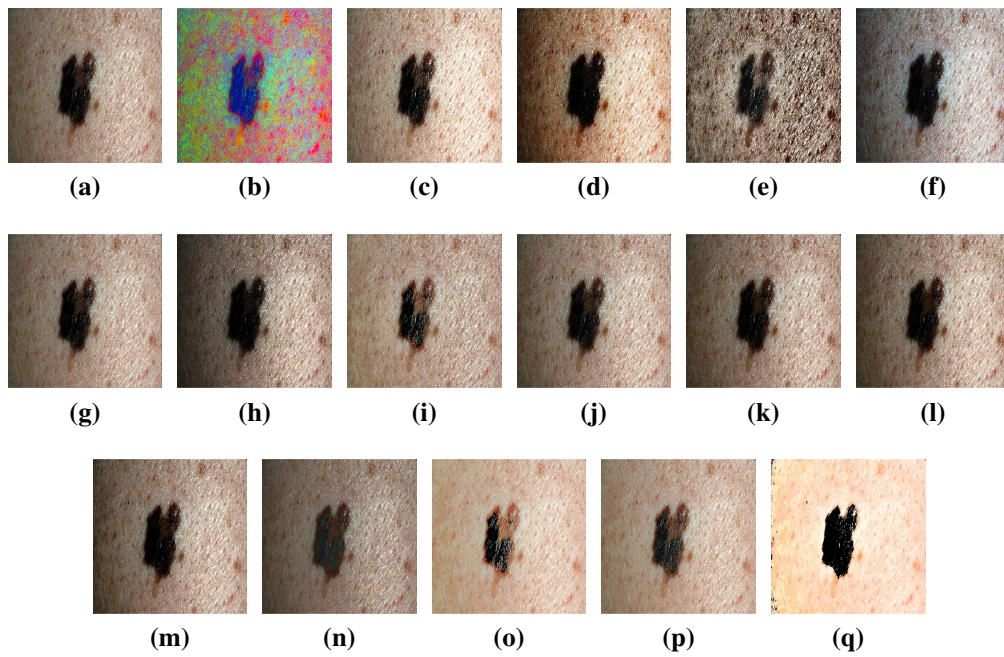


Figure 4.7: CE results on the test image-3: (a) Input image, (b) DS, (c) FGLEI, (d) HE, (e) CLAHE, (f) GEM, (g) CEHCM, (h) SMIRANK, (i) FIE, (j) CVCEA, (k) LDR, (l) WAHE, (m) RICE, (n) SHE, (o) FLM, (p) AECLII, and (q) EMST.

The effectiveness of the proposed EMST algorithm for lesion segmentation from dermatological macro-images is justified by comparing the segmentation accuracy of the segmented results obtained by OT on images enhanced with the proposed EMST and the 15 existing CE methods. It is found that the segmentation accuracy of the segmented results obtained by OT on images enhanced with the proposed EMST significantly improved when compared against the segmented results produced by OT on images enhanced with the discussed 15 CE methods. The consistency of the observations drawn out from the results on three representative test images is confirmed on the rest of the 47 test images. The numerical and statistical results on all 50 test images are reported.

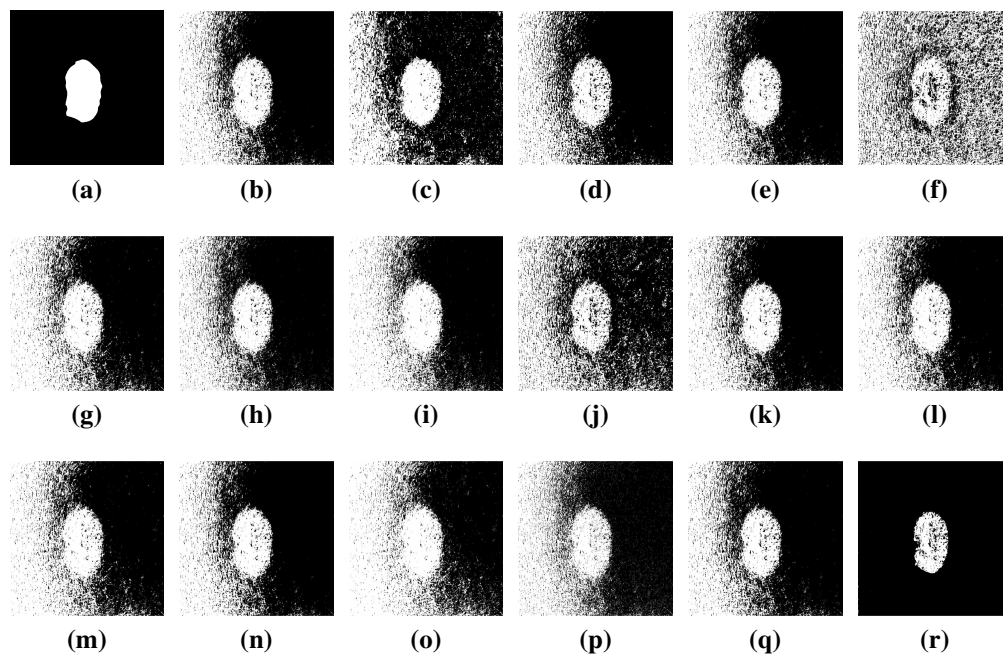


Figure 4.8: Segmented images obtained by OT on test image-1 enhanced with different CE methods: (a) Expert segmented ground-truth, (b) Unenhanced, (c) DS, (d) FGLEI, (e) HE, (f) CLAHE, (g) GEM, (h) CEHCM, (i) SMIRANK, (j) FIE, (k) CVCEA, (l) LDR, (m) WAHE, (n) RICE, (o) SHE, (p) FLM, (q) AECLII, and (r) EMST.

Fig. 4.8 to Fig. 4.10 exhibit segmented results obtained by OT on three representative test images enhanced with the proposed EMST and the 15 existing CE methods discussed. Applying OT on the original test image-1 yielded an over-segmented result, which can be observed from Fig. 4.8 (b). A significant portion of the background is falsely segmented as the lesion. The segmented results obtained by OT after enhancing test image-1 with DS (Fig. 4.8 (c)), FGLEI (Fig. 4.8 (d)), HE (Fig. 4.8 (e)), GEM (Fig. 4.8 (g)), CEHCM (Fig. 4.8 (h)), SMIRANK (Fig. 4.8 (i)), FIE (Fig. 4.8 (j)), CVCEA (Fig. 4.8 (k)), LDR (Fig. 4.8 (l)), WAHE (Fig. 4.8 (m)), RICE (Fig. 4.8 (n)), SHE (Fig. 4.8 (o)), FLM (Fig. 4.8 (p)), and AECLII (Fig. 4.8 (q))

produced over-segmented output, which is similar to the segmented results obtained by OT on the original test image-1 as shown in Fig. 4.8 (b). Fig. 4.8 (f) shows the segmented result obtained by OT on test image-1 after applying CLAHE. The output result produced is intolerable over-segmentation. Compared to the 15 CE methods discussed, the segmented output image obtained by OT on test image-1 enhanced with the proposed EMST offers comparatively good segmentation accuracy. Fig. 4.8 (r) show good consensus with the expert segmented ground-truth in Fig. 4.8 (a).

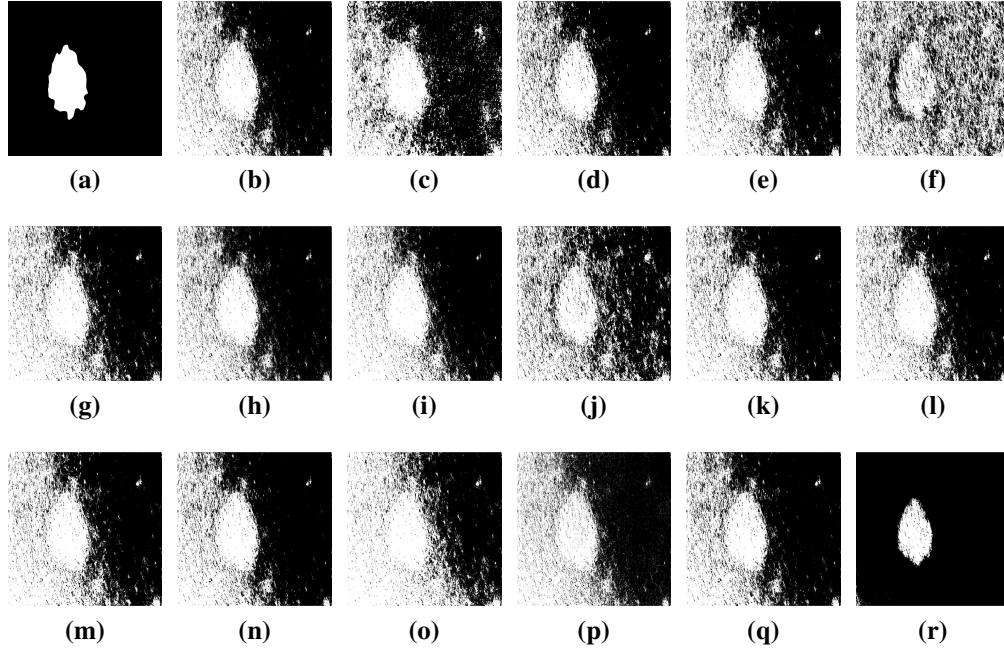


Figure 4.9: Segmented images produced by OT on test image-2 enhanced with different CE methods: (a) Expert segmented ground-truth, (b) Unenhanced, (c) DS, (d) FGLEI, (e) HE, (f) CLAHE, (g) GEM, (h) CEHCM, (i) SMIRANK, (j) FIE, (k) CVCEA, (l) LDR, (m) WAHE, (n) RICE, (o) SHE, (p) FLM, (q) AECLII, and (r) EMST.

Applying OT on the original test image-2 produced an over-segmented result, which is visible in Fig. 4.9 (b). The segmented results obtained by OT after enhancing test image-2 with DS (Fig. 4.9 (c)), FGLEI (Fig. 4.9 (d)), HE (Fig. 4.9 (e)), GEM (Fig. 4.9 (g)), CEHCM (Fig. 4.9 (h)), SMIRANK (Fig. 4.9 (i)), FIE (Fig. 4.9 (j)), CVCEA (Fig. 4.9 (k)), LDR (Fig. 4.9 (l)), WAHE (Fig. 4.9 (m)), RICE (Fig. 4.9 (n)), SHE (Fig. 4.9 (o)), FLM (Fig. 4.9 (p)), and AECLII (Fig. 4.9 (q)) produced over-segmented output, which are similar to the segmented output produced on the original test image-2 as shown in Fig. 4.9 (b). Fig. 4.9 (f) shows the segmented result after applying CLAHE. The output result produced is intolerable over-segmentation. Compared to the 15 CE methods available in literature, the segmented output image obtained by OT on test image-2 enhanced with the proposed EMST (Fig. 4.9 (r)) offers comparatively good overlap with the expert segmented ground-truth shown in

Fig. 4.9 (a).

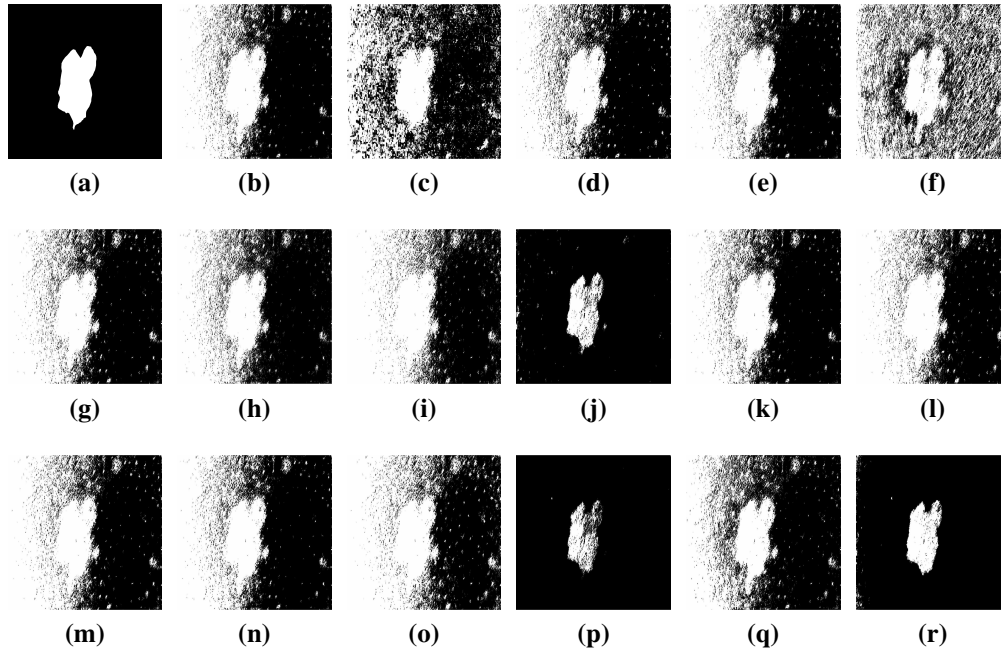


Figure 4.10: Segmented images produced by OT on test image-3 enhanced with different CE methods: (a) Expert segmented ground-truth, (b) Unenhanced, (c) DS, (d) FGLEI, (e) HE, (f) CLAHE, (g) GEM, (h) CEHCM, (i) SMIRANK, (j) FIE, (k) CVCEA, (l) LDR, (m) WAHE, (n) RICE, (o) SHE, (p) FLM, (q) AECLII, and (r) EMST.

When OT is performed on the original test image-3, the output image obtained is over-segmented, as shown in Fig. 4.10 (b). The segmented results obtained by OT after enhancing test image-3 with DS (Fig. 4.10 (c)), FGLEI (Fig. 4.10 (d)), HE (Fig. 4.10 (e)), GEM (Fig. 4.10 (g)), CEHCM (Fig. 4.10 (h)), SMIRANK (Fig. 4.10 (i)), CVCEA (Fig. 4.10 (k)), LDR (Fig. 4.10 (l)), WAHE (Fig. 4.10 (m)), RICE (Fig. 4.10 (n)), SHE (Fig. 4.10 (o)), and AECLII (Fig. 4.10 (q)) produced over-segmented outputs, which is similar to the segmented output produced on the original image as shown in Fig. 4.10 (b). Fig. 4.10 (f) shows over-segmented result after applying CLAHE. FIE (Fig. 4.10 (j)) yielded slightly under-segmented output, some portion of the lesion is missed. The FLM (Fig. 4.10 (p)) produced an under-segmented result, and the segmented output missed a significant portion of the lesion. Compared to other CE methods, the segmentation result produced after enhancing test image-3 with the proposed EMST shown in Fig. 4.10 (r) offers comparatively good overlap with the expert segmented ground-truth in Fig. 4.10 (a). All the 15 CE techniques discussed failed to improve the segmentation accuracy of segmented results obtained by OT on images enhanced with these CE techniques. The proposed EMST-based CE algorithm significantly improved the segmentation accuracy of segmented results obtained by OT on images enhanced with it. The

consistency can be observed on all 3 test images and found valid on all 50 test images.

4.2.2 Quantitative analysis

The DSC is computed for segmented results obtained by OT on the original macro-images and the enhanced macro-images with different CE methods discussed against the manually segmented ground-truth for objective evaluation. Table 4.4 shows the DSC value for the segmented results obtained by OT on images enhanced with different CE methods discussed on 50 test images. It is observed that the segmentation results obtained by OT on images enhanced with the proposed EMST-based CE method yielded the highest DSC. High DSC value demonstrates the ability of the EMST-based CE method to enhance the dermatological macro-images, which can significantly improve the segmentation accuracy of the OT algorithm to produce segmented results that have good consensus with the ground-truth created by the clinicians. The minimum standard deviation (SD) of DCS in Table 4.4 exhibited by segmented results obtained by OT reveals that EMST-based CE improved the contrast between lesion and background consistently for all 50 test images.

Table 4.4: DSC between the ground-truth and the segmented results obtained by OT on images enhanced with different CE methods.

CE methods	Test image-1	Test image-2	Test image-3	Average DSC on 50 test images (Mean $\pm$ SD)
Unenhanced	0.31	0.33	0.30	0.68 ± 0.34
DS (2021) [96]	0.35	0.33	0.33	0.61 ± 0.31
FGLEI (2021) [59]	0.31	0.34	0.30	0.64 ± 0.34
HE (2018) [47]	0.31	0.31	0.29	0.40 ± 0.27
CLAHE (2016) [101]	0.17	0.20	0.23	0.30 ± 0.20
GEM (2014) [104]	0.27	0.30	0.29	0.38 ± 0.24
CEHCM (2020) [105]	0.31	0.32	0.30	0.67 ± 0.34
SMIRANK (2016) [106]	0.27	0.30	0.26	0.40 ± 0.28
FIE (2016) [107]	0.26	0.29	0.85	0.59 ± 0.34
CVCEA (2011) [108]	0.31	0.32	0.30	0.45 ± 0.27
LDR (2013) [109]	0.29	0.31	0.28	0.47 ± 0.31
WAHE (2009) [110]	0.27	0.30	0.28	0.39 ± 0.26
RICE (2015) [111]	0.31	0.33	0.30	0.61 ± 0.34
SHE (2014) [112]	0.25	0.26	0.26	0.33 ± 0.19
FLM (2016) [113]	0.29	0.30	0.73	0.47 ± 0.29
AECLII (2019) [114]	0.31	0.33	0.33	0.68 ± 0.34
EMST	0.86	0.85	0.88	0.81 ± 0.17

4.3 CONTRIBUTIONS

The main contributions of this chapter can be summarized as follows:

1. An EfficientNet-based modified sigmoid transform (EMST) is proposed for enhancing the contrast between lesion and background regions on the dermatological macro-images.
2. Compared to the traditional sigmoid transform, provisions for fixing the boundary that divides the pixel values into lesion and background regions between which contrast needs to be improved are available in the modified sigmoid transform.
3. Crossover point that divides the macro-image into lesion and background. It is predicted using a modified EfficientNet regressor to exclude manual intervention and subjectivity.
4. A significant improvement in DSC is observed for processed macro-images by EMST.

CHAPTER 5

DTP-NET: A CONVOLUTIONAL NEURAL NETWORK MODEL TO PREDICT THRESHOLD FOR LOCALIZING THE LESIONS ON DERMATOLOGICAL MACRO-IMAGES

Being highly zoomed views, the dermatological macro-images contain only lesions and the background regions. Consequently, localizing lesions on these images becomes a thresholding problem. However, threshold prediction algorithms in literature do not offer adequate accuracy and fail to retain consistent performance on different dermatological macro-images. Furthermore, DL-based segmentation models commonly employed for skin lesion segmentation utilize encoder-decoder architectures, resulting in computationally intensive image-to-image regression tasks.

To address these challenges, deep threshold prediction network (DTP-Net), a deep convolutional neural network (CNN) model specifically designed for predicting the optimal segmentation threshold from grayscale versions of dermatological macro-images. The proposed DTP-Net performs image-to-scalar regression, where original images are given as inputs, and a simple column vector containing scalar values representing the optimum intensity threshold is used as targets during training. The computational expense for training the image-to-scalar regressor is less compared to that of the image-to-image regressors.

5.1 METHODOLOGY

Fig. 5.1 shows a schematic of the threshold prediction using the DTP-Net with the performance evaluation process. The DTP-Net is designed to predict the optimal segmentation threshold value from grayscale macro-images. The accuracy of the predicted threshold is evaluated through two methods. Firstly, the squared error between the predicted and optimal threshold values is calculated for each input image. The root mean squared error (RMSE) is then computed, representing the square root of the mean of squared errors, and serves as the initial benchmark index. Secondly, the lesion is segmented using binary thresholding with the threshold value predicted by the DTP-Net. The Dice similarity coefficient (DSC) [166] between the ground-truth and segmented images is employed as an additional benchmark index. The primary objective of the proposed DTP-Net is to accurately predict the threshold value, thereby facilitating pre-

cise lesion segmentation in macro-images.

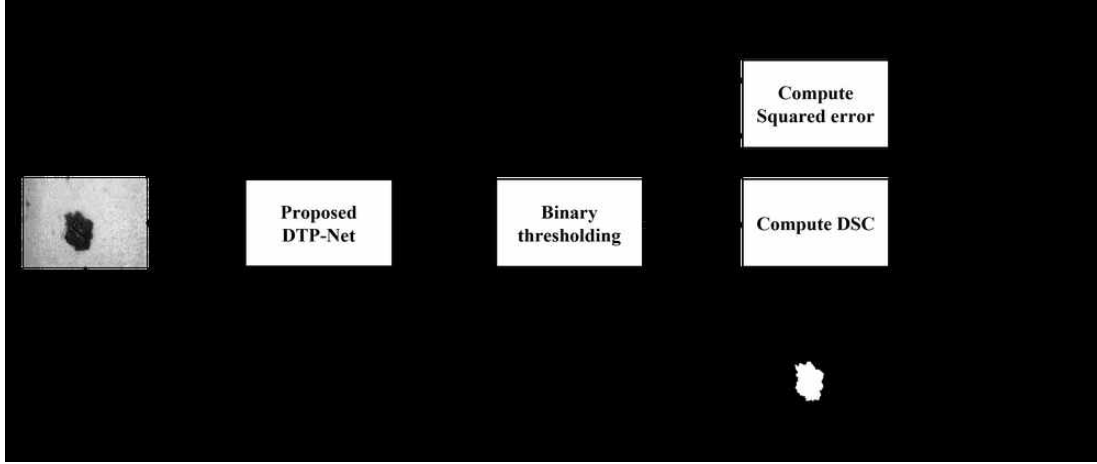


Figure 5.1: A schematic representation of the threshold prediction using DTP-Net.

5.1.1 Network architecture

A CNN is trained to regress the threshold value from a given grayscale image. The architecture of the DTP-Net is shown in Fig. 5.2. There are 18 layers with learnable weights: 11 convolutional layers, 6 batch normalization layers, and 1 fully connected (FC) layer. The DTP-Net consists of convolution layers followed by batch normalization, rectified linear unit (ReLU) as the activation function, and max pooling layers [147]. The convolution is performed by a 3×3 kernel with a stride of ‘1’ and padding ‘same’. The max pooling operation is done by a 2×2 kernel with a stride of ‘2’. The DTP-Net is designed to predict the optimal threshold value from the input image, and it has an image-to-scalar regressor architecture. So, an output regression layer is introduced immediately following the FC layer. The architectural details, including layer specifications and trainable parameters, are presented in Table 5.1. It provides a comprehensive overview of each layer’s configuration. The proposed DTP-Net has a total of 641,256 trainable parameters.

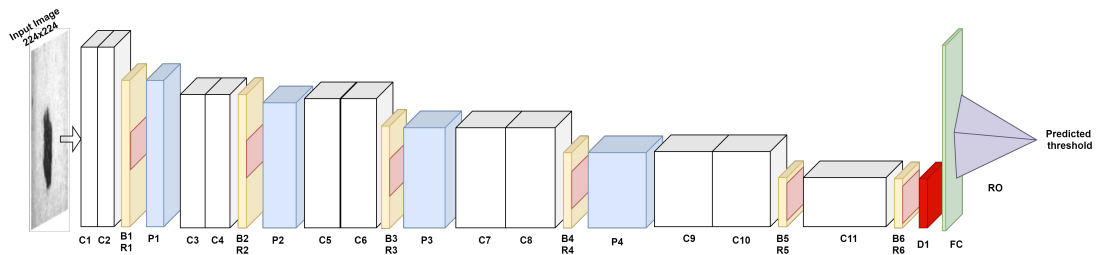


Figure 5.2: The architecture of the proposed DTP-Net. The architectural details are described in Table 5.1.

Table 5.1: Architectural details of the proposed DTP-Net.

Layers	Activations	Kernel size	Stride	Parameters
Input image	224×224×1	—	—	0
Convolution (C1)	224×224×8	3	1	80
Convolution (C2)	224×224×8	3	1	584
Batch norm. (B1)	224×224×8	—	—	16
ReLU (R1)	224×224×8	—	—	0
Max pool (P1)	112×112×8	2	2	0
Convolution (C3)	112×112×16	3	1	1168
Convolution (C4)	112×112×16	3	1	2320
Batch norm. (B2)	112×112×16	—	—	32
ReLU (R2)	112×112×16	—	—	0
Max pool (P2)	56×56×16	2	2	0
Convolution (C5)	56×56×32	3	1	4640
Convolution (C6)	56×56×32	3	1	9248
Batch norm. (B3)	56×56×32	—	—	64
ReLU (R3)	56×56×32	—	—	0
Max pool (P3)	28×28×32	2	2	0
Convolution (C7)	28×28×64	3	1	18496
Convolution (C8)	28×28×64	3	1	36928
Batch norm. (B4)	28×28×64	—	—	128
ReLU (R4)	28×28×64	—	—	0
Max pool (P4)	14×14×64	2	2	0
Convolution (C9)	14×14×128	3	1	73856
Convolution (C10)	14×14×128	3	1	147584
Batch norm. (B5)	14×14×128	—	—	256
ReLU (R5)	14×14×128	—	—	0
Convolution (C11)	14×14×256	3	1	295168
Batch norm. (B6)	14×14×256	—	—	512
ReLU (R6)	14×14×256	—	—	0
Dropout (D1)	14×14×256	—	—	0
Fully connected (FC)	1×1×1	—	—	50176
Regression output (RO)	1×1×1	—	—	0

Total trainable parameters: 641,256

The successive convolution layers used in the DTP-Net helps to capture abstract features at local to global scales of the input image. In max pooling operation, the feature map generated by the convolution layer is divided into pooling regions. The highest value from each pooling region is selected based on the belief that the highest value represents the prominent feature value [190]. The FC layer is available before the regression layer. All the neurons in the FC layer are connected to all the neurons in the preceding layer. The FC layer combines all the features learned by the layers before it to recognize broad patterns in the input image. The batch normalization layers help to accelerate network training and reduce the influence of weight initialization [191].

In the DTP-Net, a batch normalization layer is introduced between the convolution and ReLU layers in all the stages. It normalizes all inputs (x_i) in a mini-batch by computing the mean (μ_b) and variance (σ_b^2) independently. The normalized activations are computed as,

$$\hat{x}_i = \left(\frac{x_i - \mu_b}{\sqrt{\sigma_b^2 + \epsilon}} \right) \quad (5.1)$$

In Eq. (5.1), ϵ is a constant that provides numerical stability when the variance is minimal. It is commonly set to a very small value, such as 10^{-5} . After network training is completed, fixed mean and variance are computed from training data, which is utilized to normalize the validation data for making predictions using the trained network. The dropout layer is introduced in the network architecture to reduce model overfitting [192]. In the DTP-Net, the dropout layer is added before the FC layer.

In a context where all feature values in the pooling region are equally high, selecting only the highest feature value will compromise some of the discriminant features, adversely affecting the network performance [190]. Allday et al. [193] have reported that removing the final max pooling layer of AlexNet increased the classification accuracy by 2.88%. The performance of the DTP-Net with and without the max pooling layer between C10 layer and the FC layer is evaluated. It is empirically observed that the DTP-Net without max pooling layer in the last stage offers better prediction accuracy.

5.1.2 Training

The grayscale versions of the dermatological macro-images and the corresponding optimal threshold values are given as input to the DTP-Net for training. The dermatological macro-images are RGB color images that are converted to grayscale using the luminosity model [194]. This model is designed based on human visual perception, where humans are more sensitive to green than other colors, so in this model, green color channel is assigned with the highest weight, followed by red and blue color channels, as represented in Eq. (5.2).

$$I = 0.299R + 0.587G + 0.114B \quad (5.2)$$

In Eq. (5.2), I denotes grayscale image, R , G , and B indicate red, green, and blue color channels.

On the converted grayscale image, the threshold value is varied from the minimum value ('0') to the maximum value ('255') of the grayscale. The DSC [166] between

the segmented output and the corresponding ground-truth is computed for each threshold value. The threshold value for which DSC is maximum is taken as the optimal threshold value of that image.

Fig. 5.3 shows the change in DSC with the threshold value on three sample images. It can be observed from Fig. 5.3 (d), a maximum DSC of 0.79 is obtained on image-1 at a threshold value of 82. Hence, it can be concluded that the optimal threshold value of image-1 is 82. In image-2, the maximum DSC obtained is 0.89 at a threshold value of 60, which is evident in Fig. 5.3 (e). Similarly, in Fig. 5.3 (f) the optimal threshold value of image-3 is 84, and the maximum DSC obtained at the optimal threshold value is 0.94. Table 5.2 shows the maximum DSC score and the corresponding optimal threshold value of the three test images.

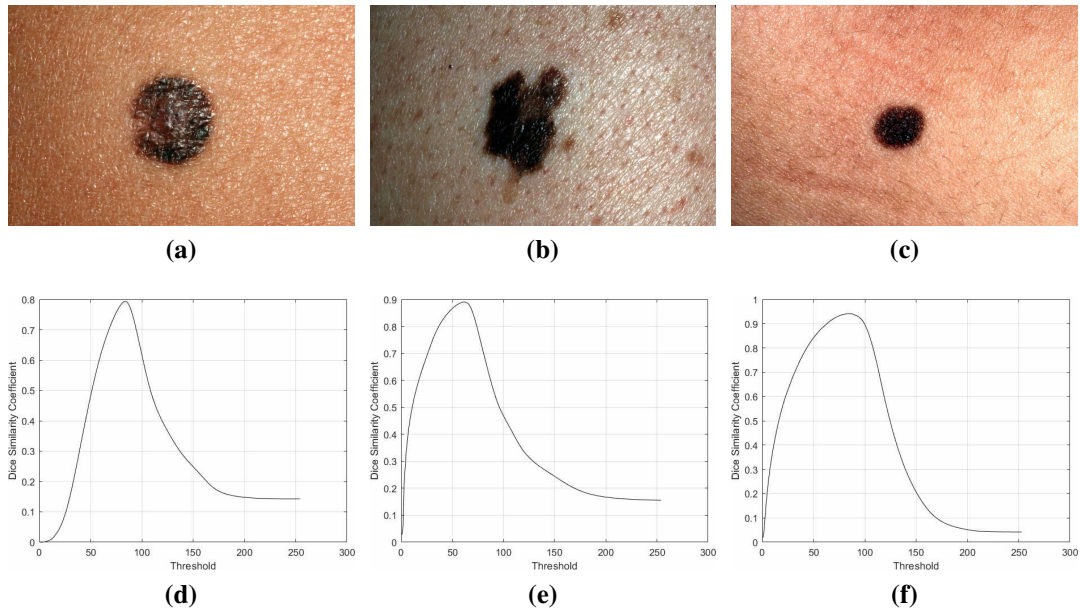


Figure 5.3: Optimal threshold value for three test images: (a)-(c) sample images, (d)-(f) graph showing variation of DSC with the change in threshold value.

Table 5.2: Maximum DSC score and the corresponding optimal threshold value of the three test images.

Test image	Maximum DSC	Optimal threshold
Image-1	0.79	82
Image-2	0.89	60
Image-3	0.94	84

A custom dataset is created by merging macro-images of melanoma and nevus lesions from the University of Waterloo skin cancer database [20], MED-NODE [162], SD-260 [163], and SD-198 [164] databases. For evaluating the performance of the DTP-Net, a seven-fold cross-validation technique is performed, with each fold con-

sisting of approximately 75 images. Six folds are used for training the regressor, the remaining one for testing. Images in each fold are randomly selected. Approximately 455 images are utilized for the training purpose, the remaining 75 for testing the performance of the model. The average performance of the seven-fold test is reported in this study.

The DTP-Net is trained with a set of grayscale versions of macro-images from the custom dataset, with images resized to 224×224 pixels and the corresponding set of optimal threshold values as the target. For the proposed image-to-scalar regressor, half of the sum of the squared error [179] is used as the loss function during training.

$$loss = \frac{1}{2} \sum_{i=1}^N (A_i - P_i)^2 \quad (5.3)$$

In this context, A_i is the actual threshold value corresponding to the i^{th} input image, and P_i is the threshold value predicted by the DTP-Net. In Eq. (5.3), N indicates the number of images used for training. The normalization factor ‘2’ is used in the loss function to minimize the influence of squaring the error on the derivative used in the gradient descent algorithm. The DTP-Net’s network weights are initialized using Xavier’s method [195]. The SGDM optimizer with a learning rate of 0.0001 and momentum of 0.9 are used during the training process. The image augmentation [139] and dropout [192] techniques are implemented to enhance the training efficiency of the DTP-Net while reducing overfitting.

5.1.2.1 Dropout

Dropout is a powerful and computationally efficient method to overcome the problem of overfitting in deep CNN models when it is trained with limited data [192]. In this method, the output neuron in an assigned layer is randomly removed with probability (p). Hence, it eliminates the participation of those ‘dropout’ neurons during the forward propagation and backward propagation. Finally, the dropout train the ensemble of all different network architecture that share weights in each epoch. This method forces all neurons to learn more robust features, making the CNN model robust with good generalization capability. In this work, dropout layer with $p = 0.2$ before FC layer is used.

5.1.2.2 Image augmentation

Various geometric transformations [139] are applied to the training images. The training images undergo vertical and horizontal flips. Additionally, each image is rotated by a random angle θ within the range of $[-45^\circ, 45^\circ]$. Moreover, the training images

are scaled or zoomed to 105% of their original size to introduce further diversity into the dataset. Each image in the training phase is augmented to get nine sub-images, increasing the total count of training images to 4,550.

5.2 RESULTS AND DISCUSSION

At the beginning of this section, the accuracy of the threshold values predicted by the DTP-Net is compared with that of the state-of-the-art threshold estimation algorithms, namely Otsu's thresholding (OT) [115], valley-emphasized OT (VEOT) [121], isodata thresholding (IT) [122], histogram slope difference distribution-based threshold (HSDDT) [125], minimum error thresholding (MET) [127], Poisson's distribution-based minimum error thresholding (PDMET) [128], Kapur's maximum entropy threshold (KMET) [129], entropy-weighted OT (EOT) [130], minimum cross-entropy thresholding (MCET) [131], type-2 fuzzy-based thresholding (Type-2-FT) [133], and fuzzy entropy thresholding (FET) [135]. Following this, the segmentation results obtained from the DTP-Net are compared with aforesaid 11 thresholding algorithms and 4 state-of-the-art DL-based skin lesion segmentation methods in terms of segmentation accuracy and computational time.

The DTP-Net and all the compared methods are implemented in MATLAB R2022a on a desktop computer operating with Windows 10 OS with an NVIDIA GeForce GTX 1650 GPU. The system works with an AMD RYZEN 5 processor with a maximum clock frequency of 3.7 GHz.

To visualize the accuracy of the threshold values predicted by the DTP-Net, scatter plots for training and testing phases are shown in Fig. 5.4. In Fig. 5.4 (a), majority of the predicted threshold values during the training phase lie on the trend line of the plot showing the best fit to the data. The scatter plot for the training phase indicates that the DTP-Net can predict the threshold values very close to the optimal threshold values of the images used for training. The prediction accuracy of the DTP-Net for the testing phase can be observed from Fig. 5.4 (b). It is seen that the majority of the data points lie on the trend line, and a few data points fall around the trend line. The scatter plot for the testing phase points that the DTP-Net can predict the threshold values close to the actual optimal threshold values for the 75 test images.

Gradient-weighted class activation mapping (Grad-CAM) is an explainable technique that is used to visualize the predictions made by deep CNNs [196]. Fig. 5.5 shows the Grad-CAM maps produced by the DTP-Net for the three test images. The Grad-CAM map shows that the DTP-Net focuses on the background region and can identify the boundary between the lesion and the background. Fig. 5.5 intuitively suggests that the DTP-Net utilizes the intensity difference between lesion and background

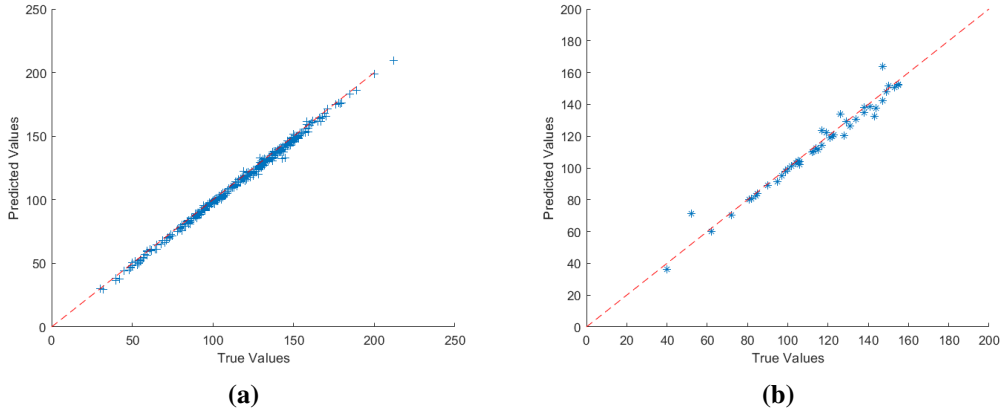


Figure 5.4: Scatter plots of training and testing phases: (a) training, and (b) testing. Points close to the red line indicates better prediction.

to predict the threshold value.

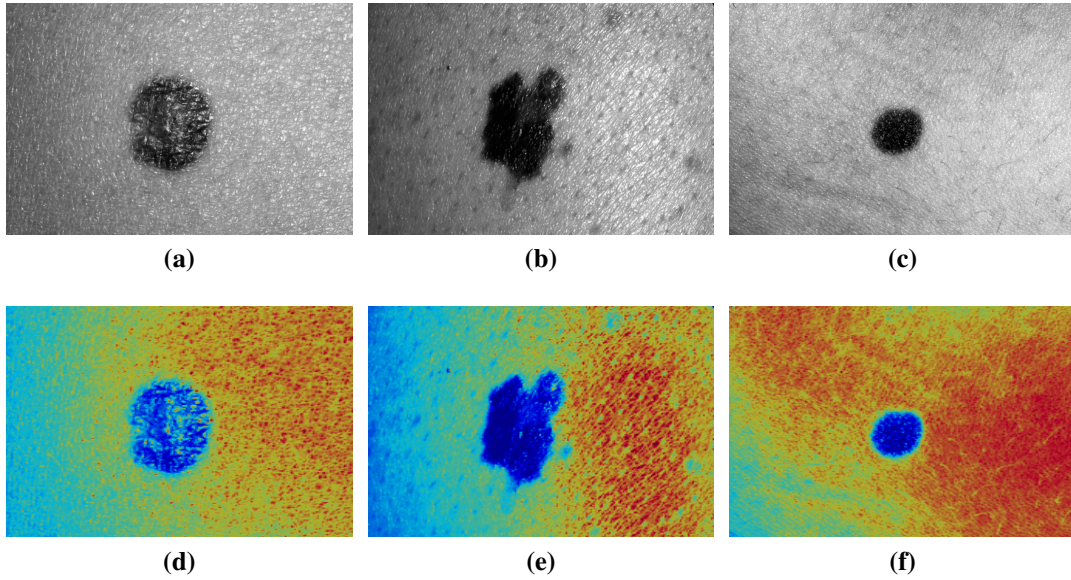


Figure 5.5: Visualization of activations in the DTP-Net, where (a)-(c) are input grayscale images and (d)-(f) Grad-CAM outputs of the DTP-Net.

To analyze the effect of image size on the performance of the DTP-Net, the performance of the DTP-Net is compared with three different image sizes: 112×112 , 224×224 , and 448×448 pixels. The learning rate and mini-batch size are fixed as 0.0001 and 16, respectively. The SGDM is adopted as the optimizer for training, and the DTP-Net is trained for 150 epochs. Table 5.3 shows that RMSE is reduced from 09.377 to 05.621 when image size is increased from 112×112 to 224×224 pixels. Again on further increasing the image size to 448×448 pixels, it is observed that the RMSE increased to 11.566. When the image size is increased to 448×448 pixels, the performance is reduced. The degradation can be explained by the fact that training the

model becomes more challenging when the number of pixels in the training images doubles. Expanding the model depth by adding more layers may improve the performance of the model when trained with images of size 448×448 pixels, which will increase the computational expenses. These data suggest that an image size of 224×224 pixels produced an excellent trade-off between the accuracy of the predicted threshold value and the computational expenses.

Table 5.3: Influence of image resizing on the performance of the DTP-Net.

Image size	RMSE	Training time/epoch (Sec.)
112×112	09.377	29
224×224	05.621	56
448×448	11.566	284

This study investigated the impact of image augmentation on performance by applying various geometric transformations to the dataset. Two separate experiments were conducted to examine the influence of image augmentation on the model's performance. Initially, the DTP-Net was trained using only 455 images from the custom dataset, resulting in an RMSE of 14.406 and a training time of 05s per epoch. Subsequently, the training data was artificially expanded by applying different geometric transformations to the 455 images, resulting in a total of 4,550 training images. These augmented images were then used to train the DTP-Net.

Based on the results presented in Table 5.4, it is evident that image augmentation significantly enhanced performance, reducing the RMSE from 14.406 to 05.621. However, this improvement came at the cost of an increased training time per epoch, taking 56 seconds due to the augmented training image set, which is ten times larger than the set used in training the model without image augmentation. Applying image augmentation techniques generated a more diverse training dataset, thereby leading to improved performance of the DTP-Net.

Table 5.4: Performance of DTP-Net with image augmentation.

Augmentation	RMSE	Training time/epoch (Sec.)
No	14.406	05
Yes	05.621	56

The macro-image and the corresponding threshold value predicted by the DTP-Net are given as input to a binary thresholding algorithm for evaluating the segmentation performance of the DTP-Net. The binary thresholding algorithm replaces the pixels in the grayscale image with a '1' if the pixel intensity value is less than the predicted

threshold value; otherwise, a '0' is assigned. Thus, a binary mask is created and is compared with the expert-segmented ground-truth.

The segmentation results obtained by the 11 state-of-the-art thresholding algorithms and the DTP-Net are shown only for three test images taken from the test dataset. The consistency of the qualitative and quantitative observations drawn from the three test images is confirmed on the remaining 72 test images. The quantitative results of 75 test images are reported.

The RMSE and DSC exhibited by various threshold estimation methods OT, VEOT, IT, HSDDT, MET, PDMET, KMET, EOT, MCET, Type-2-FT, FET, and the proposed DTP-Net on 75 test images are shown in Table 5.5. The DTP-Net exhibited an RMSE value of 5.621 on 75 test images, which is the lowest among all the threshold estimation methods in Table 5.5. The HSDDT, Type-2-FT, and FET have unacceptably high RMSE values, indicating an erroneous prediction of threshold values. The RMSE exhibited by the DTP-Net is lesser than all the discussed threshold estimation methods. Lesser value of the RMSE reflects that the threshold values predicted by the DTP-Net are close to the optimum.

Table 5.5: Average of RMSE and DSC score obtained by various threshold estimation methods on 75 test images.

Method	Average RMSE	DSC (Mean $\pm$ SD)
OTSU (1979) [115]	34.048	0.75 ± 0.275
VEOT (2020) [121]	37.482	0.71 ± 0.286
IT (1978) [122]	33.766	0.75 ± 0.276
HSDDT (2016) [125]	50.039	0.72 ± 0.239
MET (1986) [127]	35.707	0.76 ± 0.298
PDMET (1993) [128]	39.263	0.76 ± 0.250
KMET (1985) [129]	31.494	0.78 ± 0.191
EOT (2018) [130]	29.116	0.79 ± 0.195
MCET (1993) [131]	34.795	0.74 ± 0.268
Type-2-FT (2009) [133]	47.385	0.67 ± 0.292
FET (2019) [135]	43.569	0.65 ± 0.334
DTP-Net	05.621	0.88 ± 0.100

The thresholding algorithms utilize the predicted threshold values to produce the segmented images. Therefore, the segmentation efficiency of the thresholding algorithms were quantitatively evaluated using DSC score. From Table 5.5, it can be observed that among the discussed methods, the segmented images produced by DTP-Net exhibited the highest DSC score of 0.88 ± 0.100 , which is the maximum possible DSC score on these 75 test images. A high DSC score shows a good overlap of the segmented results with the ground-truth. A high DSC score displayed by the DTP-Net

shows that the predicted threshold value by the DTP-Net can localize the lesions accurately. The standard deviation (SD) of the DSC score of DTP-Net is significantly low compared to 11 state-of-the-art threshold estimation methods. A low SD of the DSC score exhibited by the DTP-Net evidently indicates that the threshold values predicted by the DTP-Net can offer consistent segmentation results on all 75 test images.

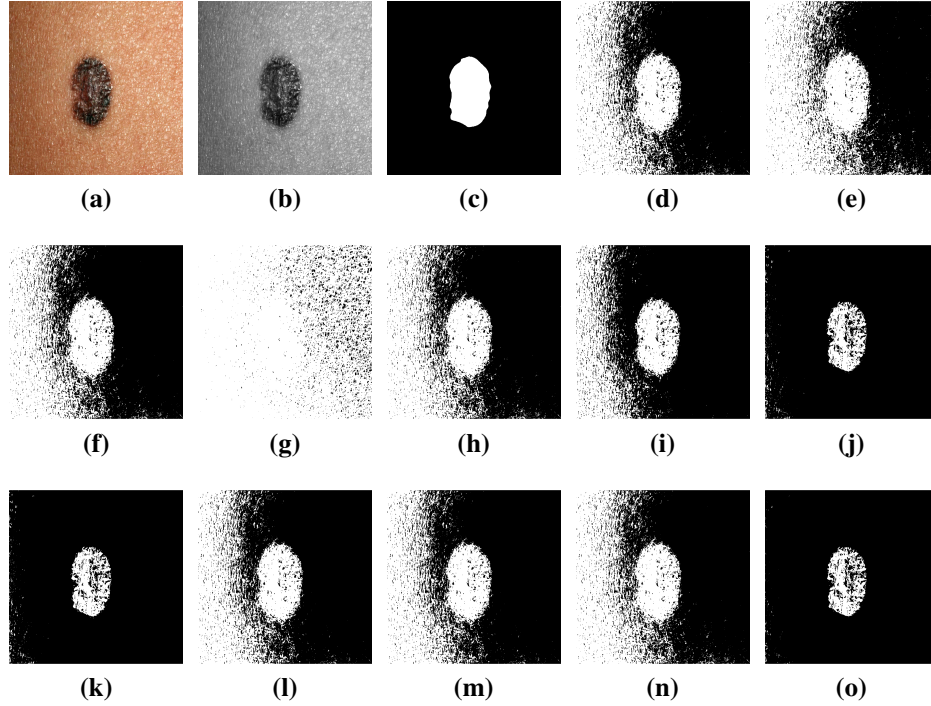


Figure 5.6: The areas of lesions segmented by various thresholding methods on test image-1: (a) Original image, (b) Grayscale version, (c) Ground-truth, (d) OT, (e) VEOT, (f) IT, (g) HSDDT, (h) MET, (i) PDMET, (j) KMET, (k) EOT, (l) MCET, (m) Type-2-FT, (n) FET, and (o) DTP-Net.

Fig. 5.6 to Fig. 5.8 shows the segmentation results of 11 discussed thresholding methods and the proposed DTP-Net on the three test images. Fig. 5.6 shows the segmentation results on test image-1, OT (Fig. 5.6 (d)), VEOT (Fig. 5.6 (e)), IT (Fig. 5.6 (f)), MET (Fig. 5.6 (h)), PDMET (Fig. 5.6 (i)), MCET (Fig. 5.6 (l)), Type-2-FT (Fig. 5.6 (m)), and FET (Fig. 5.6 (n)) produce over-segmented results. A notable portion of the background is falsely segmented as the lesion in over-segmented results. Compared to other methods, HSDDT (Fig. 5.6 (g)) result is intolerably over-segmented; the complete background is falsely segmented as the lesion. The KMET (Fig. 5.6 (j)) and EOT (Fig. 5.6 (k)) produced better segmentation results, but some of the background region at the bottom left corner is falsely segmented as the lesion. From different thresholding methods, DTP-Net (Fig. 5.6 (d)) produced a comparatively better segmentation result with the ground-truth (Fig. 5.6 (c)) for test image-1 (Fig. 5.6 (a)).

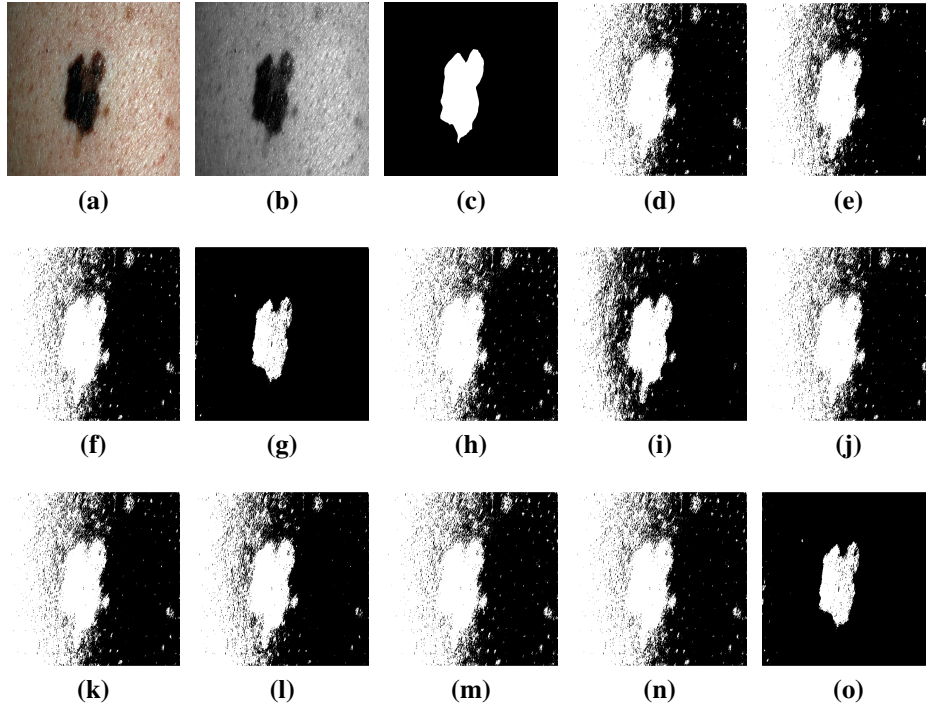


Figure 5.7: The areas of lesions segmented by various thresholding methods on test image-2: (a) Original image, (b) Grayscale version, (c) Ground-truth, (d) OT, (e) VEOT, (f) IT, (g) HSDDT, (h) MET, (i) PDMET, (j) KMET, (k) EOT, (l) MCET, (m) Type-2-FT, (n) FET, and (o) DTP-Net.

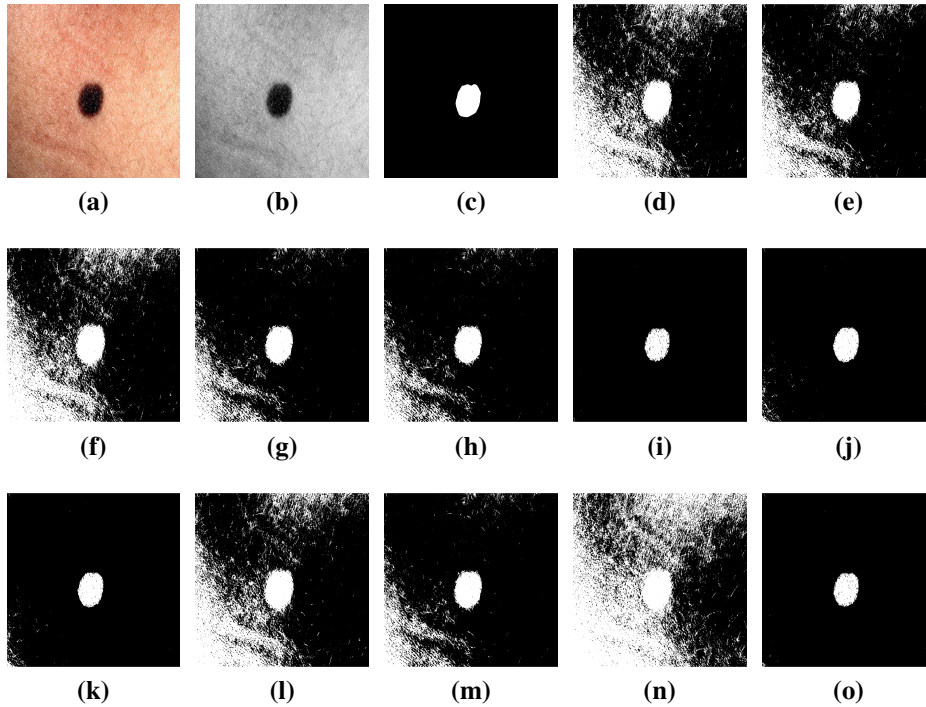


Figure 5.8: The areas of lesions segmented by various thresholding methods on test image-3: (a) Original image, (b) Grayscale version, (c) Ground-truth, (d) OT, (e) VEOT, (f) IT, (g) HSDDT, (h) MET, (i) PDMET, (j) KMET, (k) EOT, (l) MCET, (m) Type-2-FT, (n) FET, and (o) DTP-Net.

From Fig. 5.7, it can be observed that OT (Fig. 5.7 (d)), VEOT (Fig. 5.7 (e)), IT (Fig. 5.7 (f)), MET (Fig. 5.7 (h)), PDMET (Fig. 5.7 (i)), KMET (Fig. 5.7 (j)), EOT (Fig. 5.7 (k)), MCET (Fig. 5.7 (l)), Type-2-FT (Fig. 5.7 (m)), and FET (Fig. 5.7 (n)) resulted in over-segmentation. The segmented results produced by these methods resulted in false segmentation of the whole background region on the left side. The HS-DDT (Fig. 5.7 (g)) and DTP-Net (Fig. 5.7 (o)) offer comparatively better segmentation results on test image-2 (Fig. 5.7 (a)).

Fig. 5.8 shows the segmented outputs on test image-3. The FET (Fig. 5.8 (n)) produced intolerably over-segmentation; most of the background region is falsely segmented as the lesion. The OT (Fig. 5.8 (d)), VEOT (Fig. 5.8 (e)), IT (Fig. 5.8 (f)), and MCET (Fig. 5.8 (l)) also yielded over-segmented results; a good portion of the background skin is falsely segmented as the lesion. The bottom left corner of the image is segmented as the lesion in HSDDT (Fig. 5.8 (g)), MET (Fig. 5.8 (h)), and Type-2-FT (Fig. 5.8 (m)). The KMET (Fig. 5.8 (j)) and EOT (Fig. 5.8 (k)) methods could segment the lesions accurately, but a small portion of the background in the bottom left corner of the image got falsely segmented. The PDMET (Fig. 5.8 (i)) and DTP-Net (Fig. 5.8 (o)) reported satisfactory segmentation results on test image-3.

It can be observed from Fig. 5.6, 5.7, and 5.8 that only the DTP-Net yielded consistently better segmentation results against the expert-segmented ground-truth than various thresholding methods on the three test images. This observation is valid on all 75 test images considered for this experiment.

The performance of the segmentation results and computational measurements of the combined DTP-Net and morphological postprocessing (DTP-Net+MP) approach are compared with state-of-the-art DL-based lesion segmentation methods, including FCN-8s [158], U-Net [159], SegNet [160], and DeeplabV3+ [161].

The morphological postprocessing (MP) technique is used to refine the binary segmentation result. The MP includes a simple dilation and hole filling operation carried out using a morphological structuring element in the shape of a disk, with a kernel size set at 20 units. These models are trained using a custom dataset prepared with images resized to 224×224 pixels. The learning rate and mini-batch size are fixed at 0.0001 and 16, respectively. The SGDM optimizer is adopted for training the models for 150 epochs, with early stopping implemented if the model loss does not reduce for 15 epochs. The segmentation accuracy of these models is evaluated on 75 test images, ensuring no overlap between the training and testing images.

Quantitatively, Table 5.6 summarizes the segmentation results and the computational measurements of the FCN-8s, U-Net, SegNet, DeepLabV3+, and DTP-Net+MP.

The DSC exhibited by FCN-8s, U-Net, SegNet, DeepLabV3+, and DTP-Net+MP are 0.87 ± 0.063 , 0.83 ± 0.132 , 0.85 ± 0.088 , 0.88 ± 0.108 , and 0.92 ± 0.079 respectively. The DeepLabV3+ and DTP-Net+MP performed better in terms of DSC than other DL-based segmentation methods discussed.

Table 5.6: Comparison of DTP-Net+MP with existing DL-based skin lesion segmentation frameworks.

Method	DSC (Mean $\pm$ SD)	Learnable parameters in millions	Training time/epoch (Sec.)	Inference time/image (Sec)
FCN-8s (2020) [158]	0.87 ± 0.063	135.4	497	0.164
U-Net (2017) [159]	0.83 ± 0.132	30.4	118	0.088
SegNet (2022) [160]	0.85 ± 0.088	29.6	159	0.147
DeepLabV3+ (2020) [161]	0.88 ± 0.108	42.2	348	0.120
DTP-Net+MP	0.92 ± 0.079	0.6	56	0.054

The training time per epoch and the test time per image for the DL-based segmentation methods are shown in Table 5.6. It can be observed from Table 5.6 that the computational time for training the DTP-Net+MP is lesser than all the discussed DL-based segmentation methods. Our proposed DTP-Net took 56s per epoch, whereas the DeepLabV3+, which displayed similar DSC, took six times more training time per epoch (348s per epoch). The key reason for less training time exhibited by the DTP-Net is due to the difference in the target given during training. In general, for training the DL models with encoder-decoder architecture employed for performing segmentation tasks [158–161, 197, 198]. Original images are used as inputs to the model, and manually segmented ground-truth images are used as targets. For training the DTP-Net, original images are used as inputs, and optimum values of intensity threshold are used as targets. In DL-based segmentation models with encoder-decoder architecture, the target is a set of images. However, for the DTP-Net, the target is a simple column vector of scalar values that are optimum values of the intensity threshold. Also, the proposed DTP-Net has a significantly less number of learnable parameters compared to other DL-based segmentation methods shown in Table 5.6. Less learning parameters reduces the training and inference time. Consequently, the training of the DTP-Net is computationally more feasible than that of the DL-based image segmentation models with encoder-decoder architecture. The proposed framework is viable to apply in automated skin cancer diagnostic techniques since the proposed DTP-Net+MP took less than a fraction of a second to segment the potential lesion from a dermatological macro-image.

Fig. 5.9 shows the segmented results of the DTP-Net+MP, FCN-8s, U-Net, SegNet, and DeepLabV3+ on the three test images. The segmentation results of the

DTP-Net+MP shown in Fig. 5.9 are post-processed via simple dilation and hole filling operations. It has been observed that the FCN-8s produced slightly over-segmented results on test image-1. On the test image-2, FCN-8s produced relatively high over-segmentation. Certain portions of the background region are falsely segmented as the lesion. The U-Net produced over-segmented results on all three test images, which is evident in Fig. 5.9. The SegNet yielded under-segmented results on test image-1 and test image-2. A portion of the lesion is falsely segmented as background. The DeepLabV3+ produced slightly over-segmented outputs on test image-1 and test image-2. From Fig. 5.9, it is evident that only the DTP-Net+MP yielded consistently better segmentation results that agree with the expert-segmented ground-truth than various DL-based approaches namely FCN-8s, U-Net, SegNet, and DeepLabV3+, on the three test images.

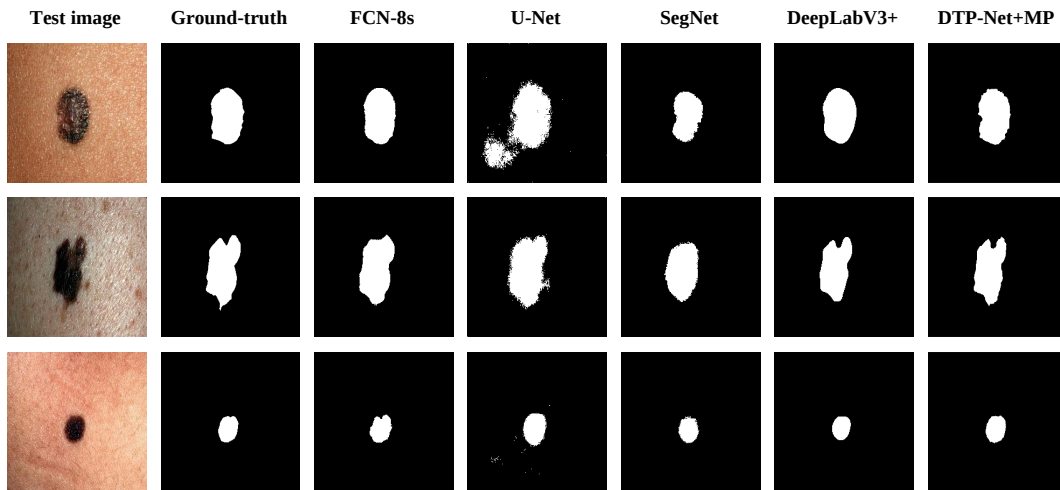


Figure 5.9: Comparison of lesion segmentation results of the DTP-Net+MP against FCN-8s, U-Net, SegNet, and DeepLabV3+ for the three test images.

The proposed DTP-Net demonstrates accurate threshold value prediction and consistent lesion segmentation with reduced computational requirements during training and testing. The segmentation framework based on DTP-Net outperforms popular thresholding methods such as OT, VEOT, IT, HSDDT, MET, PDMET, KMET, EOT, MCET, Type-2-FT, FET, as well as DL-based approaches including FCN-8s, U-Net, SegNet, and DeepLabV3+. This superior performance is evident across a comprehensive set of test images in terms of segmentation accuracy and reduced computational time for both training and testing stages.

5.3 CONTRIBUTIONS

This chapter makes the following key contributions:

1. The DTP-Net is proposed for predicting threshold values from dermatological macro-images. The network addresses the challenge of threshold estimation, which is crucial for accurately separating lesions from the background.
2. Implementing efficient training procedures to overcome limitations associated with the limited number of training images ensures robust and efficient training of the DTP-Net.
3. Demonstration of the superior performance of DTP-Net by achieving lower RMSE values compared to 11 state-of-the-art threshold estimation algorithms. This highlights the effectiveness of DTP-Net in accurately predicting the segmentation threshold values.
4. Quantitative analysis showing that the areas of skin lesions segmented using the predicted threshold values from DTP-Net exhibit a relatively better overlap with the manually contoured ground-truth.
5. The DTP-Net+MP method is computationally more feasible than the DL-based skin lesion segmentation models with encoder-decoder architecture like FCN-8s, U-Net, SegNet, and DeepLabV3+.

Overall, this chapter introduces and validates DTP-Net as a reliable and effective method for threshold estimation in dermatological macro-images, offering improved segmentation accuracy with computationally feasible results compared to DL-based segmentation methods. The DTP-Net exhibits potential applicability in automated skin cancer detection systems.

CHAPTER 6

AN INNOVATIVE APPROACH FOR SKIN LESION SEGMENTATION

This chapter comprehensively discusses an innovative approach to address the challenging task of segmenting lesions from dermatological macro-images. The main objective of this approach is to identify and delineate lesions accurately. To achieve this goal, the proposed work incorporates two preprocessing stages, each serving a critical role in improving the quality of the input image and enabling accurate segmentation. These preprocessing stages are essential for enhancing the visibility and clarity of the lesions and improving the segmentation accuracy of the segmentation algorithms.

The first preprocessing stage consists of illumination equalization using the counter exponential transform (IECET) algorithm, as discussed in Chapter 3. This algorithm aims to equalize the uneven background illumination that occurs during image acquisition of the macro-images. The second preprocessing stage incorporates an EfficientNet-based modified sigmoid transform (EMST), explained in Chapter 4. The EMST algorithm selectively enhances the contrast between the lesion and the background skin. It effectively enhances the edge information in an image, making it easier to extract precise lesion boundaries during the segmentation step.

IECET and EMST algorithms are used to enhance the quality of the input image, which helps in accurate and reliable lesion segmentation. By addressing issues related to uneven lighting and poor contrast between the lesion and background, the suggested methodology aims to overcome the limitations associated with traditional segmentation approaches and achieve improved segmentation performance with reduced computational complexity.

In the subsequent sections of this chapter, the detailed implementation and integration of the preprocessing stages, along with the subsequent lesion segmentation using the deep threshold prediction network (DTP-Net) with morphological postprocessing (MP), are discussed. Experimental results and evaluations are presented to demonstrate the effectiveness and superiority of the proposed approach in accurately identifying and delineating lesions from dermatological macro-images.

6.1 METHODOLOGY

Fig. 6.1 illustrates the schematic of the innovative approach for segmenting lesions from dermatological macro-images. It comprises two main stages. In the first stage, the input macro-images undergo preprocessing using the IECET and EMST algorithms to enhance image quality. In the second stage, the quality-enhanced macro-images are fed into the lesion segmentation block to obtain the binary segmented output images. The IECET algorithm effectively addresses the challenge of uneven background illumination commonly observed in dermatological macro-images. By employing a counter exponential transform (CET), the IECET algorithm significantly reduces the impact of uneven lighting, resulting in improved image quality. The EMST algorithm selectively improves the contrast between the lesion and the surrounding background skin. It effectively enhances the saliency and discriminability of the lesions, thereby facilitating DTP-Net to segment the lesions accurately.

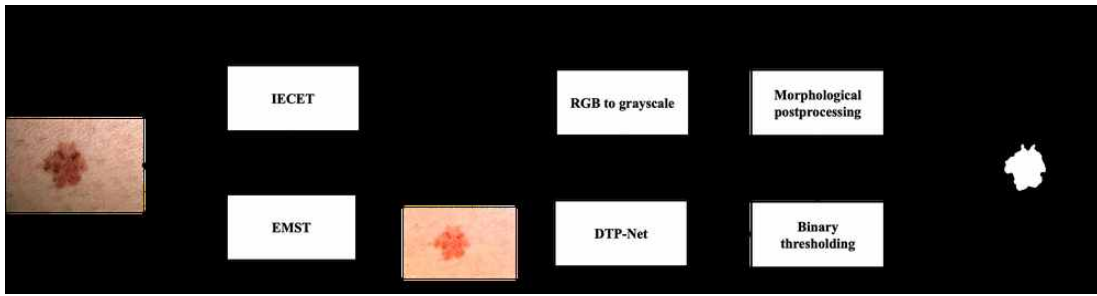


Figure 6.1: A schematic representation for skin lesion segmentation from macro-images.

The quality-enhanced macro-image obtained from the preprocessing stage is passed to the lesion segmentation block. The macro-image in RGB color space is first converted to grayscale and given as input to the DTP-Net, which predicts the optimal segmentation threshold. The binary thresholding algorithm then uses the predicted segmentation threshold value to segment the lesion from the macro-image. As discussed in the previous chapter, the segmented lesion is further refined using morphological postprocessing (MP). The MP includes a simple dilation and hole filling operation carried out using a morphological structuring element in the shape of a disk, with a kernel size set at 20 units. This results in a refined binary segmented output image. The DTP-Net+MP model has been specifically developed to address the challenges associated with lesion segmentation, leveraging the power of the deep image-to-scalar regressor. This approach is more computationally feasible than existing DL-based segmentation models, which uses image-to-image regressors.

6.2 RESULTS AND DISCUSSION

The performance of the proposed lesion segmentation approach, as described in Section 6.1, is demonstrated using five test images, as depicted in Fig. 6.2 (a). To further validate the proposed pipeline, 30 test images are selected from the University of Waterloo skin cancer database [20]. These images are specifically chosen due to the presence of uneven background illumination and poor contrast between the lesion and the surrounding skin, making them challenging cases for accurate segmentation.

In Fig. 6.2, column (a) displays five test images, column (b) presents the expert segmented ground-truth, column (c) exhibits the segmented results of DTP-Net+MP without any preprocessing, column (d) illustrates the segmented results of DTP-Net+MP with IECET, and finally, the last column (e) shows the segmented results of DTP-Net+MP with preprocessing using IECET and EMST. Fig. 6.2 (d) shows the segmented results of DTP-Net+MP on test images equalized using IECET. It is observed that these results display a higher degree of overlap with the expert-segmented ground-truth. On the other hand, the segmentation outcome of DTP-Net+MP applied directly to the test images, as shown in Fig. 6.2 (c), demonstrates a lower degree of overlap with the expert-segmented ground-truth. Fig. 6.2 (e) demonstrates a notable improvement in segmentation accuracy due to the preprocessing with the IECET and EMST algorithms. Compared to the segmentation results without any preprocessing (shown in Fig. 6.2 (c)), it is clear that combining these two preprocessing techniques significantly enhances the performance of the DTP-Net+MP in accurately segmenting the lesions from macro-images.

Quantitatively, the segmentation results in terms of DSC achieved by the DTP-Net+MP model with and without preprocessing are summarized in Table 6.1. DSC values obtained for the five test images and the average DSC computed over 30 test images are reported. Without preprocessing, the DTP-Net+MP model achieves a DSC of 0.72 for image-1, 0.85 for image-2, 0.81 for image-3, 0.81 for image-4, 0.65 for image-5, and an average DSC of 0.84 ± 0.19 over the 30 test images. However, when the input image undergoes preprocessing with the IECET algorithm alone, it achieved good improvement in segmentation accuracy in terms of DSC score on the first four test images. The DSC values for image-1, image-2, image-3, and image-4 increase to 0.83, 0.92, 0.86, and 0.82, respectively. Moreover, the average DSC for the 30 test images rises to 0.86 ± 0.12 . In image-5, the DSC score is slightly reduced after applying the IECET algorithm. This reduction in DSC is due to the lowering of contrast between the lesion and background skin. This can be observed from the last row, fourth column of Fig. 6.2.

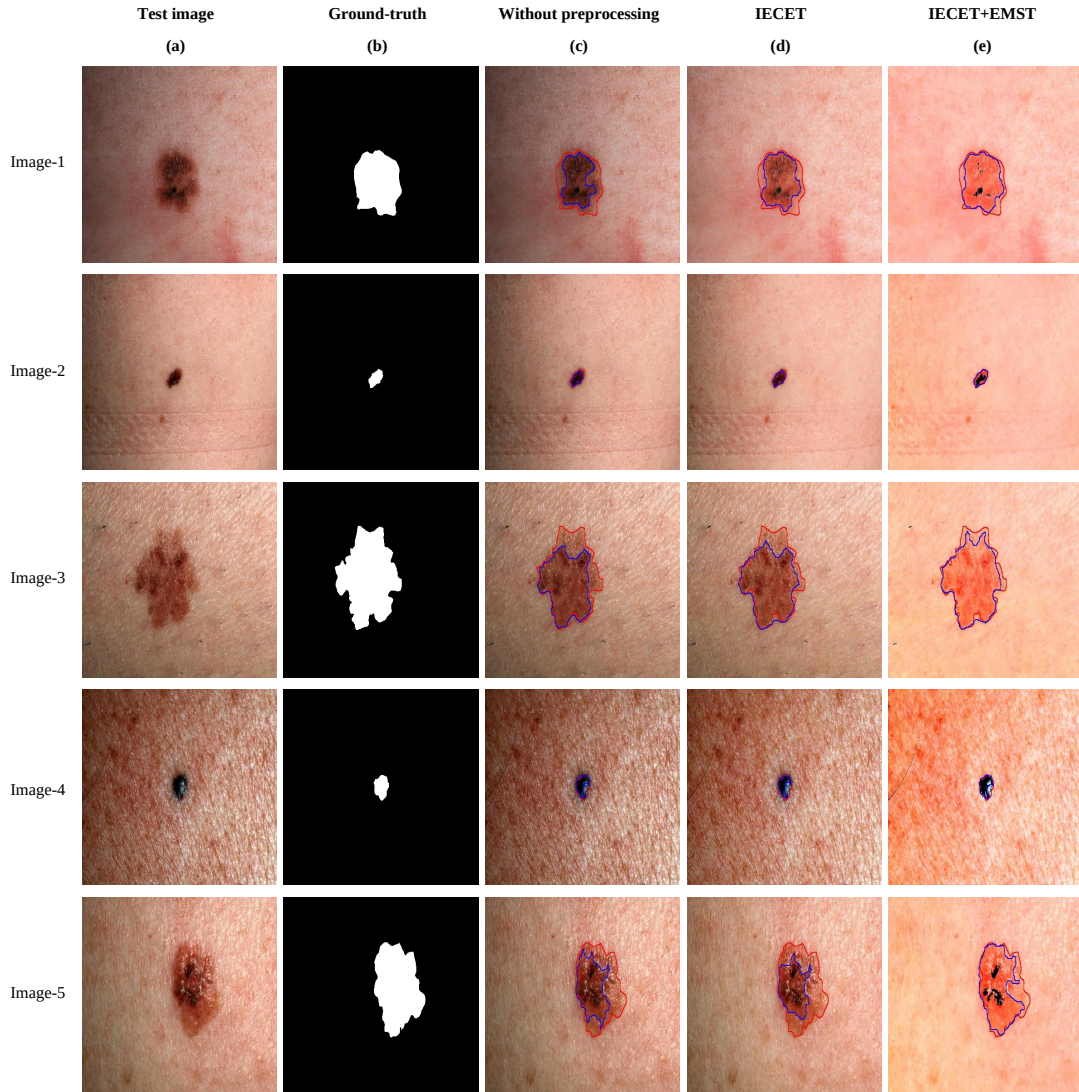


Figure 6.2: Effect of preprocessing on lesion segmentation using DTP-Net+MP: red contour represents expert-segmented ground-truth, while the blue contour represents DTP-Net+MP segmentation.

Further, applying the IECET and EMST algorithms for preprocessing leads to a significant improvement in segmentation accuracy. The DSC values for image-1, image-2, image-3, image-4, and image-5 increase from 0.72 to 0.86, from 0.85 to 0.95, from 0.81 to 0.91, from 0.81 to 0.87, and from 0.65 to 0.88, respectively. Moreover, the average DSC obtained by performing segmentation using the DTP-Net+MP for the 30 images shows a significant improvement. The DSC value rises from 0.84 ± 0.19 (without preprocessing) to 0.89 ± 0.09 (with preprocessing by IECET and EMST algorithms). Table 6.1 shows that the preprocessing techniques with IECET and EMST significantly improved the segmentation performance of the DTP-Net+MP method.

The results show the significant positive impact of the developed preprocessing algorithms on the segmentation accuracy of the DTP-Net+MP model. The DTP-Net is designed to accurately predict the threshold value from dermatological macro-images of skin lesions and perform accurate segmentation. The preprocessing stages effectively aid in selecting the optimal segmentation threshold value for the DTP-Net model. Overall, these findings strongly support the consistent performance exhibited by the proposed approach for accurately segmenting skin lesions from dermatological macro-images.

Table 6.1: Lesion segmentation performance for DTP-Net+MP algorithm (with and without preprocessing stages).

Preprocessing	Test image-1	Test image-2	Test image-3	Test image-4	Test image-5	Average DSC for 30 test images (Mean $\pm$ SD)
Without	0.72	0.85	0.81	0.81	0.65	0.84 ± 0.19
IECET	0.83	0.92	0.86	0.82	0.60	0.86 ± 0.12
IECET+EMST	0.86	0.95	0.91	0.87	0.88	0.89 ± 0.09

6.3 SUMMARY

This chapter presents the proposed innovative approach for skin lesion segmentation from dermatological macro-images. It incorporates preprocessing stages to address common challenges associated with uneven background illumination and poor image contrast of macro-images. The IECET algorithm is employed to equalize uneven background illumination, while the EMST algorithm enhances the contrast between the lesion and the background skin. Through these preprocessing stages, the quality of the macro-images are significantly improved for better segmentation.

The preprocessed image is subsequently fed into the DTP-Net+MP model, designed explicitly for lesion segmentation. Combining these preprocessing stages and the DTP-Net+MP model achieved accurate and efficient lesion segmentation.

Thus, the proposed lesion segmentation approach holds great promise for achieving highly accurate and efficient lesion segmentation, which can be incorporated into automated software tools used to analyze dermatological macro-images.

CHAPTER 7

CONCLUSIONS AND FUTURE SCOPE

This chapter is organized as follows: Section 7.1 presents concluding remarks, highlighting the key findings and contributions of this thesis. Section 7.2 explores future research directions based on the outcomes and limitations identified in this research work. It discusses potential areas for further exploration and improvements.

7.1 CONCLUSIONS

To address the issue of background illumination, IECET was developed. In order to eliminate subjectivity and manual intervention associated with the selection of the operational parameter SCP in the IECET, a modified ResNet regressor was used to automate its selection. The equalized images produced by the IECET showed higher PSNR values w.r.t the pristine images than those shown by the existing state-of-the-art illumination correction methods for macro-images. A significant improvement in segmentation accuracy of DL-based segmentation methods was observed when the macro-images were equalized with the IECET. Comparatively, the output images obtained from DeepLabV3+, FCN, and U-Net using the IECET exhibited higher DSC values than those generated by the HF, WPCCA, GCCA, and STCRD. One distinctive feature of IECET is that it is fully automated, as the modified ResNet facilitates the adaptive selection of the SCP. As the IECET uses the counter exponential transform to equalize the background lightning and the modified ResNet was used to automate the selection of SCP, the proposed algorithm is partially explainable.

The EMST network was introduced to enhance the contrast between the lesion and the background skin. Compared to the traditional sigmoid transform, EMST has provisions for fixing the boundary that divides the pixel values into lesion and background regions, between which contrast needs to be improved. The boundary pixel value, ‘crossover point’, divides the macro-image into lesion and background. It was predicted using a modified EfficientNet regressor to exclude the manual intervention and subjectivity. A significant improvement in the segmentation accuracy of OT was observed when the macro-images were preprocessed with the EMST, outperforming 15 state-of-the-art CE techniques, such as DS, FGLEI, HE, CLAHE, GEM, CEHCM, SMIRANK, FIE, CVCEA, LDR, WAHE, RICE, SHE, FLM, and AECLII.

Furthermore, a deep neural network named deep threshold prediction network (DTP-Net) was proposed for predicting the threshold value from dermatological macro images. Several efficient training procedures were implemented to overcome the network training challenges with limited training images. The DTP-Net could learn the difference between the lesion and background in the intensity space and accurately predict the threshold to separate the lesion from the background. The DTP-Net proposed in this thesis exhibited low values of RMSE compared to 11 state-of-the-art threshold estimation algorithms, namely OT, VEOT, IT, HSDDT, MET, PDMET, KMET, EOT, MCET, Type-2 FT, and FET. To evaluate the effectiveness of the DTP-Net, macro-images, and corresponding predicted threshold values were given as input to a binary thresholding algorithm. The segmented results produced by the DTP-Net exhibited a higher DSC than all the discussed threshold estimation algorithms. The DTP-Net, followed by morphological postprocessing (DTP-Net+MP), offered computational feasibility superior to DL-based skin lesion segmentation models with encoder-decoder architecture like FCN-8s, U-Net, SegNet, and DeepLabV3+. It was observed that the skin lesions segmented using the predicted threshold values from the DTP-Net displayed relatively better overlap with the manually contoured ground-truth prepared by experts when compared to alternative approaches.

In the last part of the thesis, an innovative approach for skin lesion segmentation was proposed. The integrated preprocessing stage with the IECET and EMST methods, followed by DTP-Net+MP for lesion segmentation. Segmentation on preprocessed macro-images using DTP-Net+MP significantly improved the average DSC. Combining preprocessing stages with the DTP-Net+MP segmentation framework resulted in computationally feasible and accurate lesion segmentation from dermatological macro-images of skin lesions.

Three works presented in this thesis contribute to the development of automated tools for analyzing dermatological macro-images. The proposed algorithms and techniques address key challenges related to uneven illumination, CE, and the computationally feasible segmentation of lesions from macro-images. These findings hold great promise for advancements in automated tools for diagnosing skin cancer.

7.2 FUTURE WORK

Works presented in this thesis showcase significant improvements, but there are still ample opportunities for further research and exploration, leading to better performance and increased robustness. Therefore, further developments of the proposed algorithms for quality enhancement and segmentation on macro-images are suggested.

1. DL models capable of selectively equalizing the nonuniformity of lighting in the background while preserving the pixel values in the lesions, addressing the issue of moderately enhancing the lesions by the proposed IECET algorithm, can be developed.
2. Explicitly designed segmentation algorithm for dermatological macro-images to compute the ideal crossover point for training the modified EfficientNet regressor may improve the performance of the EMST algorithm.
3. Color images may be used in place of the grayscale image as input to the DTP-Net for localizing the skin lesions, which may yield better results.
4. Create a database with lesions on darker skin to enable qualitative and quantitative evaluation of the algorithm's performance and broaden the representation of diverse skin types.

Appendices

APPENDIX A

A.1 DERIVATION OF RGB TO HSV COLOR SPACE

The RGB image is transformed to HSV color space as described in Eq. (A.1.1) to (A.1.5). The value component at a pixel location is the maximum of the gray level values from all three color channels at that location. The value component of the input image with M rows and N columns in HSV color space, at an arbitrary location (m, n) , $m = 1, 2, 3, \dots, M$ and $n = 1, 2, 3, \dots, N$, $V_i(m, n)$ can be computed from the contextual grey levels in red R_i , green G_i , and blue B_i color channels of the input image, as given in Eq. (A.1.1). At any arbitrary pixel location, the value component is the maximum of the grey scale intensities in red, green, and blue channels as expressed in Eq. (A.1.1). Before initiating the conversion, the intensity values of the individual color channels should be normalized to [0 1] [172].

$$V_i(m, n) = \max \{R_i(m, n), G_i(m, n), B_i(m, n)\}, \quad (\text{A.1.1})$$

The saturation component $S_i(m, n)$ at a pixel location (m, n) , $m = 1, 2, 3, \dots, N$ and $n = 1, 2, 3, \dots, N$ can be calculated using the value component $V_i(m, n)$ obtained from Eq. (A.1.1) and the minimum of the gray level values from all three color channels at that location. The saturation component $S_i(m, n)$ at a pixel location (m, n) is computed using Eq. (A.1.2) [172].

$$S_i(m, n) = \frac{V_i(m, n) - I_{\min}(m, n)}{V_i(m, n)} \quad (\text{A.1.2})$$

The minimum of the gray level values from all three color channels $I_{\min}(m, n)$ at a location (m, n) is [172],

$$I_{\min}(m, n) = \min \{R_i(m, n), G_i(m, n), B_i(m, n)\} \quad (\text{A.1.3})$$

The hue component $H_i(m, n)$ at a pixel location (m, n) is given by [172],

$$H_i(m, n) = \begin{cases} 5 + b_i(m, n); & \text{if } V_i(m, n) = R_i(m, n) \text{ and } G_i(m, n) = I_{min}(m, n) \\ 1 - g_i(m, n); & \text{if } V_i(m, n) = R_i(m, n) \text{ and } G_i(m, n) \neq I_{min}(m, n) \\ 1 + r_i(m, n); & \text{if } V_i(m, n) = G_i(m, n) \text{ and } B_i(m, n) = I_{min}(m, n) \\ 3 - b_i(m, n); & \text{if } V_i(m, n) = G_i(m, n) \text{ and } B_i(m, n) \neq I_{min}(m, n) \\ 3 + g_i(m, n); & \text{if } V_i(m, n) = B_i(m, n) \text{ and } R_i(m, n) = I_{min}(m, n) \\ 5 - r_i(m, n); & \text{if } V_i(m, n) = B_i(m, n) \text{ and } R_i(m, n) \neq I_{min}(m, n) \end{cases} \quad (\text{A.1.4})$$

In Eq. (A.1.4), the values of the intermediate variables $r_i(m, n)$, $g_i(m, n)$, and $b_i(m, n)$ are given by [172],

$$\begin{aligned} r_i(m, n) &= \frac{V_i(m, n) - R_i(m, n)}{V_i(m, n) - I_{min}(m, n)}, \\ g_i(m, n) &= \frac{V_i(m, n) - G_i(m, n)}{V_i(m, n) - I_{min}(m, n)}, \\ b_i(m, n) &= \frac{V_i(m, n) - B_i(m, n)}{V_i(m, n) - I_{min}(m, n)} \end{aligned} \quad (\text{A.1.5})$$

The hue component H_i is normalized to a range [0 1] by dividing the values in the component with a scalar value of 6.

A.2 DERIVATION OF HSV TO RGB COLOR SPACE

The HSV color space image are converted back to RGB color space using Eq. (A.2.6)-(A.2.12) [172]. Let H_o , S_o , and V_o be the normalized hue, saturation, and value components of the enhanced output image in HSV color space. R_o , G_o , and B_o are the normalized red, green, and blue color channels of the enhanced output image in RGB color space.

$$\begin{aligned} R_o(m, n) &= \gamma(m, n) * r_o(m, n), \\ G_o(m, n) &= \gamma(m, n) * g_o(m, n), \\ B_o(m, n) &= \gamma(m, n) * b_o(m, n) \end{aligned} \quad (\text{A.2.6})$$

In Eq. (A.2.6), the intermediate temporary variable $\gamma(m, n)$ is [172],

$$\gamma(m, n) = \frac{V_o(m, n)}{\max \{ \max \{ r_o \}, \max \{ g_o \}, \max \{ b_o \} \}} \quad (\text{A.2.7})$$

The variables, r_o , g_o , and b_o in Eq. (A.2.7) are [172],

$$r_o(m, n) = \begin{cases} 1; \text{ if } F(m, n) = 0 \text{ or } F(m, n) = 6 \\ B(m, n); \text{ if } F(m, n) = 1 \\ A(m, n); \text{ if } F(m, n) = 2 \\ A(m, n); \text{ if } F(m, n) = 3 \\ C(m, n); \text{ if } F(m, n) = 4 \\ 1; \text{ if } F(m, n) = 5 \end{cases} \quad (\text{A.2.8})$$

$$g_o(m, n) = \begin{cases} D(m, n); \text{ if } F(m, n) = 0 \text{ or } F(m, n) = 6 \\ 1; \text{ if } F(m, n) = 1 \\ 1; \text{ if } F(m, n) = 2 \\ B(m, n); \text{ if } F(m, n) = 3 \\ A(m, n); \text{ if } F(m, n) = 4 \\ A(m, n); \text{ if } F(m, n) = 5 \end{cases} \quad (\text{A.2.9})$$

$$b_o(m, n) = \begin{cases} A(m, n); \text{ if } F(m, n) = 0 \text{ or } F(m, n) = 6 \\ A(m, n); \text{ if } F(m, n) = 1 \\ D(m, n); \text{ if } F(m, n) = 2 \\ 1; \text{ if } F(m, n) = 3 \\ 1; \text{ if } F(m, n) = 4 \\ B(m, n); \text{ if } F(m, n) = 5 \end{cases} \quad (\text{A.2.10})$$

Unknown temporary variables in Eq. (A.2.8) to Eq. (A.2.10) are [172],

$$\begin{aligned}
D(m, n) &= \hat{H}_o(m, n) - F(m, n), \\
A(m, n) &= 1 - S_o(m, n), \\
B(m, n) &= 1 - [S_o(m, n) * D(m, n)], \\
C(m, n) &= 1 - [S_o(m, n) * (1 - D(m, n))]
\end{aligned} \tag{A.2.11}$$

In Eq. (A.2.11), $\hat{H}_o(m, n)$ and $F(m, n)$ is given by,

$$\begin{aligned}
\hat{H}_o(m, n) &= 6 * H_o(m, n) \\
F(m, n) &= \varphi(H_o(m, n))
\end{aligned} \tag{A.2.12}$$

The operator $\varphi(.)$ in Eq. (A.2.12) scales down its argument to the nearest integer value [172].

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PUBLICATIONS

Journals

1. **Vipin, V.**, Joseph, J., Das, M. V, and Nath, M. K., “An EfficientNet-based modified sigmoid transform for enhancing dermatological macro-images of melanoma and nevi skin lesions”, *Computer Methods and Programs in Biomedicine, Elsevier*, 222, 106935 (1-14), July 2022. (Indexing: SCI & Scopus, IF: 6.1).
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4. **Vipin, V.**, Raj, N. I., Nath, M. K., and Stephen, N., “A deep neural network using modified EfficientNet for skin cancer detection in dermoscopic images”, *Decision Analytics Journal, Elsevier*, 08, 100278 (1-11), September 2023. (Indexing: Scopus).
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5. **Vipin, V.**, Nath, M. K., Joseph, J., and Das, M. V. “An automated illumination correction algorithm based on counter exponential transform and modified ResNet for macroscopic images of the skin lesions”, *Image and Vision Computing, Elsevier*. (Indexing: SCI & Scopus, IF: 4.7)
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6. **Vipin, V.**, Nath, M. K., Joseph, J., and Das, M. V., “GoogLeNet-based image to scalar regressor network for quality assessment of illumination correction in macroscopic images of skin lesions”, *Pattern Recognition, Elsevier*. (Indexing: SCI & Scopus, IF: 8)
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7. **Vipin, V.**, Raj, N. I., and Nath, M. K., “Classification of skin lesions using deep layered fine-tuned convolutional neural networks”, *Biomedical Engineering Advances, Elsevier*.
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Conferences

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